Grealy et al. 2026
R Code with model outputs (with age)

```
#Set working directory

>setwd("~/Desktop/MAC/Environomics_FSP/Analysis/Proof_of_concept_paper/JUL
Y22/R")

#Load packages

>library(lme4)

>library(performance)

>library(lmerTest)

>library(emmeans)

>library(glmmTMB)

>library(ggplot2)

>library(tidyr)

#-------------------------------------
#Evenness of sequencing depth
#-------------------------------------
  #Experiment 3 and 4
    #lmer

#The within-pool co-efficient of variation (CV) was calculated for each
sample.

#The co-efficient of variation (CV) is calculated as the standard
deviation of within-pool read proportions divided by the mean of the
within-pool read proportions.

#A CV of 0 means samples were sequenced exactly evenly.

#Load dataset

>evenness_cv_exp34<-read.csv("evenness_cv_exp34.csv",header=TRUE)

>head(evenness_cv_exp34)

Sequencing    Batch Treatment wn_batch_std        CV
1    shotgun AG_P1P1C        5S    1463625.8 0.7171633
2    shotgun AG_P1P2C        5S     553628.5 0.2282091
3    shotgun AG_P1P3C        5S    1125369.3 0.6254850
4    shotgun AG_P1P4C        5S    1514070.9 0.5562136
5    shotgun AG_P1P5C        2M     977926.8 0.3067907
6    shotgun AG_P1P6C        2M     690503.9 0.2762951

#We performed a linear mixed model in R v4.2.3 with the CV as the response
variable, with capture or shotgun (“Sequencing”) and library preparation
input and size (“Treatment”) as fixed effects, and pool (“Batch”) as a
random effect.
```

```
>cv.mod<-lmer(CV ~ Treatment * Sequencing + (1|Batch),
data=evenness_cv_exp34,REML=TRUE)

>summary(cv.mod)

Linear mixed model fit by REML. t-tests use Satterthwaite's method
['lmerModLmerTest']
Formula: CV ~ Treatment * Sequencing + (1 | Batch)
   Data: evenness_cv_exp34

REML criterion at convergence: -23

Scaled residuals:
     Min       1Q   Median       3Q      Max
-1.71194 -0.40314 -0.08144  0.51172  1.54638

Random effects:
 Groups   Name        Variance Std.Dev.
 Batch    (Intercept) 0.006832 0.08266
 Residual             0.009516 0.09755
Number of obs: 36, groups:  Batch, 24

Fixed effects:
                             Estimate Std. Error        df t value
Pr(>|t|)
(Intercept)                  0.296736   0.063930 24.734298   4.642
9.66e-05 ***
Treatment2M                  0.015745   0.090411 24.734298   0.174
0.863
Treatment40M                 0.067600   0.090411 24.734298   0.748
0.462
Treatment5M                 -0.052148   0.090411 24.734298  -0.577
0.569
Treatment5S                  0.016354   0.078298 24.734298   0.209
0.836
Sequencingshotgun           -0.031614   0.121974 15.836068  -0.259
0.799
Treatment2M:Sequencingshotgun   0.009019   0.151687 15.005065   0.059
0.953
Treatment40M:Sequencingshotgun  0.040858   0.144084 14.620578   0.284
0.781
Treatment5M:Sequencingshotgun   0.074470   0.151687 15.005065   0.491
0.631
Treatment5S:Sequencingshotgun   0.227040   0.137635 15.327258   1.650
0.119
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Correlation of Fixed Effects:
            (Intr) Trtm2M Trt40M Trtm5M Trtm5S Sqncng Tr2M:S T40M:S Tr5M:S
Treatment2M -0.707
Treatmnt40M -0.707  0.500
Treatment5M -0.707  0.500  0.500
Treatment5S -0.816  0.577  0.577  0.577
Sqncngshtgn -0.305  0.216  0.216  0.216  0.249
Trtmnt2M:Sq  0.245 -0.347 -0.173 -0.173 -0.200 -0.804
Trtmnt40M:S  0.258 -0.183 -0.365 -0.183 -0.211 -0.847  0.681
Trtmnt5M:Sq  0.245 -0.173 -0.173 -0.347 -0.200 -0.804  0.647  0.681
Trtmnt5S:Sq  0.270 -0.191 -0.191 -0.191 -0.331 -0.886  0.713  0.750  0.713
```

```
#Model estimated means and 95% confidence intervals were calculated:

>emm<- emmeans(cv.mod, ~Treatment*Sequencing, lmer.df="satterthwaite")

>emm

Treatment Sequencing emmean     SE   df lower.CL upper.CL
 20M       capture     0.297 0.0639 24.7   0.1650    0.428
 2M        capture     0.312 0.0639 24.7   0.1807    0.444
 40M       capture     0.364 0.0639 24.7   0.2326    0.496
 5M        capture     0.245 0.0639 24.7   0.1128    0.376
 5S        capture     0.313 0.0452 24.7   0.2199    0.406
 20M       shotgun     0.265 0.1190 22.8   0.0185    0.512
 2M        shotgun     0.290 0.0864 25.4   0.1121    0.468
 40M       shotgun     0.374 0.0722 25.9   0.2252    0.522
 5M        shotgun     0.287 0.0864 25.4   0.1097    0.465
 5S        shotgun     0.509 0.0611 25.4   0.3828    0.634

Degrees-of-freedom method: satterthwaite
Confidence level used: 0.95

#Further simplify the model by removing the interaction between
“Treatment” and “Sequencing”

>cv.mod2<-lmer(CV ~ Treatment + Sequencing + (1|Batch),
data=evenness_cv_exp34,REML=TRUE)

>anova(cv.mod, cv.mod2)

Data: evenness_cv_exp34
Models:
cv.mod2: CV ~ Treatment + Sequencing + (1 | Batch)
cv.mod: CV ~ Treatment * Sequencing + (1 | Batch)
        npar     AIC     BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
cv.mod2    8 -36.135 -23.466 26.067   -52.135
cv.mod    12 -36.004 -17.002 30.002   -60.004 7.8694  4    0.09648 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

#The interaction is non-signficiant. Further simplify the model by
removing the “Sequencing” term:

>cv.mod3<-lmer(CV ~ Treatment + (1|Batch),
data=evenness_cv_exp34,REML=TRUE)

>anova(cv.mod2, cv.mod3)

Data: evenness_cv_exp34
Models:
cv.mod3: CV ~ Treatment + (1 | Batch)
cv.mod2: CV ~ Treatment + Sequencing + (1 | Batch)
        npar     AIC     BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
cv.mod3    7 -34.589 -23.505 24.295   -48.589
cv.mod2    8 -36.135 -23.466 26.067   -52.135 3.5453  1    0.05972 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

#“Sequencing” is not a significant factor. Further simplify the model by removing “Treatment”:

```
>cv.mod4<-lmer(CV ~ (1|Batch), data=evenness_cv_exp34,REML=TRUE)

>anova(cv.mod3, cv.mod4)

Data: evenness_cv_exp34
Models:
cv.mod4: CV ~ (1 | Batch)
cv.mod3: CV ~ Treatment + (1 | Batch)
        npar     AIC     BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
cv.mod4    3 -39.072 -34.322 22.536   -45.072
cv.mod3    7 -34.589 -23.505 24.295   -48.589 3.5171  4     0.4753

#”Treatment” is not a significant factor. The random factor “Batch” contributes marginally to variance.

>VarCorr(cv.mod4)

Groups   Name        Std.Dev.
 Batch    (Intercept) 0.091664
 Residual             0.100915

>summary(cv.mod4)

Linear mixed model fit by REML. t-tests use Satterthwaite's method
['lmerModLmerTest']
Formula: CV ~ (1 | Batch)
   Data: evenness_cv_exp34

REML criterion at convergence: -39.6

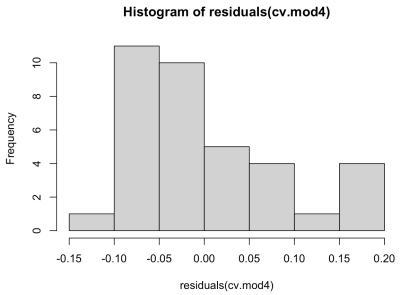
Scaled residuals:
    Min      1Q  Median      3Q     Max
-1.0445 -0.5705 -0.2375  0.4171  1.9749

Random effects:
 Groups   Name        Variance Std.Dev.
 Batch    (Intercept) 0.008402 0.09166
 Residual             0.010184 0.10091
Number of obs: 36, groups:  Batch, 24

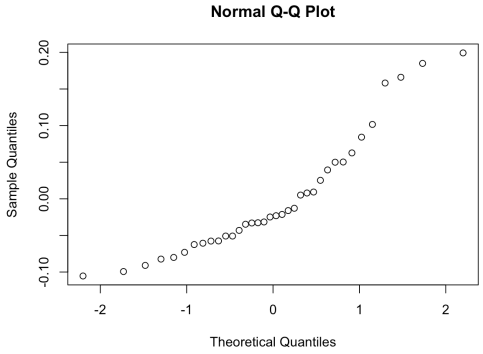
Fixed effects:
            Estimate Std. Error       df t value Pr(>|t|)
(Intercept)  0.32783    0.02552 24.92729   12.84 1.72e-12 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

#Model residuals were approximately normally distributed according to the Q-Q plot, however, the Shapiro-Wilk test indicated significant deviations from normality:

>hist(residuals(cv.mod4))
```




```
>qqnorm(residuals(cv.mod4))
```




```
>shapiro.test(residuals(cv.mod4))

Shapiro-Wilk normality test

data:  residuals(cv.mod4)
W = 0.89113, p-value = 0.001969

#As such, we performed a glmm with a beta family distribution and logit
link transformation (below).

#-------------------------------------

  #Experiment 3 and 4
   #glmm

#We performed a generalised linear mixed model in R v4.2.3 with the CV as
the response variable, with capture or shotgun (“Sequencing”) and library
preparation input and size (“Treatment”) as fixed effects, and pool
(“Batch”) as a random effect:

>cv.mod5<-glmmTMB(CV ~ Treatment * Sequencing + (1|Batch),
family=beta_family(link="logit"), data=evenness_cv_exp34)

#The interaction term between Treatment and Sequencing was tested:

>cv.mod6<-glmmTMB(CV ~ Treatment + Sequencing + (1|Batch),
family=beta_family(link="logit"), data=evenness_cv_exp34)

>anova(cv.mod5, cv.mod6)

Data: evenness_cv_exp34
Models:
cv.mod6: CV ~ Treatment + Sequencing + (1 | Batch), zi=~0, disp=~1
```

```
cv.mod5: CV ~ Treatment * Sequencing + (1 | Batch), zi=~0, disp=~1
        Df     AIC     BIC logLik deviance  Chisq Chi Df Pr(>Chisq)
cv.mod6  8 -40.689 -28.021 28.345  -56.689
cv.mod5 12 -40.287 -21.285 32.143  -64.287 7.5974      4     0.1075
```

#The interaction between Treatment and Sequencing was non-significant and was removed from the model. Next, the Treatment effect was tested:

```
>cv.mod7<-glmmTMB(CV ~ Sequencing + (1|Batch),
family=beta_family(link="logit"), data=evenness_cv_exp34)

>anova(cv.mod6, cv.mod7)

Data: evenness_cv_exp34
Models:
cv.mod7: CV ~ Sequencing + (1 | Batch), zi=~0, disp=~1
cv.mod6: CV ~ Treatment + Sequencing + (1 | Batch), zi=~0, disp=~1
        Df     AIC     BIC logLik deviance  Chisq Chi Df Pr(>Chisq)
cv.mod7  4 -45.198 -38.864 26.599  -53.198
cv.mod6  8 -40.689 -28.021 28.345  -56.689 3.4911      4     0.4792
```

#Treatment was a non-significant term and could be removed from the model. Next, the Sequencing effect was tested:

```
>cv.mod8<-glmmTMB(CV ~ (1|Batch), family=beta_family(link="logit"),
data=evenness_cv_exp34)

>anova(cv.mod7, cv.mod8)

Data: evenness_cv_exp34
Models:
cv.mod8: CV ~ (1 | Batch), zi=~0, disp=~1
cv.mod7: CV ~ Sequencing + (1 | Batch), zi=~0, disp=~1
        Df     AIC     BIC logLik deviance  Chisq Chi Df Pr(>Chisq)
cv.mod8  3 -43.606 -38.856 24.803  -49.606
cv.mod7  4 -45.198 -38.864 26.599  -53.198 3.5921      1    0.05805 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

#Sequencing was a non-significant term and could be removed from the model.

```
>summary(cv.mod8)

Family: beta  ( logit )
Formula:          CV ~ (1 | Batch)
Data: evenness_cv_exp34

      AIC       BIC    logLik -2*log(L)  df.resid
    -43.6     -38.9      24.8     -49.6        33

Random effects:

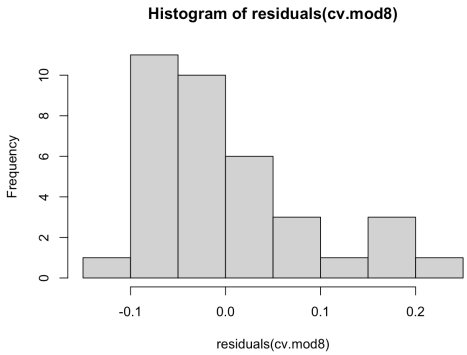
Conditional model:
 Groups Name        Variance Std.Dev.
 Batch  (Intercept) 0.1503   0.3876
Number of obs: 36, groups:  Batch, 24

Dispersion parameter for beta family (): 22.9
```

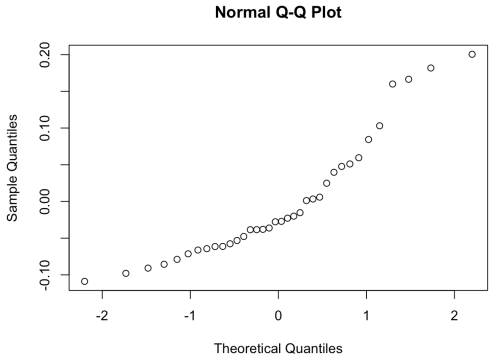
```
Conditional model:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -0.7306     0.1125  -6.494 8.38e-11 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

#Model residuals remain approximately normally distributed according to the Q-Q plot, however, the Shapiro-Wilk test still indicated significant deviations from normality:

```
>hist(residuals(cv.mod8))
```




```
>qqnorm(residuals(cv.mod8))
```




```
>shapiro.test(residuals(cv.mod8))

Shapiro-Wilk normality test

data:  residuals(cv.mod8)
W = 0.88449, p-value = 0.00132
```

```
#-----------------------------------
  #Evenness
    #Experiment 5

>evenness_cv_exp5<-read.csv("evenness_cv_exp5.csv",header=TRUE)
```

#The within-pool co-efficient of variation (CV) was calculated for each sample. We performed a linear model in R v4.2.3 with the CV as the response variable, with capture conditions (“Treatment”) as a fixed effect. Model residuals were normally distributed. Pool could not be

included as a random effect because there was only one observation per pool:

```
>cv.mod<-lm(CV ~ Treatment, data= evenness_cv_exp5)

>summary(cv.mod)

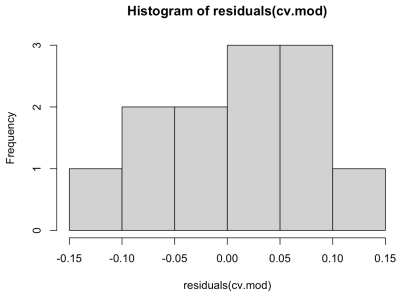
Call:
lm(formula = CV ~ Treatment, data = evenness_cv_exp5)

Residuals:
      Min        1Q    Median        3Q       Max
-0.113918 -0.046231  0.002676  0.059374  0.111015

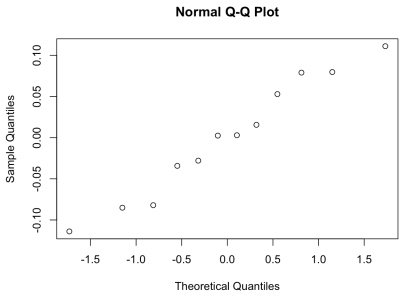
Coefficients:
                  Estimate Std. Error t value Pr(>|t|)
(Intercept)        0.522606   0.032448  16.106 6.07e-08 ***
Treatment32rxn    -0.007668   0.056201  -0.136    0.894
Treatmenthalf_bait -0.002553   0.056201  -0.045    0.965
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.07948 on 9 degrees of freedom
Multiple R-squared:  0.002064,      Adjusted R-squared:  -0.2197
F-statistic: 0.009307 on 2 and 9 DF,  p-value: 0.9907

>hist(residuals(cv.mod))
```




```
>qqnorm(residuals(cv.mod))
```




```
>shapiro.test(residuals(cv.mod))

Shapiro-Wilk normality test

data:  residuals(cv.mod)
W = 0.95943, p-value = 0.7756
```

#Model estimated means and 95% confidence intervals were calculated:

```
>emm<- emmeans(cv.mod, ~Treatment, lmer.df="satterthwaite")

>emm

Treatment emmean     SE df lower.CL upper.CL
 16rxn      0.523 0.0324  9    0.449    0.596
 32rxn      0.515 0.0459  9    0.411    0.619
 half_bait  0.520 0.0459  9    0.416    0.624

Confidence level used: 0.95

#-----------------------------------
#Enrichment (Aim1)
#-----------------------------------
#Target recovery
#lmer

>exp1<-read.csv("exp1_w_year.csv", header=TRUE)

>head(exp1)

                                    Sample Year Age Treatment     Batch
Recovery Chlp
1                Brachyscome_barkerae_AG003 2011   9   shotgun AG_P1P11C
0.5  0.9
2              Brachyscome_basaltica_AG004 2013   7   shotgun AG_P1P11C
0.3  2.0
3 Brachyscome_ciliaris_var_lanuginosa_AG010 2016   4   shotgun AG_P1P12C
0.6  0.7
4             Brachyscome_melanocarpa_AG035 2010  10   shotgun AG_P1P12C
0.5  0.4
5                Brachyscome_obovata_AG041 2010  10   shotgun AG_P1P12C
0.4  1.1
6              Brachyscome_papillosa_AG043 2011   9   shotgun AG_P1P12C
0.5  1.0
```

#We performed a linear mixed model with “Treatment” (shotgun or capture) as a fixed effect, “Batch” (pool) as a random effect, and “Sample nested in Batch” as a random effect. “Sample nested in Batch” was removed as an effect because it contributed 0 to variance. Note that phylogenetic effects were not included i.e., tribe and genus because this experiment only covered one tribe and two genera.

```
>exp1.mod<-lmer(Recovery ~ Treatment * Year + (1 | Batch) + (1|
Batch:Sample), data=exp1)

>VarCorr(exp1.mod)

Groups       Name        Std.Dev.
 Batch:Sample (Intercept) 0.0000
 Batch        (Intercept) 1.2931
 Residual                 3.9509
```

#Remove Batch:Sample term as it contributes 0 to variance.

```
>exp1.mod2<-lmer(Recovery ~ Treatment * Year + (1 | Batch), data=exp1)
```

```
>anova(exp1.mod,exp1.mod2)

Data: exp1
Models:
exp1.mod2: Recovery ~ Treatment * Year + (1 | Batch)
exp1.mod: Recovery ~ Treatment * Year + (1 | Batch) + (1 | Batch:Sample)
          npar AIC    BIC logLik -2*log(L) Chisq Df Pr(>Chisq)
exp1.mod2    6 368 380.95   -178       356
exp1.mod     7 370 385.11   -178       356     0  1          1

#Test removing Batch:

>exp1.mod3<-lm(Recovery ~ Treatment * Year, data=exp1)

>anova(exp1.mod2,exp1.mod3)

Data: exp1
Models:
exp1.mod3: Recovery ~ Treatment * Year
exp1.mod2: Recovery ~ Treatment * Year + (1 | Batch)
          npar   AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp1.mod3    5 366.9 377.70 -178.45     356.9
exp1.mod2    6 368.0 380.95 -178.00     356.0 0.8999  1     0.3428

#Batch is also a non-significant term and can be removed from the model.
#Test removing the interaction between Treatment and Year

>exp1.mod4<-lm(Recovery ~ Treatment + Year, data=exp1)

>anova(exp1.mod3,exp1.mod4)

Analysis of Variance Table

Model 1: Recovery ~ Treatment * Year
Model 2: Recovery ~ Treatment + Year
  Res.Df    RSS Df Sum of Sq      F Pr(>F)
1     60 989.88
2     61 996.54 -1   -6.6526 0.4032 0.5278

#The interaction between treatment and year is non-significant and can be
removed from the model.
#Test removing Year:

>exp1.mod5<-lm(Recovery ~ Treatment, data=exp1)

>anova(exp1.mod4,exp1.mod5)

Analysis of Variance Table

Model 1: Recovery ~ Treatment + Year
Model 2: Recovery ~ Treatment
  Res.Df     RSS Df Sum of Sq      F Pr(>F)
1     61  996.54
2     62 1001.88 -1   -5.3387 0.3268 0.5697

#Year is non-significant and can be removed from the model.

>AIC(exp1.mod5)
```

```
[1] 363.6718
```

```
>summary(exp1.mod5)

Call:
lm(formula = Recovery ~ Treatment, data = exp1)

Residuals:
    Min      1Q  Median      3Q     Max
-6.6000 -2.2750 -0.0625  0.1375 14.2000

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)       14.4000     0.7106   20.26   <2e-16 ***
Treatmentshotgun -13.9375     1.0050  -13.87   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 4.02 on 62 degrees of freedom
Multiple R-squared:  0.7562,  Adjusted R-squared:  0.7523
F-statistic: 192.3 on 1 and 62 DF,  p-value: < 2.2e-16
```

#Model estimated means and 95% confidence intervals were calculated:

```
>emm<- emmeans(exp1.mod5, ~Treatment, lmer.df="satterthwaite")
```

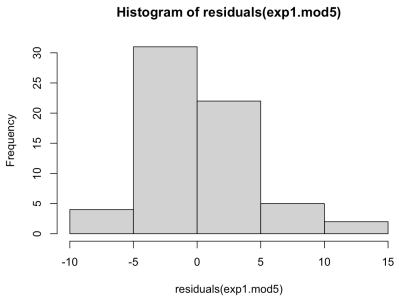
```
>emm

Treatment emmean    SE df lower.CL upper.CL
 capture   14.400 0.711 62   12.979    15.82
 shotgun    0.463 0.711 62   -0.958     1.88

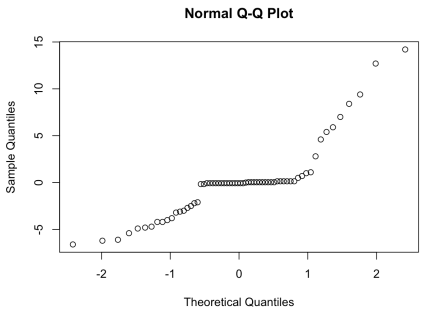
Confidence level used: 0.95
```

#Model residuals were approximately normally distributed according to the histogram and Q-Q plot, however, the Shapiro-Wilk test indicated significant deviations from normality:

```
>hist(residuals(exp1.mod5))
```




```
>qqnorm(residuals(exp1.mod5))
```




```
>shapiro.test(residuals(exp1.mod5))

Shapiro-Wilk normality test

data:  residuals(exp1.mod5)
W = 0.82325, p-value = 2.762e-07

#-----------------------------------

#Target recovery
#glmm

#Change target recovery to a proportion

>exp1$Recovery_prop <- exp1$Recovery / 100

#We performed a glmm with a beta family distribution and logit link
transformation in R v4.2.3 with the proportion of target recovered as the
response variable (“Recovery”, with capture or shotgun (“Sequencing”) and
library preparation input and size (“Treatment”) as fixed effects, and
pool (“Batch”) as a random effect:

>exp1.mod6<-glmmTMB(Recovery_prop ~ Treatment * Year + (1 | Batch) + (1|
Batch:Sample), family=beta_family(link="logit"), data=exp1)

#Test removing Sample nested within Batch:

>exp1.mod7<-glmmTMB(Recovery_prop ~ Treatment * Year + (1 | Batch),
family=beta_family(link="logit"), data=exp1)

>anova(exp1.mod6, exp1.mod7)

Data: exp1
Models:
exp1.mod7: Recovery_prop ~ Treatment * Year + (1 | Batch), zi=~0, disp=~1
exp1.mod6: Recovery_prop ~ Treatment * Year + (1 | Batch) + (1 |
Batch:Sample), zi=~0, disp=~1
          Df     AIC     BIC logLik deviance  Chisq Chi Df Pr(>Chisq)
exp1.mod7  6 -345.60 -332.65 178.80  -357.60
exp1.mod6  7 -400.34 -385.22 207.17  -414.34 56.737      1   4.98e-14 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

#Sample nested within Batch contributes significantly to variance and
cannot be removed from the model. Test removing the Batch effect:

>exp1.mod8<-glmmTMB(Recovery_prop ~ Treatment * Year + (1| Batch:Sample),
family=beta_family(link="logit"), data=exp1)
```

```
>anova(exp1.mod6, exp1.mod8)

Data: exp1
Models:
exp1.mod8: Recovery_prop ~ Treatment * Year + (1 | Batch:Sample), zi=~0,
disp=~1
exp1.mod6: Recovery_prop ~ Treatment * Year + (1 | Batch) + (1 |
Batch:Sample), zi=~0, disp=~1
          Df     AIC     BIC logLik deviance  Chisq Chi Df Pr(>Chisq)
exp1.mod8  6 -400.84 -387.89 206.42  -412.84
exp1.mod6  7 -400.34 -385.22 207.17  -414.34 1.4969      1     0.2211

#Batch is non-significant and can be removed from the model.
#Test removing the interaction term between treatment and year:

>exp1.mod9<-glmmTMB(Recovery_prop ~ Treatment + Year + (1| Batch:Sample),
family=beta_family(link="logit"), data=exp1)

>anova(exp1.mod8, exp1.mod9)

Data: exp1
Models:
exp1.mod9: Recovery_prop ~ Treatment + Year + (1 | Batch:Sample), zi=~0,
disp=~1
exp1.mod8: Recovery_prop ~ Treatment * Year + (1 | Batch:Sample), zi=~0,
disp=~1
          Df     AIC     BIC logLik deviance  Chisq Chi Df Pr(>Chisq)
exp1.mod9  5 -399.67 -388.88 204.84  -409.67
exp1.mod8  6 -400.84 -387.89 206.42  -412.84 3.1704      1    0.07498 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

#The interaction term between treatment and year is non-significant and
can be removed from the model.
#Test removing Year:

>exp1.mod10<-glmmTMB(Recovery_prop ~ Treatment + (1| Batch:Sample),
family=beta_family(link="logit"), data=exp1)

>anova(exp1.mod9, exp1.mod10)

Data: exp1
Models:
exp1.mod10: Recovery_prop ~ Treatment + (1 | Batch:Sample), zi=~0, disp=~1
exp1.mod9: Recovery_prop ~ Treatment + Year + (1 | Batch:Sample), zi=~0,
disp=~1
           Df     AIC     BIC logLik deviance  Chisq Chi Df Pr(>Chisq)
exp1.mod10  4 -401.39 -392.75 204.69  -409.39
exp1.mod9   5 -399.67 -388.88 204.84  -409.67 0.2821      1     0.5953

#Year is insignificant and can be removed from the model.

summary(exp1.mod10)

Family: beta  ( logit )
Formula:          Recovery_prop ~ Treatment + (1 | Batch:Sample)
Data: exp1
```

```
      AIC       BIC    logLik -2*log(L)  df.resid
   -401.4    -392.8     204.7    -409.4        60

Random effects:

Conditional model:
 Groups       Name        Variance Std.Dev.
 Batch:Sample (Intercept) 0.1645   0.4055
Number of obs: 64, groups:  Batch:Sample, 32

Dispersion parameter for beta family (): 1.29e+03

Conditional model:
                 Estimate Std. Error z value Pr(>|z|)
(Intercept)      -1.84171    0.07318  -25.17   <2e-16 ***
Treatmentshotgun -3.61038    0.07454  -48.44   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
>AIC(exp1.mod10)
```

```
[1] -401.3876
```

```
>emm<- emmeans(exp1.mod10, ~Treatment, type="response")
```

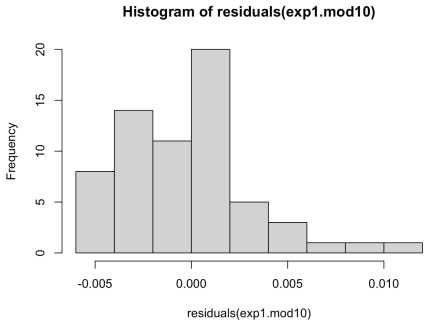
```
>emm
```

```
Treatment response       SE  df asymp.LCL asymp.UCL
 capture    0.13685 0.008640 Inf   0.12077   0.15469
 shotgun    0.00427 0.000437 Inf   0.00349   0.00522

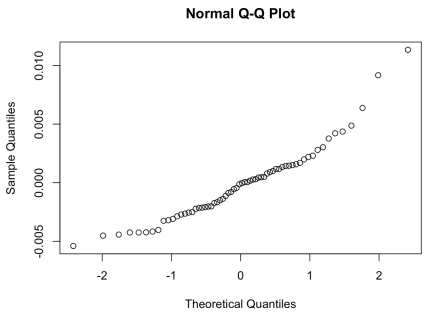
Confidence level used: 0.95
Intervals are back-transformed from the logit scale
```

```
#Model residuals remain approximately normally distributed according to
the Q-Q plot, however, the Shapiro-Wilk test still indicated significant
deviations from normality:
```

```
>hist(residuals(exp1.mod10))
```




```
>qqnorm(residuals(exp1.mod10))
```




```
>shapiro.test(residuals(exp1.mod10))

Shapiro-Wilk normality test

data:  residuals(exp1.mod10)
W = 0.93213, p-value = 0.00166

#-----------------------------------

#Target Recovery
#Non-parametric

#The absolute difference in the percentage of reads on target between the
shotgun and capture datasets was also calculated for each sample (n = 32):

>wide <- pivot_wider(exp1, id_cols = c(Sample,Batch),names_from =
Treatment, values_from = Recovery)

>wide$Diff <-wide$shotgun - wide$capture

#A non-parametric Wilcoxon signed rank test with continuity correction was
performed in R v4.2.3:

>wilcox.test(wide$shotgun, wide$capture, exact=FALSE)

Wilcoxon rank sum test with continuity correction

data:  wide$shotgun and wide$capture
W = 0, p-value = 4.509e-12
alternative hypothesis: true location shift is not equal to 0

#-----------------------------------

#Chloroplast recovery
#lmer

#We also performed a linear mixed model with chloroplast recovery (“Chlp”)
as the response variable, “Treatment” (shotgun or capture) as a fixed
effect, “Batch” (pool) as a random effect, and “Sample nested in Batch” as
a random effect. “Sample nested in Batch” was removed as an effect because
it contributed 0 to variance.

>exp1.mod11<-lmer(Chlp ~Treatment * Age + (1|Batch) + (1|Batch:Sample),
data=exp1)

#Test removing Sample nested within Batch:
```

```
>exp1.mod12<-lmer(Chlp ~Treatment * Age + (1|Batch), data=exp1)

>anova(exp1.mod11, exp1.mod12)

Data: exp1
Models:
exp1.mod12: Chlp ~ Treatment * Age + (1 | Batch)
exp1.mod11: Chlp ~ Treatment * Age + (1 | Batch) + (1 | Batch:Sample)
           npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp1.mod12    6 27.992 40.945 -7.9960    15.992
exp1.mod11    7  1.917 17.029  6.0415   -12.083 28.075  1  1.167e-07 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

#Sample nested within Batch is a significant term and cannot be removed
from the model. Next test removing Batch:

>exp1.mod13<-lmer(Chlp ~Treatment * Age + (1|Batch:Sample), data=exp1)

>anova(exp1.mod11, exp1.mod13)

Data: exp1
Models:
exp1.mod13: Chlp ~ Treatment * Age + (1 | Batch:Sample)
exp1.mod11: Chlp ~ Treatment * Age + (1 | Batch) + (1 | Batch:Sample)
           npar      AIC    BIC logLik -2*log(L) Chisq Df Pr(>Chisq)
exp1.mod13    6 -0.08305 12.870 6.0415   -12.083
exp1.mod11    7  1.91695 17.029 6.0415   -12.083     0  1          1

#Batch can be removed from the model.
#Test removing the interacton term between treatment and year.

>exp1.mod14<-lmer(Chlp ~Treatment + Age + (1|Batch:Sample), data=exp1)

>anova(exp1.mod13, exp1.mod14)

Data: exp1
Models:
exp1.mod14: Chlp ~ Treatment + Age + (1 | Batch:Sample)
exp1.mod13: Chlp ~ Treatment * Age + (1 | Batch:Sample)
           npar     AIC   BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp1.mod14    5  4.4357 15.23 2.7822   -5.5643
exp1.mod13    6 -0.0830 12.87 6.0415  -12.0830 6.5187  1    0.01067 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

#The interaction between Treatment and Age is significant and cannot be
removed from the model.

>summary(exp1.mod14)

Linear mixed model fit by REML. t-tests use Satterthwaite's method
['lmerModLmerTest']
Formula: Chlp ~ Treatment + Age + (1 | Batch:Sample)
   Data: exp1

REML criterion at convergence: 12.3

Scaled residuals:
```

```
    Min      1Q  Median      3Q     Max
-1.4281 -0.5158  0.0108  0.4134  3.4579

Random effects:
 Groups       Name        Variance Std.Dev.
 Batch:Sample (Intercept) 0.05961  0.2442
 Residual                 0.02240  0.1497
Number of obs: 64, groups:  Batch:Sample, 32

Fixed effects:
                 Estimate Std. Error       df t value Pr(>|t|)
(Intercept)      0.495209   0.102428 32.069391   4.835 3.20e-05 ***
Treatmentshotgun  0.268750   0.037416 31.000000   7.183 4.48e-08 ***
Age             -0.012844   0.004981 29.999999  -2.578   0.0151 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Correlation of Fixed Effects:
            (Intr) Trtmnt
Trtmntshtgn -0.183
Age         -0.869  0.000
```

```
>AIC(exp1.mod14)

[1] 22.29215
```

#Model estimated means and 95% confidence intervals were calculated:

```
>emm<- emmeans(exp1.mod14, ~Treatment, lmer.df="satterthwaite")

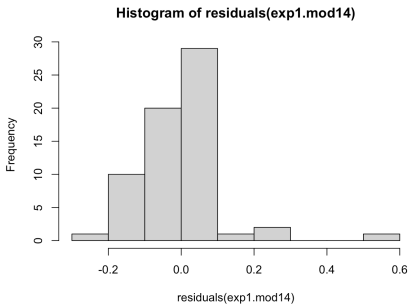
>emm

Treatment emmean     SE   df lower.CL upper.CL
 capture    0.266 0.0506 39.3    0.163    0.368
 shotgun    0.534 0.0506 39.3    0.432    0.637

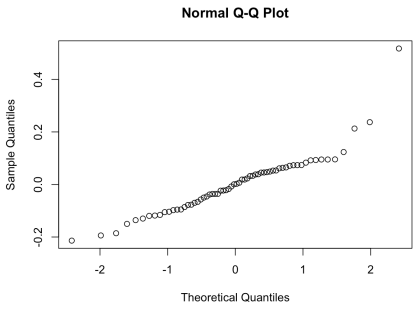
Degrees-of-freedom method: satterthwaite
Confidence level used: 0.95
```

#Model residuals indicated significant deviations from normality:

```
>hist(residuals(exp1.mod14))
```




```
>qqnorm(residuals(exp1.mod14))
```




```
>shapiro.test(residuals(exp1.mod14))


      Shapiro-Wilk normality test

data:  residuals(exp1.mod14)
W = 0.89759, p-value = 6.499e-05

#-----------------------------------

#Chloroplast recovery
#glmm

>exp1$Chlp_prop <- exp1$Chlp / 100

#We performed a glmm with a beta family distribution and logit link
transformation in R v4.2.3 with the proportion of chloroplast recovered as
the response variable (“Chlp_prop”, as the response variable, “Treatment”
(shotgun or capture) as a fixed effect, “Batch” (pool) as a random effect,
and “Sample nested in Batch” as a random effect:

>exp1.mod15<-glmmTMB(Chlp_prop ~Treatment * Age + (1|Batch) +
(1|Batch:Sample), family=beta_family(link="logit"),  data=exp1)

#Test removing Sample nested in Batch from the model:

>exp1.mod16<-glmmTMB(Chlp_prop ~Treatment * Age + (1|Batch),
family=beta_family(link="logit"),  data=exp1)

>anova(exp1.mod15, exp1.mod16)

Data: exp1
Models:
exp1.mod16: Chlp_prop ~ Treatment * Age + (1 | Batch), zi=~0, disp=~1
exp1.mod15: Chlp_prop ~ Treatment * Age + (1 | Batch) + (1 |
Batch:Sample), zi=~0, disp=~1
           Df     AIC     BIC logLik deviance Chisq Chi Df Pr(>Chisq)
exp1.mod16  6 -602.34 -589.39 307.17  -614.34
exp1.mod15  7 -677.47 -662.36 345.74  -691.47 77.13      1  < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

#Sample nested in Batch cannot be removed from the model. Test removing
Batch:

>exp1.mod17<-glmmTMB(Chlp_prop ~Treatment * Age + (1|Batch:Sample),
family=beta_family(link="logit"),  data=exp1)

>anova(exp1.mod15, exp1.mod17)
```

```
Data: exp1
Models:
exp1.mod17: Chlp_prop ~ Treatment * Age + (1 | Batch:Sample), zi=~0,
disp=~1
exp1.mod15: Chlp_prop ~ Treatment * Age + (1 | Batch) + (1 |
Batch:Sample), zi=~0, disp=~1
           Df     AIC     BIC logLik deviance Chisq Chi Df Pr(>Chisq)
exp1.mod17  6 -679.47 -666.52 345.74  -691.47
exp1.mod15  7 -677.47 -662.36 345.74  -691.47     0      1          1

#Batch can be removed from the model.
#Test removing the interaction between Treatment and Age.

>exp1.mod18<-glmmTMB(Chlp_prop ~Treatment + Age + (1|Batch:Sample),
family=beta_family(link="logit"),  data=exp1)

>anova(exp1.mod17, exp1.mod18)

Data: exp1
Models:
exp1.mod18: Chlp_prop ~ Treatment + Age + (1 | Batch:Sample), zi=~0,
disp=~1
exp1.mod17: Chlp_prop ~ Treatment * Age + (1 | Batch:Sample), zi=~0,
disp=~1
           Df     AIC     BIC logLik deviance  Chisq Chi Df Pr(>Chisq)
exp1.mod18  5 -680.81 -670.02 345.41  -690.81
exp1.mod17  6 -679.47 -666.52 345.74  -691.47 0.6568      1     0.4177

#The interaction between treatment and age is non-significant and can be
removed from the model.
#Test removing Age.

>exp1.mod19<-glmmTMB(Chlp_prop ~Treatment + (1|Batch:Sample),
family=beta_family(link="logit"),  data=exp1)

>anova(exp1.mod18, exp1.mod19)

Data: exp1
Models:
exp1.mod19: Chlp_prop ~ Treatment + (1 | Batch:Sample), zi=~0, disp=~1
exp1.mod18: Chlp_prop ~ Treatment + Age + (1 | Batch:Sample), zi=~0,
disp=~1
           Df     AIC     BIC logLik deviance  Chisq Chi Df Pr(>Chisq)
exp1.mod19  4 -676.94 -668.30 342.47  -684.94
exp1.mod18  5 -680.81 -670.02 345.41  -690.81 5.8752      1    0.01536 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

#Age is significant and cannot be removed from the model.

>AIC(exp1.mod18)

[1] -680.8142

>summary(exp1.mod18)

Family: beta  ( logit )
Formula:          Chlp_prop ~ Treatment + Age + (1 | Batch:Sample)
```

```
Data: exp1

      AIC       BIC    logLik -2*log(L)  df.resid
   -680.8    -670.0     345.4    -690.8        59

Random effects:

Conditional model:
 Groups       Name        Variance Std.Dev.
 Batch:Sample (Intercept) 0.3378   0.5812
Number of obs: 64, groups:  Batch:Sample, 32

Dispersion parameter for beta family (): 1.48e+04

Conditional model:
                 Estimate Std. Error z value Pr(>|z|)
(Intercept)      -5.63117    0.22498 -25.030   <2e-16 ***
Treatmentshotgun  0.70212    0.03463  20.277   <2e-16 ***
Age              -0.02827    0.01111  -2.545   0.0109 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
>emm<- emmeans(exp1.mod18, ~Treatment, type="response")

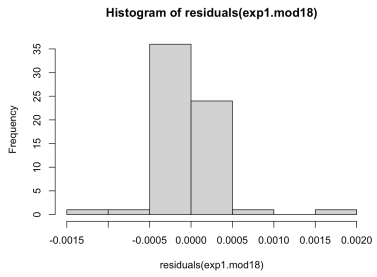
>emm

Treatment response       SE  df asymp.LCL asymp.UCL
 capture    0.00216 0.000231 Inf   0.00175   0.00266
 shotgun    0.00435 0.000455 Inf   0.00354   0.00534

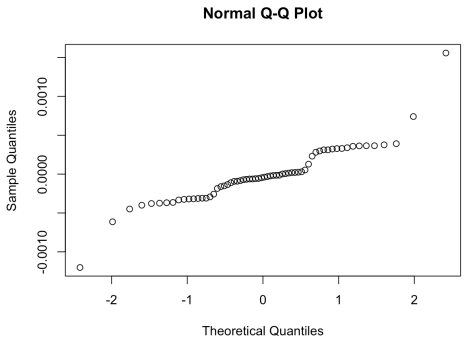
Confidence level used: 0.95
Intervals are back-transformed from the logit scale

#Model residuals still indicate significant deviations from normality:

>hist(residuals(exp1.mod18))
```




```
>qqnorm(residuals(exp1.mod18))
```




```
>shapiro.test(residuals(exp1.mod18))

Shapiro-Wilk normality test

data:  residuals(exp1.mod18)
W = 0.89149, p-value = 3.879e-05

#-----------------------------------

#Chloroplast recovery
#Non-parametric

#A non-parametric Wilcoxon signed rank test with continuity correction was
also performed in R v4.2.3:

>wide <- pivot_wider(exp1, id_cols = c(Sample,Batch),names_from =
Treatment, values_from = Chlp)

>wilcox.test(wide$shotgun, wide$capture, exact=FALSE)

Wilcoxon rank sum test with continuity correction

data:  wide$shotgun and wide$capture
W = 774.5, p-value = 0.0003575
alternative hypothesis: true location shift is not equal to 0

#-----------------------------------
#Between replicates (Aim 2)
#-----------------------------------
#Target Recovery
#Non-parametric

>exp2<-read.csv("exp2_w_year.csv",header=TRUE)
head(exp2)

                    Sample Year Age      Genus       Tribe   DayPair
DayPair2 Replicate
1 Angianthus_tomentosus_AG330 2012   8 Angianthus Gnaphalieae      same
sameD2         1
2 Angianthus_tomentosus_AG330 2012   8 Angianthus Gnaphalieae      same
sameD2         2
3  Angianthus_uniflorus_AG331 1995  25 Angianthus Gnaphalieae different
different         1
4  Angianthus_uniflorus_AG331 1995  25 Angianthus Gnaphalieae different
different         2
```

```
5       Bellis_perennis_AG279 2015   5     Bellis    Astereae      same
sameD1         1
6       Bellis_perennis_AG279 2015   5     Bellis    Astereae      same
sameD1         2
  Day     Batch Recovery     Chlp
1  D2 AG_P2P15C 45.50774 99.62795
2  D2 AG_P2P16C 45.26136 99.67648
3  D1 AG_P2P18C 36.25747 96.17707
4  D2 AG_P2P17C 30.44207 99.53629
5  D1  AG_P2P1C 41.68444 89.34541
6  D1  AG_P2P2C 21.65349 93.33010
```

#A non-parametric Wilcoxon signed rank test with continuity correction was performed in R v4.2.3 to compare target recovery between replicates within a sample:

```
>wide <- pivot_wider(exp2, id_cols = c(Sample),names_from = Replicate,
values_from = Recovery)
```

```
>wide
```

```
# A tibble: 90 × 3
   Sample                                `1`   `2`
   <chr>                               <dbl> <dbl>
 1 Angianthus_tomentosus_AG330         45.5  45.3
 2 Angianthus_uniflorus_AG331          36.3  30.4
 3 Bellis_perennis_AG279               41.7  21.7
 4 Brachyscome_ciliocarpa_AG013        22.1  17.5
 5 Brachyscome_dentata_AG314           50.0  48.3
 6 Brachyscome_dichromosomatica_AG019 46.8  36.3
 7 Brachyscome_formosa_AG198           12.6  10.9
 8 Brachyscome_microcarpa_AG036         8.68  8.03
 9 Brachyscome_nivalis_AG199           22.9   7.89
10 Brachyscome_sinclairii_AG069        24.6  18.6
# ℹ 80 more rows
# ℹ Use `print(n = ...)` to see more rows
```

```
>wilcox.test(wide$'1', wide$'2', exact=FALSE)
```

```
Wilcoxon rank sum test with continuity correction

data:  wide$"1" and wide$"2"
W = 4804.5, p-value = 0.03099
alternative hypothesis: true location shift is not equal to 0
```

```
#-----------------------------------
```

#Target Recovery
#lmer

#To partition the variance components, we performed a linear mixed model with the percentage of target base-pairs recovered (“Recovery”) as the response variable with “Sample”, “Day” (day 1 or day 2) and “Batch” (pool) nested within “Day” as random effects:

```
>exp2.mod <- lmer (Recovery ~ Age + (1|Tribe) + (1|Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Day) + (1 | Day:Batch), data = exp2, REML =
TRUE)
```

```
>VarCorr(exp2.mod)

Groups             Name        Std.Dev.
 Tribe:Genus:Sample (Intercept) 7.8001e+00
 Tribe:Genus        (Intercept) 1.1741e+01
 Day:Batch          (Intercept) 6.9155e+00
 Day                (Intercept) 0.0000e+00
 Tribe              (Intercept) 3.6116e-04
 Residual                       4.9675e+00

>check_singularity(exp2.mod)

[1] TRUE

>isSingular(exp2.mod)

[1] TRUE

#Test removing Day from the model:

>exp2.mod2 <- lmer (Recovery ~ Age + (1|Tribe) + (1|Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Day:Batch), data = exp2, REML = TRUE)

>isSingular(exp2.mod2)
[
1] TRUE

>anova(exp2.mod, exp2.mod2)

Data: exp2
Models:
exp2.mod2: Recovery ~ Age + (1 | Tribe) + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Day:Batch)
exp2.mod: Recovery ~ Age + (1 | Tribe) + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Day) + (1 | Day:Batch)
          npar  AIC    BIC  logLik -2*log(L) Chisq Df Pr(>Chisq)
exp2.mod2    7 1331 1353.4 -658.52      1317
exp2.mod     8 1333 1358.6 -658.52      1317     0  1          1

#Day can be removed from the model. Test removing Tribe from the model:

>exp2.mod3 <- lmer (Recovery ~ Age + (1|Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Day:Batch), data = exp2, REML = TRUE)

>anova(exp2.mod2, exp2.mod3)

Data: exp2
Models:
exp2.mod3: Recovery ~ Age + (1 | Tribe:Genus) + (1 | Tribe:Genus:Sample) +
(1 | Day:Batch)
exp2.mod2: Recovery ~ Age + (1 | Tribe) + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Day:Batch)
          npar  AIC    BIC  logLik -2*log(L) Chisq Df Pr(>Chisq)
exp2.mod3    6 1329 1348.2 -658.52      1317
exp2.mod2    7 1331 1353.4 -658.52      1317     0  1          1

#Tribe is non-significant and can be removed from the model. Test removing
Genus nested in Tribe from the model:
```

```
>exp2.mod4 <- lmer (Recovery ~  Age + (1 | Tribe:Genus:Sample) + (1 |
Day:Batch), data = exp2, REML = TRUE)

>anova(exp2.mod3, exp2.mod4)

Data: exp2
Models:
exp2.mod4: Recovery ~ Age + (1 | Tribe:Genus:Sample) + (1 | Day:Batch)
exp2.mod3: Recovery ~ Age + (1 | Tribe:Genus) + (1 | Tribe:Genus:Sample) +
(1 | Day:Batch)
          npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp2.mod4    5 1387.7 1403.7 -688.86    1377.7
exp2.mod3    6 1329.0 1348.2 -658.52    1317.0 60.673  1  6.738e-15 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

#Genus nested in Tribe significant and cannot be removed from the model.
Test removing Sample nested within Genus and Tribe:

>exp2.mod5 <- lmer (Recovery ~ Age + (1|Tribe:Genus) + (1 | Day:Batch),
data = exp2, REML = TRUE)

>anova(exp2.mod3, exp2.mod5)

Data: exp2
Models:
exp2.mod5: Recovery ~ Age + (1 | Tribe:Genus) + (1 | Day:Batch)
exp2.mod3: Recovery ~ Age + (1 | Tribe:Genus) + (1 | Tribe:Genus:Sample) +
(1 | Day:Batch)
          npar    AIC    BIC  logLik -2*log(L) Chisq Df Pr(>Chisq)
exp2.mod5    5 1379.8 1395.7 -684.88    1369.8
exp2.mod3    6 1329.0 1348.2 -658.52    1317.0 52.71  1  3.867e-13 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

#Sample nested within Genus and Tribe is significant and cannot be removed
from the model. Test removing Batch nested in Day:

>exp2.mod6 <- lmer (Recovery ~ Age + (1|Tribe:Genus) + (1 |
Tribe:Genus:Sample), data = exp2, REML = TRUE)

>anova(exp2.mod3, exp2.mod6)

Data: exp2
Models:
exp2.mod6: Recovery ~ Age + (1 | Tribe:Genus) + (1 | Tribe:Genus:Sample)
exp2.mod3: Recovery ~ Age + (1 | Tribe:Genus) + (1 | Tribe:Genus:Sample) +
(1 | Day:Batch)
          npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp2.mod6    5 1398.4 1414.4 -694.22    1388.4
exp2.mod3    6 1329.0 1348.2 -658.52    1317.0 71.396  1  < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

#Batch nested within Day is significant and cannot be removed from the
model.
#Test removing Age:
```

```
>exp2.mod7 <- lmer (Recovery ~ (1|Tribe:Genus) + (1 | Tribe:Genus:Sample)
+ (1 | Day:Batch), data = exp2, REML = TRUE)
anova(exp2.mod3, exp2.mod7)

Data: exp2
Models:
exp2.mod7: Recovery ~ (1 | Tribe:Genus) + (1 | Tribe:Genus:Sample) + (1 |
Day:Batch)
exp2.mod3: Recovery ~ Age + (1 | Tribe:Genus) + (1 | Tribe:Genus:Sample) +
(1 | Day:Batch)
          npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp2.mod7    5 1327.1 1343.1 -658.55    1317.1
exp2.mod3    6 1329.0 1348.2 -658.52    1317.0 0.0551  1     0.8143

#Age is insignificant and can be removed from the model.

>summary(exp2.mod7)

Linear mixed model fit by REML. t-tests use Satterthwaite's method
['lmerModLmerTest']
Formula: Recovery ~ (1 | Tribe:Genus) + (1 | Tribe:Genus:Sample) + (1 |
Day:Batch)
   Data: exp2

REML criterion at convergence: 1312.9

Scaled residuals:
    Min      1Q  Median      3Q     Max
-4.6481 -0.2051  0.0154  0.2725  4.3198

Random effects:
 Groups             Name        Variance Std.Dev.
 Tribe:Genus:Sample (Intercept)  60.10    7.752
 Tribe:Genus        (Intercept) 137.41   11.722
 Day:Batch          (Intercept)  46.82    6.843
 Residual                        24.68    4.968
Number of obs: 180, groups:  Tribe:Genus:Sample, 90; Tribe:Genus, 26;
Day:Batch, 12

Fixed effects:
            Estimate Std. Error     df t value Pr(>|t|)
(Intercept)   42.139      3.302 29.710   12.76 1.37e-13 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

#The intraclass correlation was calculated using the variance components
of the final model:

>vc <- as.data.frame(VarCorr(exp2.mod7))

>vc

             grp        var1 var2      vcov     sdcor
1 Tribe:Genus:Sample (Intercept) <NA>  60.09583  7.752150
2        Tribe:Genus (Intercept) <NA> 137.41206 11.722289
3          Day:Batch (Intercept) <NA>  46.82468  6.842856
4           Residual        <NA> <NA>  24.67627  4.967521

>sigma_sample  <- vc$vcov[vc$grp == "Tribe:Genus:Sample"]
```

```
>sigma_genus <- vc$vcov[vc$grp == "Tribe:Genus"]

>sigma_batch   <- vc$vcov[vc$grp == "Day:Batch"]

>sigma_resid   <- vc$vcov[vc$grp == "Residual"]

>total_var <- sigma_genus + sigma_sample + sigma_batch + sigma_resid

>ICC <- (sigma_genus + sigma_sample) / total_var

>ICC

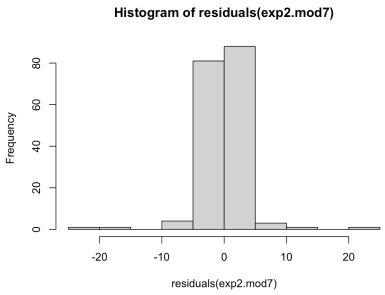
[1] 0.7342059

>AIC(exp2.mod7)

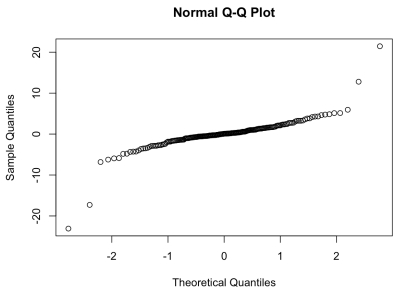
[1] 1322.89

#Model residuals are approximately normally distributed according to the
histogram and Q-Q plot, however, the Shapiro-Wilk test still indicated
significant deviations from normality:

>hist(residuals(exp2.mod7))
```




```
>qqnorm(residuals(exp2.mod7))
```




```
>shapiro.test(residuals(exp2.mod7))

Shapiro-Wilk normality test

data:  residuals(exp2.mod7)
W = 0.75714, p-value = 5.906e-16

#-----------------------------------

#Target Recovery
#glmm
```

```
#We performed a generalised linear mixed model with the proportion of
target base-pairs recovered (“Recovery_prop”) as the response variable
with “Sample”, “Day” (day 1 or day 2) and “Batch” (pool) nested within
“Day” as random effects. A beta family distribution was used with a logit
link transformation:

#Convert percentage target recovery to a proportion between 0 and 1:

>exp2$Recovery_prop <- exp2$Recovery / 100

>exp2.mod8 <- glmmTMB (Recovery_prop ~ Age + (1|Tribe) + (1|Tribe:Genus) +
(1 | Tribe:Genus:Sample) + (1 | Day) + (1 | Day:Batch),
family=beta_family(link="logit"), data = exp2)

#Test removing the random effect Tribe:

>exp2.mod9 <- glmmTMB (Recovery_prop ~ Age + (1|Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Day) + (1 | Day:Batch),
family=beta_family(link="logit"), data = exp2)

>anova(exp2.mod8, exp2.mod9)

Data: exp2
Models:
exp2.mod9: Recovery_prop ~ Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + , zi=~0, disp=~1
exp2.mod9:     (1 | Day) + (1 | Day:Batch), zi=~0, disp=~1
exp2.mod8: Recovery_prop ~ Age + (1 | Tribe) + (1 | Tribe:Genus) + (1 | ,
zi=~0, disp=~1
exp2.mod8:     Tribe:Genus:Sample) + (1 | Day) + (1 | Day:Batch), zi=~0,
disp=~1
          Df     AIC     BIC logLik deviance Chisq Chi Df Pr(>Chisq)
exp2.mod9  7 -283.02 -260.67 148.51  -297.02
exp2.mod8  8 -281.02 -255.47 148.51  -297.02     0      1          1

#Tribe is non-significant and can be removed from the model. Test removing
Genus nested within Tribe:

>exp2.mod10 <- glmmTMB (Recovery_prop ~ Age + (1 | Tribe:Genus:Sample) +
(1 | Day) + (1 | Day:Batch), family=beta_family(link="logit"), data =
exp2)

>anova(exp2.mod9, exp2.mod10)

Data: exp2
Models:
exp2.mod10: Recovery_prop ~ Age + (1 | Tribe:Genus:Sample) + (1 | Day) + ,
zi=~0, disp=~1
exp2.mod10:     (1 | Day:Batch), zi=~0, disp=~1
exp2.mod9: Recovery_prop ~ Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + , zi=~0, disp=~1
exp2.mod9:     (1 | Day) + (1 | Day:Batch), zi=~0, disp=~1
           Df     AIC     BIC logLik deviance Chisq Chi Df Pr(>Chisq)
exp2.mod10  6 -223.69 -204.53 117.84  -235.69
exp2.mod9   7 -283.02 -260.67 148.51  -297.02 61.33      1  4.827e-15 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
#Genus nested in tribe is significant and cannot be removed from the
model. Test removing Sample nested within Genus nested within Tribe:

>exp2.mod11 <- glmmTMB (Recovery_prop ~ Age + (1|Tribe:Genus) + (1 | Day)
+ (1 | Day:Batch), family=beta_family(link="logit"), data = exp2)

>anova(exp2.mod9, exp2.mod11)

Data: exp2
Models:
exp2.mod11: Recovery_prop ~ Age + (1 | Tribe:Genus) + (1 | Day) + (1 |
Day:Batch), zi=~0, disp=~1
exp2.mod9: Recovery_prop ~ Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + , zi=~0, disp=~1
exp2.mod9:     (1 | Day) + (1 | Day:Batch), zi=~0, disp=~1
           Df     AIC     BIC logLik deviance  Chisq Chi Df Pr(>Chisq)
exp2.mod11  6 -263.42 -244.27 137.71  -275.42
exp2.mod9   7 -283.02 -260.67 148.51  -297.02 21.594      1  3.369e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

#Sample nested in Genus nested in tribe is significant and cannot be
removed from the model. Test removing Day:

>exp2.mod12 <- glmmTMB (Recovery_prop ~ Age + (1|Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Day:Batch), family=beta_family(link="logit"),
data = exp2)

>anova(exp2.mod9, exp2.mod12)

Data: exp2
Models:
exp2.mod12: Recovery_prop ~ Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + , zi=~0, disp=~1
exp2.mod12:     (1 | Day:Batch), zi=~0, disp=~1
exp2.mod9: Recovery_prop ~ Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + , zi=~0, disp=~1
exp2.mod9:     (1 | Day) + (1 | Day:Batch), zi=~0, disp=~1
           Df     AIC     BIC logLik deviance Chisq Chi Df Pr(>Chisq)
exp2.mod12  6 -285.02 -265.86 148.51  -297.02
exp2.mod9   7 -283.02 -260.67 148.51  -297.02     0      1          1

#Day is non-significant and can be removed from the model. Test removing
Batch nested within Day:

>exp2.mod13 <- glmmTMB (Recovery_prop ~ Age + (1|Tribe:Genus) + (1 |
Tribe:Genus:Sample), family=beta_family(link="logit"), data = exp2)

>anova(exp2.mod12, exp2.mod13)

Data: exp2
Models:
exp2.mod13: Recovery_prop ~ Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample), zi=~0, disp=~1
exp2.mod12: Recovery_prop ~ Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + , zi=~0, disp=~1
exp2.mod12:     (1 | Day:Batch), zi=~0, disp=~1
           Df     AIC     BIC logLik deviance  Chisq Chi Df Pr(>Chisq)
exp2.mod13  5 -234.82 -218.86 122.41  -244.82
```

```
exp2.mod12  6 -285.02 -265.86 148.51  -297.02 52.193      1   5.03e-13 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

#Batch nested within Day is significant and cannot be removed from the model.
#Test removing Age:

```
>exp2.mod14 <- glmmTMB (Recovery_prop ~  (1|Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Day:Batch), family=beta_family(link="logit"),
data = exp2)
```

```
>anova(exp2.mod12, exp2.mod14)
```

```
Data: exp2
Models:
exp2.mod14: Recovery_prop ~ (1 | Tribe:Genus) + (1 | Tribe:Genus:Sample) +
, zi=~0, disp=~1
exp2.mod14:     (1 | Day:Batch), zi=~0, disp=~1
exp2.mod12: Recovery_prop ~ Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + , zi=~0, disp=~1
exp2.mod12:     (1 | Day:Batch), zi=~0, disp=~1
           Df     AIC     BIC logLik deviance Chisq Chi Df Pr(>Chisq)
exp2.mod14  5 -287.02 -271.05 148.51  -297.02
exp2.mod12  6 -285.02 -265.86 148.51  -297.02 7e-04      1      0.979
```

#Age is non-significant and can be removed from the model.

```
>summary(exp2.mod14)
```

```
Family: beta  ( logit )
Formula:          Recovery_prop ~ (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) +
    (1 | Day:Batch)
Data: exp2

      AIC       BIC    logLik -2*log(L)  df.resid
   -287.0    -271.1     148.5    -297.0       175

Random effects:

Conditional model:
 Groups             Name        Variance Std.Dev.
 Tribe:Genus        (Intercept) 0.3613   0.6011
 Tribe:Genus:Sample (Intercept) 0.1060   0.3255
 Day:Batch          (Intercept) 0.1191   0.3451
Number of obs: 180, groups:  Tribe:Genus, 26; Tribe:Genus:Sample, 90;
Day:Batch, 12

Dispersion parameter for beta family (): 44.4

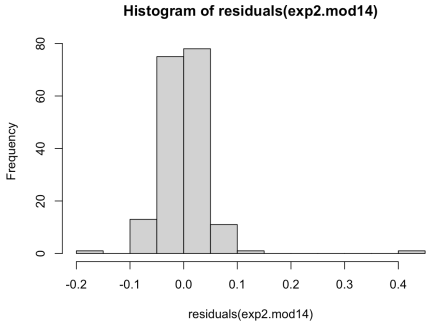
Conditional model:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -0.3899     0.1662  -2.345    0.019 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
>AIC(exp2.mod14)
```

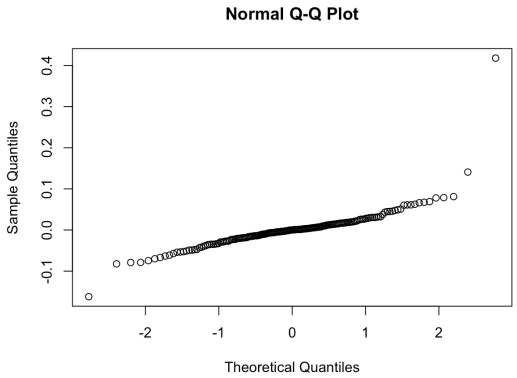
```
[1] -287.0181
```

#Model residuals are remain approximately normally distributed according to the histogram and Q-Q plot, however, the Shapiro-Wilk test still indicated significant deviations from normality:

```
>hist(residuals(exp2.mod14))
```




```
>qqnorm(residuals(exp2.mod14))
```




```
>shapiro.test(residuals(exp2.mod14))

Shapiro-Wilk normality test

data:  residuals(exp2.mod14)
W = 0.73681, p-value < 2.2e-16

#-----------------------------------

#Chloroplast Recovery
#Non-parametric

#A non-parametric Wilcoxon signed rank test with continuity correction was performed in R v4.2.3 to compare chloroplast recovery between replicates within a sample:

>wide <- pivot_wider(exp2, id_cols = c(Sample),names_from = Replicate, values_from = Chlp)

>wide

# A tibble: 90 × 3
   Sample                                `1`    `2`
```

```
   <chr>                              <dbl> <dbl>
 1 Angianthus_tomentosus_AG330         99.6  99.7
 2 Angianthus_uniflorus_AG331          96.2  99.5
 3 Bellis_perennis_AG279               89.3  93.3
 4 Brachyscome_ciliocarpa_AG013        98.9  98.7
 5 Brachyscome_dentata_AG314           98.6  98.6
 6 Brachyscome_dichromosomatica_AG019  99.6  99.8
 7 Brachyscome_formosa_AG198           28.4  33.6
 8 Brachyscome_microcarpa_AG036        83.2  89.6
 9 Brachyscome_nivalis_AG199           95.7  96.6
10 Brachyscome_sinclairii_AG069        94.2  91.9
# ℹ 80 more rows
# ℹ Use `print(n = ...)` to see more rows
```

```
>wilcox.test(wide$'1', wide$'2', exact=FALSE)

Wilcoxon rank sum test with continuity correction

data:  wide$"1" and wide$"2"
W = 4109.5, p-value = 0.866
alternative hypothesis: true location shift is not equal to 0

#-----------------------------------

#Chloroplast Recovery
#lmer

>exp2.mod15 <- lmer(Chlp ~ Age + (1|Tribe) + (1|Tribe:Genus) + (1 |
Tribe:Genus:Sample) +(1|Day)+ (1 | Day:Batch), data = exp2,REML = TRUE)

#Test removing Tribe from the model:

>exp2.mod16 <- lmer(Chlp ~  Age + (1|Tribe:Genus) + (1 |
Tribe:Genus:Sample) +(1|Day)+ (1 | Day:Batch), data = exp2,REML = TRUE)

>anova(exp2.mod15, exp2.mod16)

Data: exp2
Models:
exp2.mod16: Chlp ~ Age + (1 | Tribe:Genus) + (1 | Tribe:Genus:Sample) + (1
| Day) + (1 | Day:Batch)
exp2.mod15: Chlp ~ Age + (1 | Tribe) + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Day) + (1 | Day:Batch)
           npar    AIC    BIC logLik -2*log(L) Chisq Df Pr(>Chisq)
exp2.mod16    7 1508.4 1530.8 -747.2    1494.4
exp2.mod15    8 1510.4 1535.9 -747.2    1494.4     0  1          1

#Tribe is non-significant and can be removed from the model. Test removing
Genus nested in Tribe:

>exp2.mod17 <- lmer(Chlp ~   Age + (1 | Tribe:Genus:Sample) +(1|Day)+ (1 |
Day:Batch), data = exp2,REML = TRUE)

>anova(exp2.mod16, exp2.mod17)

Data: exp2
Models:
exp2.mod17: Chlp ~ Age + (1 | Tribe:Genus:Sample) + (1 | Day) + (1 |
Day:Batch)
```

```
exp2.mod16: Chlp ~ Age + (1 | Tribe:Genus) + (1 | Tribe:Genus:Sample) + (1
| Day) + (1 | Day:Batch)
           npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp2.mod17    6 1522.2 1541.3 -755.09    1510.2
exp2.mod16    7 1508.4 1530.8 -747.20    1494.4 15.781  1   7.11e-05 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

#Genus nested in tribe is significant and cannot be removed from the model. Test removing Sample nested in Genus nested in Tribe:

```
>exp2.mod18 <- lmer(Chlp ~  Age + (1|Tribe:Genus)  +(1|Day)+ (1 |
Day:Batch), data = exp2,REML = TRUE)

>anova(exp2.mod16, exp2.mod18)

Data: exp2
Models:
exp2.mod18: Chlp ~ Age + (1 | Tribe:Genus) + (1 | Day) + (1 | Day:Batch)
exp2.mod16: Chlp ~ Age + (1 | Tribe:Genus) + (1 | Tribe:Genus:Sample) + (1
| Day) + (1 | Day:Batch)
           npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp2.mod18    6 1632.5 1651.7 -810.27    1620.5
exp2.mod16    7 1508.4 1530.8 -747.20    1494.4 126.14  1  < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

#Sample nested in Genus nested in tribe is significant and cannot be removed from the model. Test removing Day:

```
>exp2.mod19 <- lmer(Chlp ~  Age + (1|Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Day:Batch), data = exp2,REML = TRUE)

>anova(exp2.mod16, exp2.mod19)

Data: exp2
Models:
exp2.mod19: Chlp ~ Age + (1 | Tribe:Genus) + (1 | Tribe:Genus:Sample) + (1
| Day:Batch)
exp2.mod16: Chlp ~ Age + (1 | Tribe:Genus) + (1 | Tribe:Genus:Sample) + (1
| Day) + (1 | Day:Batch)
           npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp2.mod19    6 1507.1 1526.2 -747.53    1495.1
exp2.mod16    7 1508.4 1530.8 -747.20    1494.4 0.6652  1     0.4147
```

#Day is non-significant and can be removed from the model. Test removing Batch nested in Day:

```
>exp2.mod20 <- lmer(Chlp ~  Age + (1|Tribe:Genus) + (1 |
Tribe:Genus:Sample), data = exp2,REML = TRUE)

>anova(exp2.mod19, exp2.mod20)

Data: exp2
Models:
exp2.mod20: Chlp ~ Age + (1 | Tribe:Genus) + (1 | Tribe:Genus:Sample)
exp2.mod19: Chlp ~ Age + (1 | Tribe:Genus) + (1 | Tribe:Genus:Sample) + (1
| Day:Batch)
           npar    AIC    BIC  logLik -2*log(L) Chisq Df Pr(>Chisq)
```

```
exp2.mod20    5 1505.1 1521.0 -747.53    1495.1
exp2.mod19    6 1507.1 1526.2 -747.53    1495.1     0  1          1
```

#Batch nested in Day is non-significant and can be removed from the model.
#Test removing Age:

```
>exp2.mod21 <- lmer(Chlp ~   (1|Tribe:Genus) + (1 | Tribe:Genus:Sample),
data = exp2,REML = TRUE)
```

```
>anova(exp2.mod20, exp2.mod21)
```

```
Data: exp2
Models:
exp2.mod21: Chlp ~ (1 | Tribe:Genus) + (1 | Tribe:Genus:Sample)
exp2.mod20: Chlp ~ Age + (1 | Tribe:Genus) + (1 | Tribe:Genus:Sample)
           npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp2.mod21    4 1503.3 1516.1 -747.66    1495.3
exp2.mod20    5 1505.1 1521.0 -747.53    1495.1 0.2654  1     0.6065
```

#Age is non-significant and can be removed from the model.

```
>summary(exp2.mod21)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method
['lmerModLmerTest']
Formula: Chlp ~ (1 | Tribe:Genus) + (1 | Tribe:Genus:Sample)
   Data: exp2

REML criterion at convergence: 1491.1

Scaled residuals:
    Min      1Q  Median      3Q     Max
-6.7777 -0.0290  0.0552  0.1000  6.1776

Random effects:
 Groups              Name        Variance Std.Dev.
 Tribe:Genus:Sample (Intercept) 401.09   20.027
 Tribe:Genus        (Intercept)  88.50    9.408
 Residual                        58.21    7.630
Number of obs: 180, groups:  Tribe:Genus:Sample, 90; Tribe:Genus, 26

Fixed effects:
            Estimate Std. Error     df t value Pr(>|t|)
(Intercept)   90.218      3.266 32.917   27.62   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
>AIC(exp2.mod21)
```

```
[1] 1499.138
```

#The intraclass correlation was calculated using the variance components
of the final model:

```
>VarCorr(exp2.mod21)
```

```
Groups              Name        Std.Dev.
 Tribe:Genus:Sample (Intercept) 20.0272
 Tribe:Genus        (Intercept)  9.4076
```

```
 Residual                        7.6298
```

```
>vc <- as.data.frame(VarCorr(exp2.mod21))
```

```
>vc
```

```
              grp        var1 var2      vcov     sdcor
1 Tribe:Genus:Sample (Intercept) <NA> 401.09034 20.027240
2        Tribe:Genus (Intercept) <NA>  88.50347  9.407628
3           Residual        <NA> <NA>  58.21366  7.629788
```

```
>sigma_sample  <- vc$vcov[vc$grp == "Tribe:Genus:Sample"]
```

```
>sigma_genus  <- vc$vcov[vc$grp == "Tribe:Genus"]
```

```
>total_var <- sigma_genus + sigma_sample + sigma_resid
```

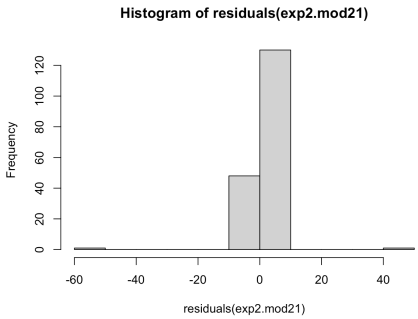
```
>ICC <- (sigma_genus + sigma_sample) / total_var
```

```
>ICC
```

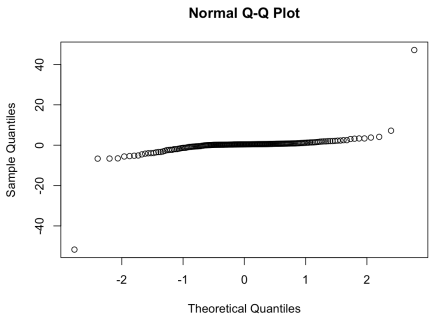
```
[1] 0.9520169
```

```
#Model residuals are approximately normally distributed according to the
histogram and Q-Q plot, however, the Shapiro-Wilk test still indicated
significant deviations from normality:
```

```
>hist(residuals(exp2.mod21))
```




```
>qqnorm(residuals(exp2.mod21))
```




```
>shapiro.test(residuals(exp2.mod21))
```

```
Shapiro-Wilk normality test
```

```
data:  residuals(exp2.mod21)
W = 0.34632, p-value < 2.2e-16

#-----------------------------------

#Chloroplast Recovery
#glmm

#We performed a generalised linear mixed model with the proportion of off-
target chloroplast base-pairs recovered (“Chlp_prop”) as the response
variable with “Sample”, “Day” (day 1 or day 2) and “Batch” (pool) nested
within “Day” as random effects:

>exp2$Chlp_prop <- exp2$Chlp / 100

>exp2.mod22 <- glmmTMB(Chlp_prop ~ Age + (1|Tribe) + (1|Tribe:Genus) + (1
| Tribe:Genus:Sample) +(1|Day)+ (1 | Day:Batch),
family=beta_family(link="logit"), data = exp2, subset = Chlp_prop != 0)

#Test removing Tribe:

>exp2.mod23 <- glmmTMB(Chlp_prop ~ Age + (1|Tribe:Genus) + (1 |
Tribe:Genus:Sample) +(1|Day)+ (1 | Day:Batch),
family=beta_family(link="logit"), data = exp2, subset = Chlp_prop != 0)

>anova(exp2.mod22, exp2.mod23)

Data: exp2
Subset: Chlp_prop != 0
Models:
exp2.mod23: Chlp_prop ~ Age + (1 | Tribe:Genus) + (1 | Tribe:Genus:Sample)
+ , zi=~0, disp=~1
exp2.mod23:     (1 | Day) + (1 | Day:Batch), zi=~0, disp=~1
exp2.mod22: Chlp_prop ~ Age + (1 | Tribe) + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + , zi=~0, disp=~1
exp2.mod22:     (1 | Day) + (1 | Day:Batch), zi=~0, disp=~1
           Df     AIC     BIC logLik deviance Chisq Chi Df Pr(>Chisq)
exp2.mod23  7 -804.25 -781.94 409.12  -818.25
exp2.mod22  8 -802.25 -776.75 409.12  -818.25     0      1          1

#Tribe is non-significant and can be removed from the model. Test removing
Genus nested in Tribe:

>exp2.mod24 <- glmmTMB(Chlp_prop ~  Age + (1 | Tribe:Genus:Sample)
+(1|Day)+ (1 | Day:Batch), family=beta_family(link="logit"), data = exp2,
subset = Chlp_prop != 0)

>anova(exp2.mod23, exp2.mod24)

Data: exp2
Subset: Chlp_prop != 0
Models:
exp2.mod24: Chlp_prop ~ Age + (1 | Tribe:Genus:Sample) + (1 | Day) + (1 |
, zi=~0, disp=~1
exp2.mod24:     Day:Batch), zi=~0, disp=~1
exp2.mod23: Chlp_prop ~ Age + (1 | Tribe:Genus) + (1 | Tribe:Genus:Sample)
+ , zi=~0, disp=~1
exp2.mod23:     (1 | Day) + (1 | Day:Batch), zi=~0, disp=~1
           Df     AIC     BIC logLik deviance Chisq Chi Df Pr(>Chisq)
```

```
exp2.mod24  6
exp2.mod23  7 -804.25 -781.94 409.12  -818.25           1
```

#Genus nested in Tribe is non-significant and can be removed from the model. Test removing Sample nested in Genus nested in Tribe:

```
>exp2.mod25 <- glmmTMB(Chlp_prop ~ Age + (1|Day)+ (1 | Day:Batch),
family=beta_family(link="logit"), data = exp2, subset = Chlp_prop != 0)

>anova(exp2.mod24, exp2.mod25)

Data: exp2
Subset: Chlp_prop != 0
Models:
exp2.mod25: Chlp_prop ~ Age + (1 | Day) + (1 | Day:Batch), zi=~0, disp=~1
exp2.mod24: Chlp_prop ~ Age + (1 | Tribe:Genus:Sample) + (1 | Day) + (1 |
, zi=~0, disp=~1
exp2.mod24:     Day:Batch), zi=~0, disp=~1
           Df     AIC     BIC logLik deviance Chisq Chi Df Pr(>Chisq)
exp2.mod25  5 -513.86 -497.92 261.93  -523.86
exp2.mod24  6
```

#Sample nested in genus nested in Tribe is non-significant and can be removed from the model. Test removing Day:

```
>exp2.mod26 <- glmmTMB(Chlp_prop ~ Age  + (1 | Day:Batch),
family=beta_family(link="logit"), data = exp2, subset = Chlp_prop != 0)

>anova(exp2.mod25, exp2.mod26)

Data: exp2
Subset: Chlp_prop != 0
Models:
exp2.mod26: Chlp_prop ~ Age + (1 | Day:Batch), zi=~0, disp=~1
exp2.mod25: Chlp_prop ~ Age + (1 | Day) + (1 | Day:Batch), zi=~0, disp=~1
           Df     AIC     BIC logLik deviance Chisq Chi Df Pr(>Chisq)
exp2.mod26  4 -515.86 -503.11 261.93  -523.86
exp2.mod25  5 -513.86 -497.92 261.93  -523.86     0      1     0.9999
```

#Day is non-significant and can be removed from the model. Test removing Batch within Day:

```
>exp2.mod27 <- glmmTMB(Chlp_prop ~ Age, family=beta_family(link="logit"),
data = exp2, subset = Chlp_prop != 0)

>anova(exp2.mod26, exp2.mod27)

Data: exp2
Subset: Chlp_prop != 0
Models:
exp2.mod27: Chlp_prop ~ Age, zi=~0, disp=~1
exp2.mod26: Chlp_prop ~ Age + (1 | Day:Batch), zi=~0, disp=~1
           Df     AIC     BIC logLik deviance Chisq Chi Df Pr(>Chisq)
exp2.mod27  3 -517.86 -508.29 261.93  -523.86
exp2.mod26  4 -515.86 -503.11 261.93  -523.86     0      1          1
```

#Batch within Day is non-significant and can be removed from the model.

```
>summary(exp2.mod27)
```

```
Family: beta  ( logit )
Formula:          Chlp_prop ~ Age
Data: exp2

      AIC       BIC    logLik -2*log(L)  df.resid
   -517.9    -508.3     261.9    -523.9       176


Dispersion parameter for beta family (): 1.98

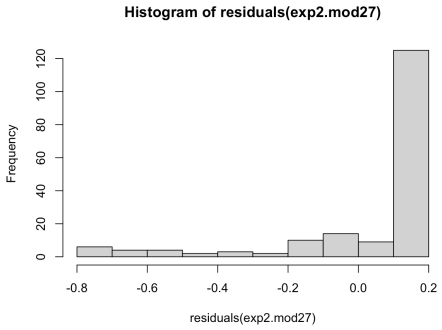
Conditional model:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  1.757867   0.160520  10.951   <2e-16 ***
Age         -0.005890   0.007174  -0.821    0.412
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
>AIC(exp2.mod27)

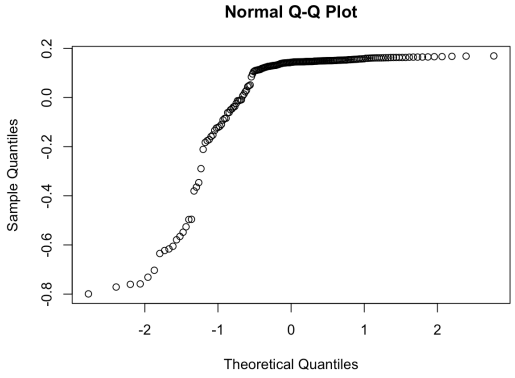
[1] -517.8555
```

#Model residuals seem to be even less normally distributed according to the histogram and Q-Q plot, and the Shapiro-Wilk test still indicated significant deviations from normality:

```
>hist(residuals(exp2.mod27))
```




```
>qqnorm(residuals(exp2.mod27))
```




```
>shapiro.test(residuals(exp2.mod27))

Shapiro-Wilk normality test
```

```
data:  residuals(exp2.mod27)
W = 0.61154, p-value < 2.2e-16

#-----------------------------------
#Between treatments (Aim 3)
#-----------------------------------

>exp3<-read.csv("exp3_w_year.csv", header=TRUE)

>head(exp3)

     Sample Year Age     Batch       Genus    Tribe     Size Input_cat
1     Brachyscome_barkerae_AG003 2011   9 AG_P1P11C Brachyscome Astereae
mini       5ng
2     Brachyscome_barkerae_AG003 2011   9 AG_P1P23C Brachyscome Astereae
standard       5ng
3    Brachyscome_basaltica_AG004 2013   7 AG_P1P11C Brachyscome Astereae
mini       5ng
4    Brachyscome_basaltica_AG004 2013   7 AG_P1P23C Brachyscome Astereae
standard       5ng
5 Brachyscome_bellidioides_AG005 1998  22  AG_P1P9C Brachyscome Astereae
mini       5ng
6 Brachyscome_bellidioides_AG005 1998  22 AG_P1P21C Brachyscome Astereae
standard       5ng
  Input_cont Treatment Composition     Chlp   Ast353  logChlp
1          5        5M           U 43.83695 50.16228 1.641840
2          5        5S           U 11.56584 63.54325 1.063177
3          5        5M           U 77.42370 49.38874 1.888874
4          5        5S           U 21.39545 54.24860 1.330321
5          5        5M           S 33.24167 35.81688 1.521683
6          5        5S           S 30.09274 59.10068 1.478462

#Target recovery
#lmer

#We performed a linear mixed model with the percentage target base-pairs
recovered (“Ast353”) as the response variable with “Size” (mini or
standard) as a fixed factor, “Tribe”, “Genus within Tribe” and “Sample
nested in Genus nested in Tribe” as a random effects, and “Batch” (pool)
as a random effect:

>exp3.mod<-lmer(Ast353 ~ Size * Age + (1|Tribe) + (1|Tribe:Genus) +
(1|Tribe:Genus:Sample) + (1|Batch), data=exp3, REML=TRUE)

#Test removing Tribe from the model:

>exp3.mod2<-lmer(Ast353 ~ Size * Age + (1|Tribe:Genus) +
(1|Tribe:Genus:Sample) + (1|Batch), data=exp3, REML=TRUE)

>anova(exp3.mod, exp3.mod2)

Data: exp3
Models:
exp3.mod2: Ast353 ~ Size * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Batch)
exp3.mod: Ast353 ~ Size * Age + (1 | Tribe) + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Batch)
          npar    AIC    BIC  logLik -2*log(L) Chisq Df Pr(>Chisq)
```

```
exp3.mod2    8 871.15 893.71 -427.58    855.15
exp3.mod     9 873.15 898.53 -427.58    855.15     0  1          1
```

#Tribe does not contribute to variance so is removed from the model.Genus within Tribe is then removed from the model:

```
>exp3.mod3<-lmer(Ast353 ~ Size * Age + (1|Tribe:Genus:Sample) + (1|Batch), data=exp3, REML=TRUE)

>anova(exp3.mod2, exp3.mod3)

Data: exp3
Models:
exp3.mod3: Ast353 ~ Size * Age + (1 | Tribe:Genus:Sample) + (1 | Batch)
exp3.mod2: Ast353 ~ Size * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Batch)
          npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp3.mod3    7 869.15 888.90 -427.58    855.15
exp3.mod2    8 871.15 893.71 -427.58    855.15 0.0033  1     0.9543
```

#Genus also does not significantly contribute to variance. The Sample term is simplified:

```
>exp3.mod4<-lmer(Ast353 ~ Size * Age + (1|Sample) + (1|Batch), data=exp3, REML=TRUE)

>anova(exp3.mod3, exp3.mod4)

Data: exp3
Models:
exp3.mod3: Ast353 ~ Size * Age + (1 | Tribe:Genus:Sample) + (1 | Batch)
exp3.mod4: Ast353 ~ Size * Age + (1 | Sample) + (1 | Batch)
          npar    AIC   BIC  logLik -2*log(L) Chisq Df Pr(>Chisq)
exp3.mod3    7 869.15 888.9 -427.58    855.15
exp3.mod4    7 869.15 888.9 -427.58    855.15     0  0
```

#Test removing the factor Sample:

```
>exp3.mod5<-lmer(Ast353 ~ Size * Age  + (1|Batch), data=exp3, REML=TRUE)

>anova(exp3.mod4, exp3.mod5)

Data: exp3
Models:
exp3.mod5: Ast353 ~ Size * Age + (1 | Batch)
exp3.mod4: Ast353 ~ Size * Age + (1 | Sample) + (1 | Batch)
          npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp3.mod5    6 967.73 984.65 -477.87    955.73
exp3.mod4    7 869.15 888.90 -427.58    855.15 100.58  1  < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

#Sample cannot be removed from the model. Test removing Batch:

```
>exp3.mod6<-lmer(Ast353 ~ Size * Age + (1|Sample), data=exp3, REML=TRUE)

>anova(exp3.mod4, exp3.mod6)

Data: exp3
```

```
Models:
exp3.mod6: Ast353 ~ Size * Age + (1 | Sample)
exp3.mod4: Ast353 ~ Size * Age + (1 | Sample) + (1 | Batch)
          npar    AIC    BIC  logLik -2*log(L) Chisq Df Pr(>Chisq)
exp3.mod6    6 941.21 958.14 -464.61    929.21
exp3.mod4    7 869.15 888.90 -427.58    855.15 74.06  1  < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
#Batch cannot be removed from the model.
#Test removing the interaction term between size and age:

>exp3.mod7<-lmer(Ast353 ~ Size + Age + (1|Sample) + (1|Batch), data=exp3,
REML=TRUE)

>anova(exp3.mod4, exp3.mod7)
```

```
Data: exp3
Models:
exp3.mod7: Ast353 ~ Size + Age + (1 | Sample) + (1 | Batch)
exp3.mod4: Ast353 ~ Size * Age + (1 | Sample) + (1 | Batch)
          npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp3.mod7    6 867.23 884.16 -427.62    855.23
exp3.mod4    7 869.15 888.90 -427.58    855.15 0.0796  1     0.7779
```

```
#The interaction between size and age is non-significant and can be
removed from the model.
#Test removing Age:

>exp3.mod8<-lmer(Ast353 ~ Size + (1|Sample) + (1|Batch), data=exp3,
REML=TRUE)

>anova(exp3.mod7, exp3.mod8)
```

```
Data: exp3
Models:
exp3.mod8: Ast353 ~ Size + (1 | Sample) + (1 | Batch)
exp3.mod7: Ast353 ~ Size + Age + (1 | Sample) + (1 | Batch)
          npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp3.mod8    5 872.87 886.97 -431.43    862.87
exp3.mod7    6 867.23 884.16 -427.62    855.23 7.6343  1   0.005727 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
#Age is significant and cannot be removed from the model.

>summary(exp3.mod7)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method
['lmerModLmerTest']
Formula: Ast353 ~ Size + Age + (1 | Sample) + (1 | Batch)
   Data: exp3

REML criterion at convergence: 849.6

Scaled residuals:
     Min       1Q   Median       3Q      Max
-1.79233 -0.44322 -0.04534  0.54131  1.60772
```

```
Random effects:
 Groups   Name        Variance Std.Dev.
 Sample   (Intercept) 111.27   10.549
 Batch    (Intercept)  32.07    5.663
 Residual              10.99    3.314
Number of obs: 124, groups:  Sample, 62; Batch, 8

Fixed effects:
             Estimate Std. Error      df t value Pr(>|t|)
(Intercept)   58.2941     4.0305 19.3526  14.463 7.86e-12 ***
Sizestandard   4.1886     4.0482  5.5512   1.035  0.34377
Age           -0.3524     0.1252 59.4296  -2.816  0.00659 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

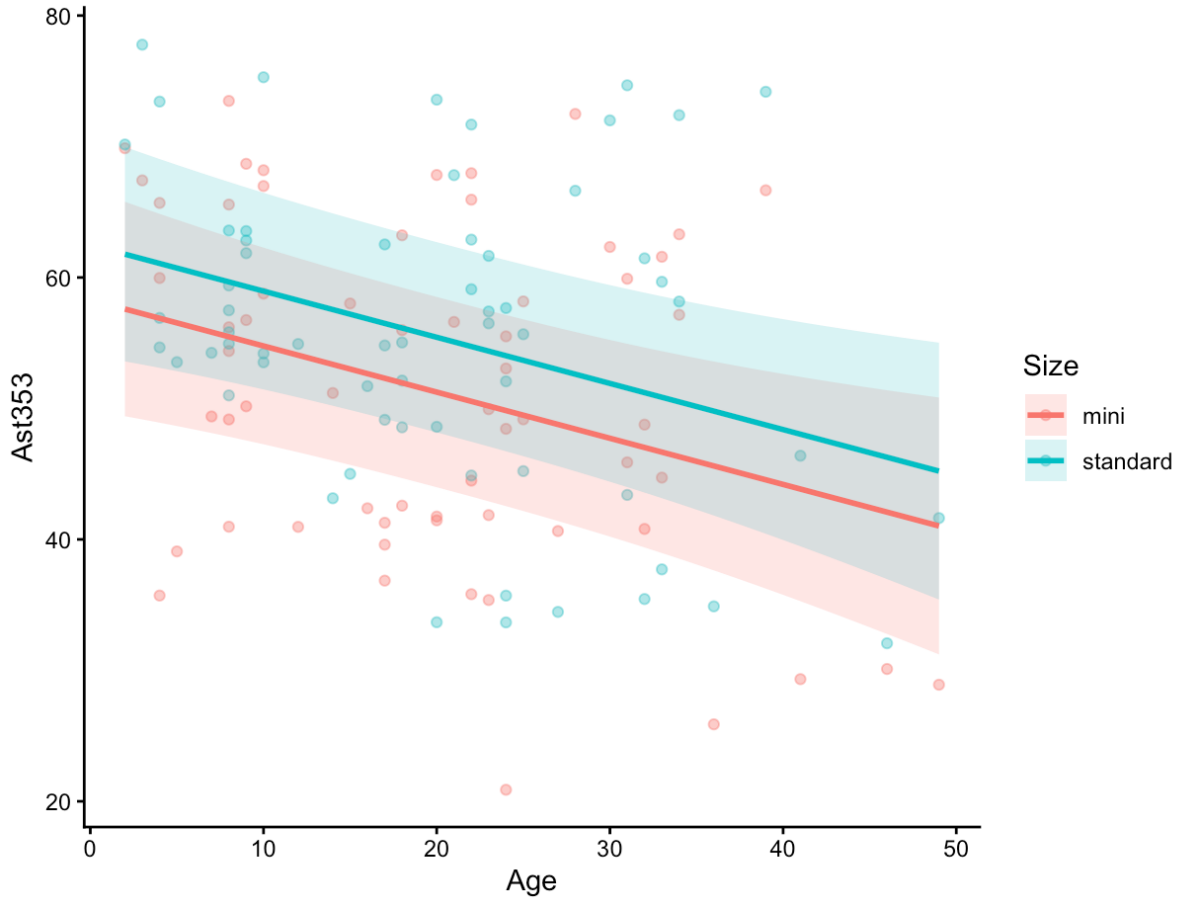
Correlation of Fixed Effects:
            (Intr) Szstnd
Sizestandrd -0.502
Age         -0.621  0.000
```

```
#Library reaction size does not have a significant effect on target
recovery.
#The model estimated marginal means were calculated for each library size
treatment:

>emm<- emmeans(exp3.mod7, ~Age | Size, at = list(Age = seq(min(exp3$Age),
max(exp3$Age), length=100)), lmer.df="satterthwaite")

>emm_df<-as.data.frame(emm)

>ggplot() +
  geom_point(data = exp3,
             aes(x = Age, y = Ast353, colour = Size),
             alpha = 0.4) +
  geom_ribbon(data = emm_df,
              aes(x = Age, ymin = lower.CL, ymax = upper.CL, fill = Size),
              alpha = 0.2) +
  geom_line(data = emm_df,
            aes(x = Age, y = emmean, colour = Size),
            linewidth = 1) +
  theme_classic()
```




```
#The variance components were estimated:

>vc <- as.data.frame(VarCorr(exp3.mod7))

>vc$percent <- vc$vcov / sum(vc$vcov) * 100

>vc
```

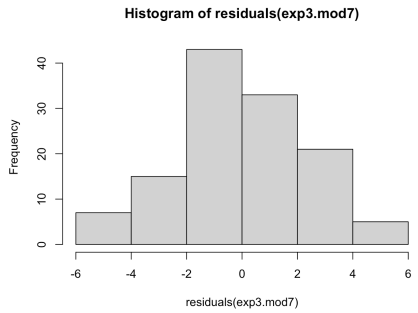
```
      grp        var1 var2      vcov     sdcor   percent
1  Sample (Intercept) <NA> 111.27098 10.548506 72.102461
2   Batch (Intercept) <NA>  32.06666  5.662743 20.778867
3 Residual        <NA> <NA>  10.98578  3.314480  7.118671
```

```
>AIC(exp3.mod7)
```

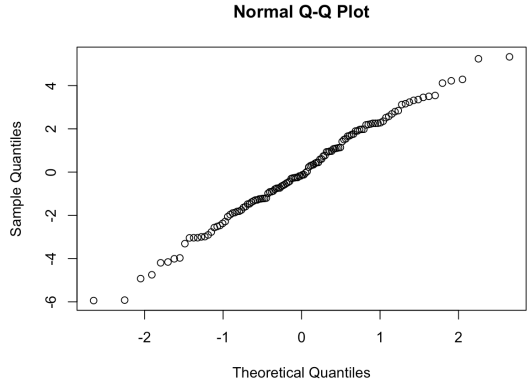
```
[1] 861.5579
```

```
#Model residuals are normally distributed according to the histogram, Q-Q
plot, and the Shapiro-Wilk test:

>hist(residuals(exp3.mod7))
```




```
>qqnorm(residuals(exp3.mod7))
```




```
>shapiro.test(residuals(exp3.mod7))

      Shapiro-Wilk normality test

data:  residuals(exp3.mod7)
W = 0.99303, p-value = 0.7978

#-----------------------------------

#Target recovery
#glmm

>exp3$Ast353_prop <- exp3$Ast353 / 100

#We performed a generalised linear mixed model with the proportion of
target base-pairs recovered (“Ast353_prop”) as the response variable with
“Size” (mini or standard) as a fixed factor, “Tribe”, “Genus within Tribe”
and “Sample nested in Genus nested in Tribe” as a random effects, and
“Batch” (pool) as a random effect:

>exp3.mod9<-glmmTMB(Ast353_prop ~ Size * Age + (1|Tribe) + (1|Tribe:Genus)
+ (1|Tribe:Genus:Sample) + (1|Batch), family=beta_family(link="logit"),
data=exp3)

#Test removing Tribe from the model:

>exp3.mod10<-glmmTMB(Ast353_prop ~ Size * Age + (1|Tribe:Genus) +
(1|Tribe:Genus:Sample) + (1|Batch), family=beta_family(link="logit"),
data=exp3)

>anova(exp3.mod9, exp3.mod10)
```

```
Data: exp3
Models:
exp3.mod10: Ast353_prop ~ Size * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + , zi=~0, disp=~1
exp3.mod10:     (1 | Batch), zi=~0, disp=~1
exp3.mod9: Ast353_prop ~ Size * Age + (1 | Tribe) + (1 | Tribe:Genus) + ,
zi=~0, disp=~1
exp3.mod9:     (1 | Tribe:Genus:Sample) + (1 | Batch), zi=~0, disp=~1
           Df    AIC     BIC logLik deviance Chisq Chi Df Pr(>Chisq)
exp3.mod10  8 -279.1 -256.54 147.55   -295.1
exp3.mod9   9 -277.1 -251.72 147.55   -295.1     0      1          1
```

#Tribe is non-significant and can be removed from the model. Test removing Genus nested within Tribe:

```
>exp3.mod11<-glmmTMB(Ast353_prop ~ Size * Age + (1|Tribe:Genus:Sample) +
(1|Batch), family=beta_family(link="logit"), data=exp3)

>anova(exp3.mod10, exp3.mod11)
```

```
Data: exp3
Models:
exp3.mod11: Ast353_prop ~ Size * Age + (1 | Tribe:Genus:Sample) + (1 |
Batch), zi=~0, disp=~1
exp3.mod10: Ast353_prop ~ Size * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + , zi=~0, disp=~1
exp3.mod10:     (1 | Batch), zi=~0, disp=~1
           Df    AIC     BIC logLik deviance Chisq Chi Df Pr(>Chisq)
exp3.mod11  7 -281.1 -261.36 147.55   -295.1
exp3.mod10  8 -279.1 -256.54 147.55   -295.1     0      1          1
```

#Genus nested in Tribe is also non-significant and can be removed from the model. Test removing Sample nested within Genus nested within Tribe:

```
>exp3.mod12<-glmmTMB(Ast353_prop ~ Size * Age  + (1|Batch),
family=beta_family(link="logit"), data=exp3)

>anova(exp3.mod11, exp3.mod12)
```

```
Data: exp3
Models:
exp3.mod12: Ast353_prop ~ Size * Age + (1 | Batch), zi=~0, disp=~1
exp3.mod11: Ast353_prop ~ Size * Age + (1 | Tribe:Genus:Sample) + (1 |
Batch), zi=~0, disp=~1
           Df     AIC     BIC  logLik deviance  Chisq Chi Df Pr(>Chisq)
exp3.mod12  6 -177.55 -160.63  94.777  -189.55
exp3.mod11  7 -281.11 -261.36 147.553  -295.11 105.55      1  < 2.2e-16
***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

#Sample nested within Genus nested in Tribe is significant and cannot be removed from the model. Test removing Batch:

```
>exp3.mod13<-glmmTMB(Ast353_prop ~ Size * Age + (1|Tribe:Genus:Sample),
family=beta_family(link="logit"), data=exp3)

>anova(exp3.mod11, exp3.mod13)
```

```
Data: exp3
Models:
exp3.mod13: Ast353_prop ~ Size * Age + (1 | Tribe:Genus:Sample), zi=~0,
disp=~1
exp3.mod11: Ast353_prop ~ Size * Age + (1 | Tribe:Genus:Sample) + (1 |
Batch), zi=~0, disp=~1
           Df     AIC     BIC logLik deviance  Chisq Chi Df Pr(>Chisq)
exp3.mod13  6 -203.37 -186.45 107.69  -215.37
exp3.mod11  7 -281.11 -261.36 147.55  -295.11 79.735      1  < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

#Batch is significant and cannot be removed from the model.
#Test removing the interaction between size and age:

```
>exp3.mod14<-glmmTMB(Ast353_prop ~ Size + Age + (1|Tribe:Genus:Sample) +
(1|Batch), family=beta_family(link="logit"), data=exp3)

>anova(exp3.mod11, exp3.mod14)
```

```
Data: exp3
Models:
exp3.mod14: Ast353_prop ~ Size + Age + (1 | Tribe:Genus:Sample) + (1 |
Batch), zi=~0, disp=~1
exp3.mod11: Ast353_prop ~ Size * Age + (1 | Tribe:Genus:Sample) + (1 |
Batch), zi=~0, disp=~1
           Df     AIC     BIC logLik deviance Chisq Chi Df Pr(>Chisq)
exp3.mod14  6 -282.93 -266.00 147.46  -294.93
exp3.mod11  7 -281.11 -261.36 147.55  -295.11  0.18      1     0.6714
```

#The interacton between size and age is insignificant and can be removed
from the model.
#Test removing age:

```
>exp3.mod15<-glmmTMB(Ast353_prop ~ Size  + (1|Tribe:Genus:Sample) +
(1|Batch), family=beta_family(link="logit"), data=exp3)

>anova(exp3.mod14, exp3.mod15)
```

```
Data: exp3
Models:
exp3.mod15: Ast353_prop ~ Size + (1 | Tribe:Genus:Sample) + (1 | Batch),
zi=~0, disp=~1
exp3.mod14: Ast353_prop ~ Size + Age + (1 | Tribe:Genus:Sample) + (1 |
Batch), zi=~0, disp=~1
           Df     AIC     BIC logLik deviance Chisq Chi Df Pr(>Chisq)
exp3.mod15  5 -277.38 -263.28 143.69  -287.38
exp3.mod14  6 -282.93 -266.00 147.46  -294.93 7.543      1   0.006024 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

#Age is significant and cannot be removed from the model.

```
>summary(exp3.mod14)
```

```
Family: beta  ( logit )
Formula:          Ast353_prop ~ Size + Age + (1 | Tribe:Genus:Sample) + (1
| Batch)
```

```
Data: exp3

      AIC       BIC    logLik -2*log(L)  df.resid
   -282.9    -266.0     147.5    -294.9       118

Random effects:

Conditional model:
 Groups             Name        Variance Std.Dev.
 Tribe:Genus:Sample (Intercept) 0.19451  0.4410
 Batch              (Intercept) 0.04499  0.2121
Number of obs: 124, groups:  Tribe:Genus:Sample, 62; Batch, 8

Dispersion parameter for beta family ():  239

Conditional model:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)   0.346294   0.159836   2.167   0.0303 *
Sizestandard  0.182128   0.151887   1.199   0.2305
Age          -0.014797   0.005221  -2.834   0.0046 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
#Library size does not significantly influence target recovery.

>emm <- emmeans(exp3.mod14, ~Size + Age, type="response")

>emm
```

```
Size      Age response     SE  df asymp.LCL asymp.UCL
 mini     19.9    0.513 0.0303 Inf     0.454     0.572
 standard 19.9    0.558 0.0299 Inf     0.499     0.616

Confidence level used: 0.95
Intervals are back-transformed from the logit scale
```

```
>pairs(emm, adjust="tukey")
```

```
contrast                                                odds.ratio    SE
df null z.ratio
 mini Age19.9354838709677 / standard Age19.9354838709677      0.833 0.127
Inf    1  -1.199
 p.value
  0.2305

Tests are performed on the log odds ratio scale
```

```
>AIC(exp3.mod14)
```

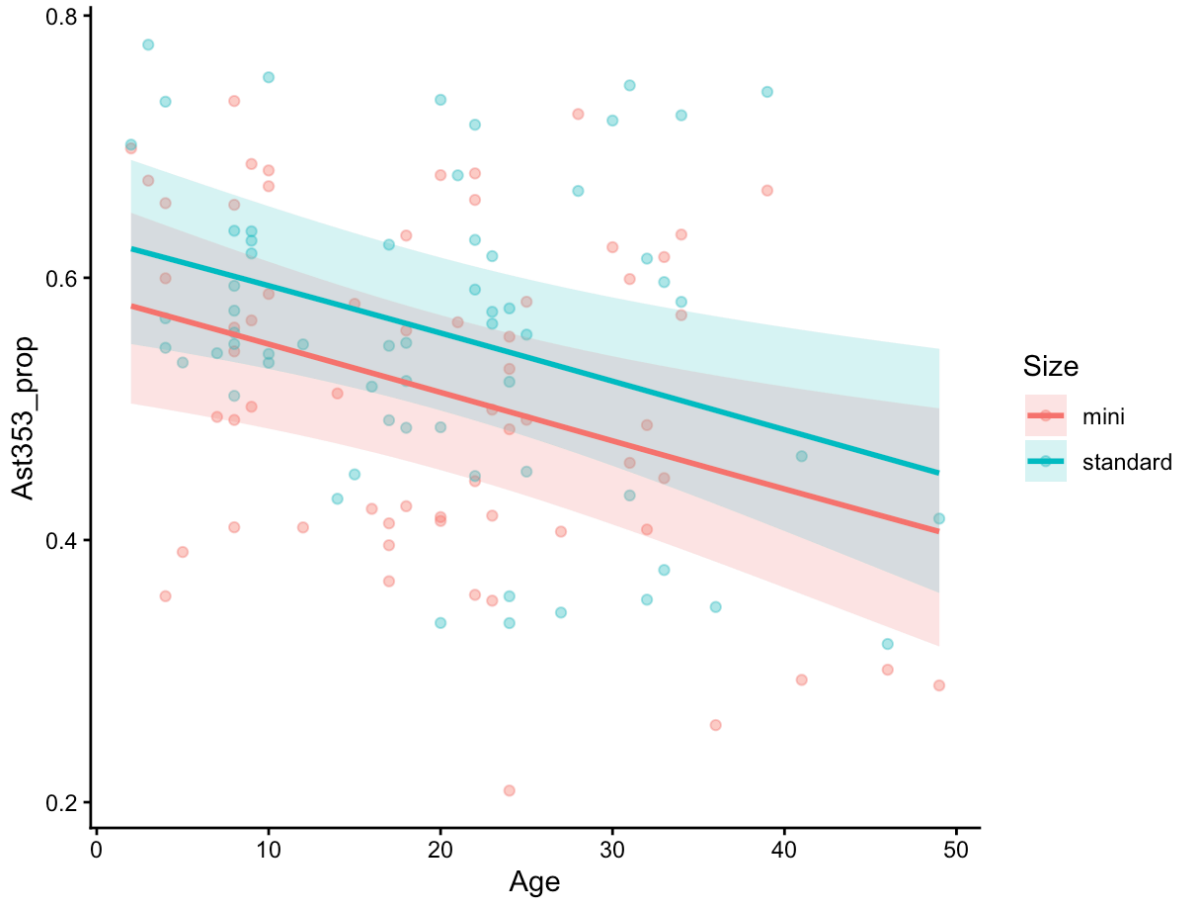
```
[1] -282.9252
```

```
>emm<- emmeans(exp3.mod14, ~Age | Size, at = list(Age = seq(min(exp3$Age),
max(exp3$Age), length=100)), type="response")

>emm_df<-as.data.frame(emm)

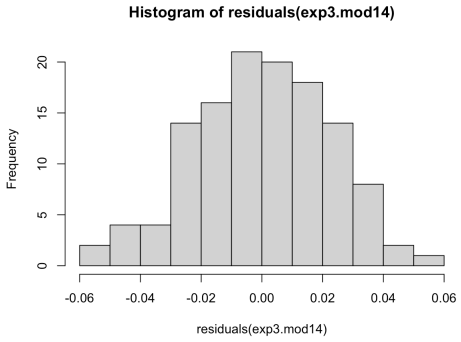
>ggplot() +
  geom_point(data = exp3,
             aes(x = Age, y = Ast353_prop, colour = Size),
```

```
            alpha = 0.4) +
  geom_ribbon(data = emm_df,
              aes(x = Age, ymin = asymp.LCL, ymax = asymp.UCL, fill =
Size),
              alpha = 0.2) +
  geom_line(data = emm_df,
            aes(x = Age, y = response, colour = Size),
            linewidth = 1) +
  theme_classic()
```

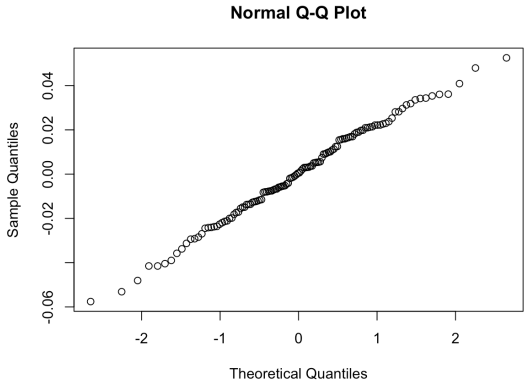


#Model residuals are now non-normally distributed according to the Shapiro-Wilk test, though the histogram and Q-Q plot show residuals are approximately normally distributed:

```
>hist(residuals(exp3.mod14))
```



```
>qqnorm(residuals(exp3.mod14))
```




```
>shapiro.test(residuals(exp3.mod14))

Shapiro-Wilk normality test

data:  residuals(exp3.mod14)
W = 0.99342, p-value = 0.832

#-----------------------------------

#Chloroplast recovery
#lmer

#We performed a linear mixed model with the percentage of chloroplast base
pairs recovered (“Chlp”) as the response variable with “Size” (mini or
standard) as a fixed factor, “Tribe”, “Genus within Tribe” and “Sample
nested in Genus nested in Tribe” as a random effects, and “Batch” (pool)
as a random effect:

>exp3.mod16<-lmer(Chlp ~ Size * Age + (1|Tribe) + (1|Tribe:Genus) +
(1|Tribe:Genus:Sample) + (1|Batch), data=exp3, REML=TRUE)

#Test removing Tribe as a factor:

>exp3.mod17<-lmer(Chlp ~ Size * Age  + (1|Tribe:Genus) +
(1|Tribe:Genus:Sample) + (1|Batch), data=exp3, REML=TRUE)

>anova(exp3.mod16, exp3.mod17)

Data: exp3
Models:
exp3.mod17: Chlp ~ Size * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Batch)
exp3.mod16: Chlp ~ Size * Age + (1 | Tribe) + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Batch)
           npar    AIC    BIC  logLik -2*log(L) Chisq Df Pr(>Chisq)
exp3.mod17    8 968.17 990.73 -476.08    952.17
exp3.mod16    9 970.17 995.55 -476.08    952.17     0  1          1

#Tribe does not significantly contribute to the model. Next, remove Genus
within Tribe:

>exp3.mod18<-lmer(Chlp ~ Size * Age  +  (1|Tribe:Genus:Sample) +
(1|Batch), data=exp3, REML=TRUE)
```

```
>anova(exp3.mod17, exp3.mod18)

Data: exp3
Models:
exp3.mod18: Chlp ~ Size * Age + (1 | Tribe:Genus:Sample) + (1 | Batch)
exp3.mod17: Chlp ~ Size * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Batch)
           npar    AIC     BIC  logLik -2*log(L) Chisq Df Pr(>Chisq)
exp3.mod18    7 992.51 1012.25 -489.25    978.51
exp3.mod17    8 968.17  990.73 -476.08    952.17 26.34  1  2.862e-07 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

#Genus within Tribe cannot be removed. Next, test removing Sample from the
model:

>exp3.mod19<-lmer(Chlp ~ Size * Age  + (1|Tribe:Genus)  + (1|Batch),
data=exp3, REML=TRUE)

>anova(exp3.mod17, exp3.mod19)

Data: exp3
Models:
exp3.mod19: Chlp ~ Size * Age + (1 | Tribe:Genus) + (1 | Batch)
exp3.mod17: Chlp ~ Size * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Batch)
           npar     AIC     BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp3.mod19    7 1007.50 1027.24 -496.75    993.50
exp3.mod17    8  968.17  990.73 -476.08    952.17 41.333  1  1.284e-10 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

#Sample cannot be removed. Next, test removing Batch from the model:

>exp3.mod20<-lmer(Chlp ~ Size * Age  + (1|Tribe:Genus) +
(1|Tribe:Genus:Sample), data=exp3, REML=TRUE)

>anova(exp3.mod17, exp3.mod20)

Data: exp3
Models:
exp3.mod20: Chlp ~ Size * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample)
exp3.mod17: Chlp ~ Size * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Batch)
           npar    AIC     BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp3.mod20    7 992.76 1012.50 -489.38    978.76
exp3.mod17    8 968.17  990.73 -476.08    952.17 26.593  1  2.511e-07 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

#Batch cannot be removed.
#Test removing the interaction between size and age:

>exp3.mod21<-lmer(Chlp ~ Size + Age  + (1|Tribe:Genus) +
(1|Tribe:Genus:Sample) + (1|Batch), data=exp3, REML=TRUE)

>anova(exp3.mod17, exp3.mod21)
```

```
Data: exp3
Models:
exp3.mod21: Chlp ~ Size + Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Batch)
exp3.mod17: Chlp ~ Size * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Batch)
           npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp3.mod21    7 974.89 994.63 -480.45    960.89
exp3.mod17    8 968.17 990.73 -476.08    952.17 8.7237  1   0.003141 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

#The interaction between size and age is significant and cannot be removed.

```
>summary(exp3.mod17)

Linear mixed model fit by REML. t-tests use Satterthwaite's method
['lmerModLmerTest']
Formula: Chlp ~ Size * Age + (1 | Tribe:Genus) + (1 | Tribe:Genus:Sample)
+      (1 | Batch)
   Data: exp3

REML criterion at convergence: 946.5

Scaled residuals:
     Min       1Q   Median       3Q      Max
-2.06047 -0.36212 -0.00672  0.38795  2.65054

Random effects:
 Groups             Name        Variance Std.Dev.
 Tribe:Genus:Sample (Intercept) 111.83   10.575
 Batch              (Intercept)  36.06    6.005
 Tribe:Genus        (Intercept) 158.45   12.588
 Residual                        44.09    6.640
Number of obs: 124, groups:  Tribe:Genus:Sample, 62; Batch, 8;
Tribe:Genus, 5

Fixed effects:
                 Estimate Std. Error       df t value Pr(>|t|)
(Intercept)       36.8558     8.1646   6.9838   4.514  0.00277 **
Sizestandard     -13.2537     4.9236   7.7361  -2.692  0.02826 *
Age               -0.4097     0.1520  75.8829  -2.696  0.00865 **
Sizestandard:Age   0.3305     0.1094  57.2634   3.021  0.00376 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Correlation of Fixed Effects:
            (Intr) Szstnd Age
Sizestandrd -0.302
Age         -0.431  0.160
Szstndrd:Ag  0.134 -0.444 -0.360

>vc <- as.data.frame(VarCorr(exp3.mod17))

>vc$percent <- vc$vcov / sum(vc$vcov) * 100

>vc
              grp        var1 var2      vcov     sdcor  percent
```

```
1 Tribe:Genus:Sample (Intercept) <NA> 111.82800 10.574876 31.91093
2              Batch (Intercept) <NA>  36.06251  6.005207 10.29070
3        Tribe:Genus (Intercept) <NA> 158.45261 12.587796 45.21560
4           Residual        <NA> <NA>  44.09481  6.640393 12.58277
```

```
>emm<- emmeans(exp3.mod17, ~Size * Age, lmer.df="satterthwaite")
```

```
>emm
```

```
Size      Age emmean   SE   df lower.CL upper.CL
 mini     19.9   28.7 7.38 5.05     9.76     47.6
 standard 19.9   22.0 7.38 5.05     3.10     40.9

Degrees-of-freedom method: satterthwaite
Confidence level used: 0.95
```

```
>pairs(emm, adjust="none")
```

```
contrast                                                estimate   SE   df
t.ratio p.value
 mini Age19.9354838709677 - standard Age19.9354838709677     6.66 4.41
5.02   1.511  0.1910

Degrees-of-freedom method: satterthwaite
```

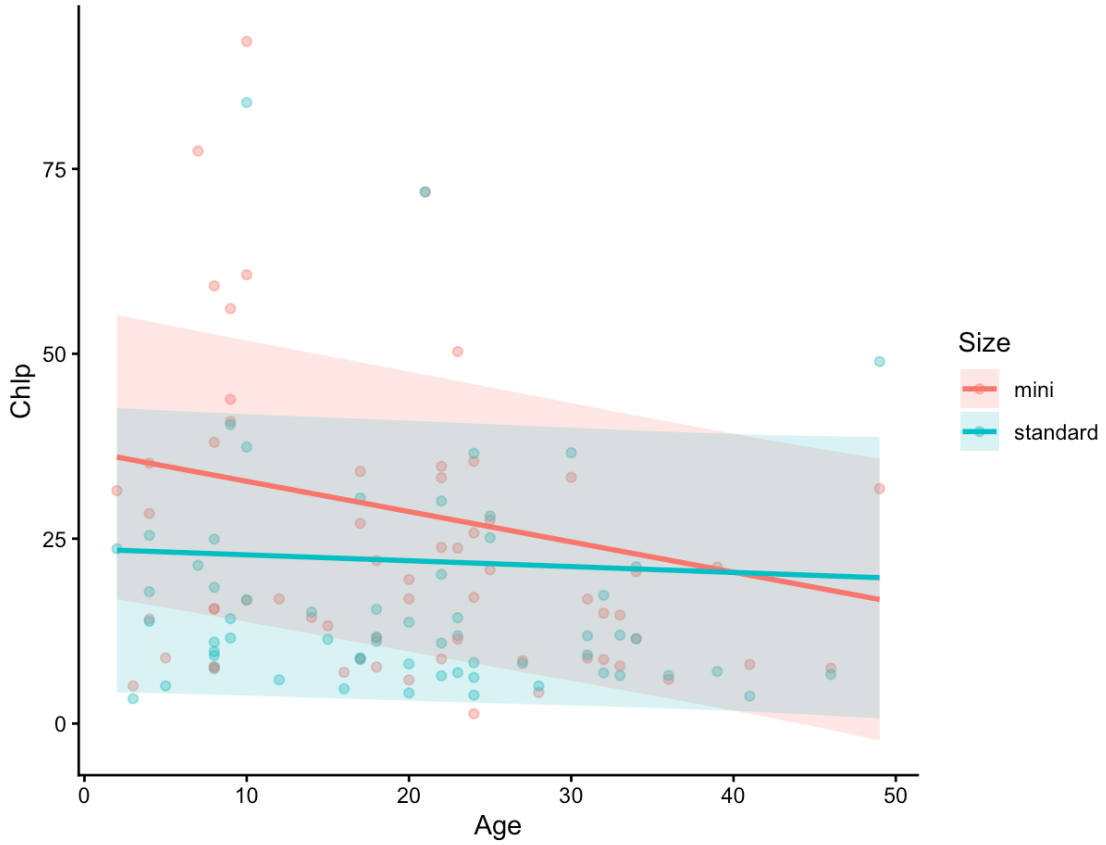
```
>AIC(exp3.mod17)
```

```
[1] 962.5325
```

```
>emm<- emmeans(exp3.mod17, ~Age | Size, at = list(Age = seq(min(exp3$Age),
max(exp3$Age), length=100)), lmer.df="satterthwaite")
```

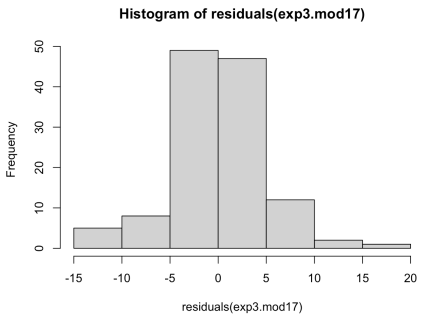
```
>emm_df<-as.data.frame(emm)
```

```
>ggplot() +
  geom_point(data = exp3,
             aes(x = Age, y = Chlp, colour = Size),
             alpha = 0.4) +
  geom_ribbon(data = emm_df,
              aes(x = Age, ymin = lower.CL, ymax = upper.CL, fill = Size),
              alpha = 0.2) +
  geom_line(data = emm_df,
            aes(x = Age, y = emmean, colour = Size),
            linewidth = 1) +
  theme_classic()
```

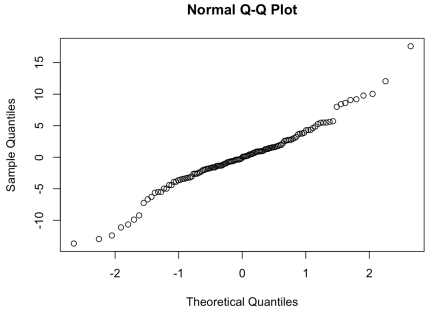


#Model residuals are approximately normally distributed according to the histogram, Q-Q plot, but the Shapiro-Wilk test indicates that statistically they are non-normal:

```
>hist(residuals(exp3.mod17))
```




```
>qqnorm(residuals(exp3.mod17))
```




```
>shapiro.test(residuals(exp3.mod17))
```

```
Shapiro-Wilk normality test

data:  residuals(exp3.mod17)
W = 0.96675, p-value = 0.003765

#-----------------------------------

#Chloroplast recovery
#glmm

#We performed a linear mixed model with the proportion of chloroplast base
pairs recovered (“Chlp_prop”) as the response variable with “Size” (mini
or standard) as a fixed factor, “Tribe”, “Genus within Tribe” and “Sample
nested in Genus nested in Tribe” as a random effects, and “Batch” (pool)
as a random effect:

>exp3$Chlp_prop <- exp3$Chlp / 100

>exp3.mod22<-glmmTMB(Chlp_prop ~ Size * Age + (1|Tribe) + (1|Tribe:Genus)
+ (1|Tribe:Genus:Sample) + (1|Batch), family=beta_family(link="logit"),
data=exp3)

#Test removing Tribe as a factor:

>exp3.mod23<-glmmTMB(Chlp_prop ~ Size * Age + (1|Tribe:Genus) +
(1|Tribe:Genus:Sample) + (1|Batch), family=beta_family(link="logit"),
data=exp3)

>anova(exp3.mod22, exp3.mod23)

Data: exp3
Models:
exp3.mod23: Chlp_prop ~ Size * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + , zi=~0, disp=~1
exp3.mod23:     (1 | Batch), zi=~0, disp=~1
exp3.mod22: Chlp_prop ~ Size * Age + (1 | Tribe) + (1 | Tribe:Genus) + (1
| , zi=~0, disp=~1
exp3.mod22:     Tribe:Genus:Sample) + (1 | Batch), zi=~0, disp=~1
           Df     AIC     BIC logLik deviance Chisq Chi Df Pr(>Chisq)
exp3.mod23  8 -290.25 -267.69 153.13  -306.25
exp3.mod22  9 -288.25 -262.87 153.13  -306.25     0      1          1

#Tribe is non-significant and can be removed from the model. Test removing
Genus nested in Tribe:

>exp3.mod24<-glmmTMB(Chlp_prop ~ Size * Age + (1|Tribe:Genus:Sample) +
(1|Batch), family=beta_family(link="logit"), data=exp3)

>anova(exp3.mod23, exp3.mod24)

Data: exp3
Models:
exp3.mod24: Chlp_prop ~ Size * Age + (1 | Tribe:Genus:Sample) + (1 |
Batch), zi=~0, disp=~1
exp3.mod23: Chlp_prop ~ Size * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + , zi=~0, disp=~1
exp3.mod23:     (1 | Batch), zi=~0, disp=~1
           Df     AIC     BIC logLik deviance  Chisq Chi Df Pr(>Chisq)
```

```
exp3.mod24  7 -266.95 -247.21 140.47  -280.95
exp3.mod23  8 -290.25 -267.69 153.13  -306.25 25.302      1  4.903e-07 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

#Genus nested in Tribe is significant and cannot be removed from the model. Test removing Sample nested in Genus nested in Tribe:

```
>exp3.mod25<-glmmTMB(Chlp_prop ~ Size * Age + (1|Tribe:Genus) + (1|Batch),
family=beta_family(link="logit"), data=exp3)

>anova(exp3.mod23, exp3.mod25)

Data: exp3
Models:
exp3.mod25: Chlp_prop ~ Size * Age + (1 | Tribe:Genus) + (1 | Batch),
zi=~0, disp=~1
exp3.mod23: Chlp_prop ~ Size * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + , zi=~0, disp=~1
exp3.mod23:     (1 | Batch), zi=~0, disp=~1
           Df     AIC     BIC logLik deviance  Chisq Chi Df Pr(>Chisq)
exp3.mod25  7 -192.07 -172.33 103.04  -206.07
exp3.mod23  8 -290.25 -267.69 153.13  -306.25 100.18      1  < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

#Sampled nested in Genus nested in Tribe is significant and cannot be removed from the model. Test removing Batch:

```
>exp3.mod26<-glmmTMB(Chlp_prop ~ Size * Age  + (1|Tribe:Genus) +
(1|Tribe:Genus:Sample), family=beta_family(link="logit"), data=exp3)

>anova(exp3.mod23, exp3.mod26)

Data: exp3
Models:
exp3.mod26: Chlp_prop ~ Size * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample), zi=~0, disp=~1
exp3.mod23: Chlp_prop ~ Size * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + , zi=~0, disp=~1
exp3.mod23:     (1 | Batch), zi=~0, disp=~1
           Df     AIC     BIC logLik deviance  Chisq Chi Df Pr(>Chisq)
exp3.mod26  7 -219.82 -200.08 116.91  -233.82
exp3.mod23  8 -290.25 -267.69 153.13  -306.25 72.432      1  < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

#Batch is significant and cannot be removed from the model.
#Test removing the interaction between size and age:

```
>exp3.mod27<-glmmTMB(Chlp_prop ~ Size + Age + (1|Tribe:Genus) +
(1|Tribe:Genus:Sample) + (1|Batch), family=beta_family(link="logit"),
data=exp3)

>anova(exp3.mod23, exp3.mod27)

Data: exp3
Models:
```

```
exp3.mod27: Chlp_prop ~ Size + Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + , zi=~0, disp=~1
exp3.mod27:     (1 | Batch), zi=~0, disp=~1
exp3.mod23: Chlp_prop ~ Size * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + , zi=~0, disp=~1
exp3.mod23:     (1 | Batch), zi=~0, disp=~1
           Df     AIC     BIC logLik deviance Chisq Chi Df Pr(>Chisq)
exp3.mod27  7 -277.69 -257.95 145.85  -291.69
exp3.mod23  8 -290.25 -267.69 153.13  -306.25 14.56      1  0.0001358 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

#The interaction betwen size and age is significant and can't be removed from the model.

```
>summary(exp3.mod23)
```

```
Family: beta  ( logit )
Formula:          Chlp_prop ~ Size * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) +
    (1 | Batch)
Data: exp3

      AIC       BIC    logLik -2*log(L)  df.resid
   -290.3    -267.7     153.1    -306.3       116

Random effects:

Conditional model:
 Groups             Name        Variance Std.Dev.
 Tribe:Genus        (Intercept) 0.4267   0.6532
 Tribe:Genus:Sample (Intercept) 0.5000   0.7071
 Batch              (Intercept) 0.1818   0.4264
Number of obs: 124, groups:  Tribe:Genus, 5; Tribe:Genus:Sample, 62;
Batch, 8

Dispersion parameter for beta family ():  118

Conditional model:
                  Estimate Std. Error z value Pr(>|z|)
(Intercept)      -0.566035   0.477335  -1.186  0.23569
Sizestandard     -0.839640   0.316208  -2.655  0.00792 **
Age              -0.023108   0.009272  -2.492  0.01270 *
Sizestandard:Age  0.017459   0.004316   4.045 5.22e-05 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
>emm <- emmeans(exp3.mod23, ~Size, type="response")
```

```
>emm
```

```
Size     response     SE  df asymp.LCL asymp.UCL
 mini        0.264 0.0827 Inf    0.1345     0.452
 standard    0.180 0.0628 Inf    0.0868     0.336

Confidence level used: 0.95
Intervals are back-transformed from the logit scale
```

```
>pairs(emm, adjust="none")
```

```
contrast        odds.ratio    SE  df null z.ratio p.value
 mini / standard       1.63 0.499 Inf    1   1.611  0.1073

Tests are performed on the log odds ratio scale
```

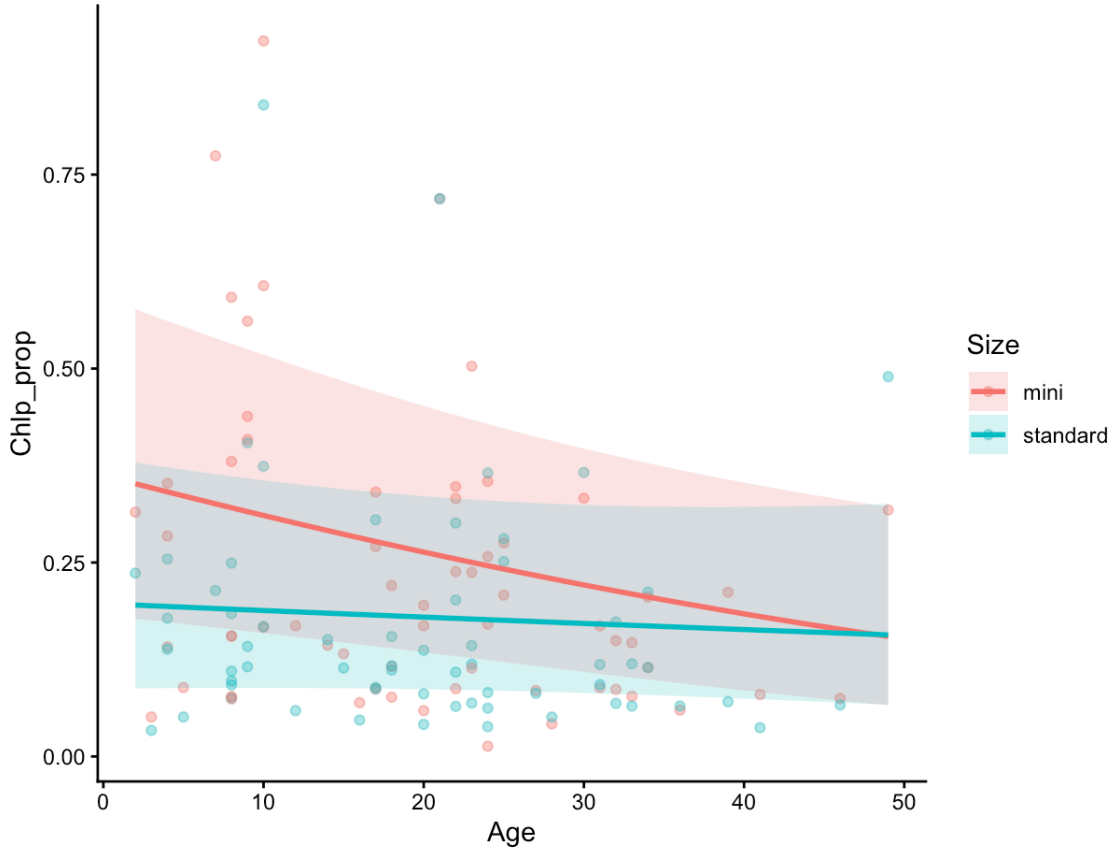
```
>AIC(exp3.mod23)

[1] -290.2527

>emm<- emmeans(exp3.mod23, ~Age | Size, at = list(Age = seq(min(exp3$Age),
max(exp3$Age), length=100)), type="response")

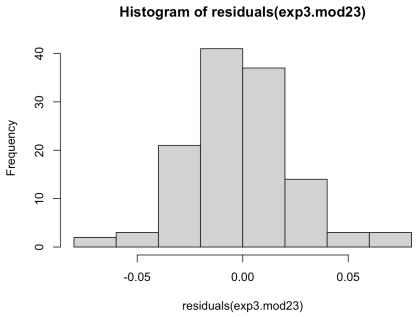
>emm_df<-as.data.frame(emm)

>ggplot() +
  geom_point(data = exp3,
             aes(x = Age, y = Chlp_prop, colour = Size),
             alpha = 0.4) +
  geom_ribbon(data = emm_df,
              aes(x = Age, ymin = asymp.LCL, ymax = asymp.UCL, fill =
Size),
              alpha = 0.2) +
  geom_line(data = emm_df,
            aes(x = Age, y = response, colour = Size),
            linewidth = 1) +
  theme_classic()
```

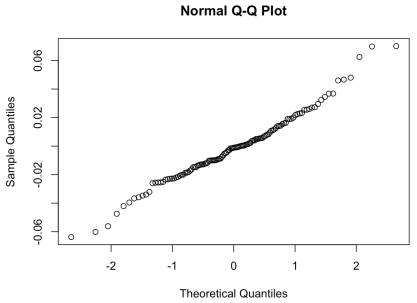


#Model residuals remain approximately normally distributed according to the histogram, Q-Q plot, but the Shapiro-Wilk test indicates that statistically they are non-normal:

```
>hist(residuals(exp3.mod23))
```




```
>qqnorm(residuals(exp3.mod23))
```




```
>shapiro.test(residuals(exp3.mod23))

Shapiro-Wilk normality test

data:  residuals(exp3.mod23)
W = 0.98141, p-value = 0.08567

#-----------------------------------
#Between treatments (Aim 4)
#-----------------------------------

#Target recovery
#lmer

>exp4<-read.csv("exp4_w_year.csv", header=TRUE)

>head(exp4)

  Sample Year Age     Batch       Genus    Tribe Size Input_cat
1  Brachyscome_barkerae_AG003 2011   9 AG_P1P15C Brachyscome Astereae mini
20ng
2  Brachyscome_barkerae_AG003 2011   9  AG_P1P7C Brachyscome Astereae mini
2ng
3  Brachyscome_barkerae_AG003 2011   9 AG_P1P19C Brachyscome Astereae mini
40ng
4  Brachyscome_barkerae_AG003 2011   9 AG_P1P11C Brachyscome Astereae mini
5ng
5 Brachyscome_basaltica_AG004 2013   7 AG_P1P15C Brachyscome Astereae mini
20ng
6 Brachyscome_basaltica_AG004 2013   7  AG_P1P7C Brachyscome Astereae mini
2ng
```

```
  Input_cont Treatment Composition     Chlp   Ast353  logChlp logAst353
1         20       20M           U 13.01628 49.34990 1.114487  1.693286
2          2        2M           U 29.85010 48.21345 1.474946  1.683168
3         40       40M           U 11.88396 56.26845 1.074961  1.750265
4          5        5M           U 43.83695 50.16228 1.641840  1.700377
5         20       20M           U 21.06654 47.04926 1.323593  1.672553
6          2        2M           U 50.15098 42.87860 1.700279  1.632241
```

#We performed a linear mixed model with the percentage target base-pairs recovered (“Ast353”) as the response variable with “Input” (2 ng, 5 ng, 20 ng, 40 ng) as a fixed, categorical factor, “Tribe”, “Genus within Tribe”, “Sample nested in tribe nested in Genus” and “Batch” (pool) as random effects:

```
>exp4.mod<-lmer(Ast353 ~ Input_cat * Age + (1|Tribe) + (1|Tribe:Genus) + (1|Tribe:Genus:Sample) + (1|Batch), data=exp4, REML=TRUE)
```

#Test removing Tribe as a factor:

```
>exp4.mod2<-lmer(Ast353 ~ Input_cat * Age + (1|Tribe:Genus) + (1|Tribe:Genus:Sample) + (1|Batch), data=exp4, REML=TRUE)

>anova(exp4.mod, exp4.mod2)
```

```
Data: exp4
Models:
exp4.mod2: Ast353 ~ Input_cat * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Batch)
exp4.mod: Ast353 ~ Input_cat * Age + (1 | Tribe) + (1 | Tribe:Genus) + (1
| Tribe:Genus:Sample) + (1 | Batch)
          npar    AIC    BIC  logLik -2*log(L) Chisq Df Pr(>Chisq)
exp4.mod2   12 1689.7 1731.8 -832.84    1665.7
exp4.mod    13 1691.7 1737.3 -832.84    1665.7     0  1          1
```

#Tribe can be removed as it contributes nothing to variance. Now test removing Genus within Tribe:

```
>exp4.mod3<-lmer(Ast353 ~ Input_cat * Age  + (1|Tribe:Genus:Sample) + (1|Batch), data=exp4, REML=TRUE)

>anova(exp4.mod2, exp4.mod3)
```

```
Data: exp4
Models:
exp4.mod3: Ast353 ~ Input_cat * Age + (1 | Tribe:Genus:Sample) + (1 |
Batch)
exp4.mod2: Ast353 ~ Input_cat * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Batch)
          npar    AIC    BIC  logLik -2*log(L) Chisq Df Pr(>Chisq)
exp4.mod3   11 1687.7 1726.3 -832.84    1665.7
exp4.mod2   12 1689.7 1731.8 -832.84    1665.7     0  1          1
```

#Genus within Tribe can be removed. Simplify the sample term:

```
>exp4.mod4<-lmer(Ast353 ~ Input_cat * Age  + (1|Sample) + (1|Batch), data=exp4, REML=TRUE)

>anova(exp4.mod3, exp4.mod4)
```

```
Data: exp4
Models:
exp4.mod3: Ast353 ~ Input_cat * Age + (1 | Tribe:Genus:Sample) + (1 |
Batch)
exp4.mod4: Ast353 ~ Input_cat * Age + (1 | Sample) + (1 | Batch)
          npar    AIC    BIC  logLik -2*log(L) Chisq Df Pr(>Chisq)
exp4.mod3   11 1687.7 1726.3 -832.84    1665.7
exp4.mod4   11 1687.7 1726.3 -832.84    1665.7     0  0
```

#Now test removing Sample from the model:

```
>exp4.mod5<-lmer(Ast353 ~ Input_cat * Age  + (1|Batch), data=exp4,
REML=TRUE)

>anova(exp4.mod4, exp4.mod5)

Data: exp4
Models:
exp4.mod5: Ast353 ~ Input_cat * Age + (1 | Batch)
exp4.mod4: Ast353 ~ Input_cat * Age + (1 | Sample) + (1 | Batch)
          npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp4.mod5   10 1972.7 2007.8 -976.35    1952.7
exp4.mod4   11 1687.7 1726.3 -832.84    1665.7 287.03  1  < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

#Sample cannot be removed from the model. Now try removing Batch:

```
>exp4.mod6<-lmer(Ast353 ~ Input_cat * Age  + (1|Sample) , data=exp4,
REML=TRUE)

>anova(exp4.mod4, exp4.mod6)
Data: exp4
Models:
exp4.mod6: Ast353 ~ Input_cat * Age + (1 | Sample)
exp4.mod4: Ast353 ~ Input_cat * Age + (1 | Sample) + (1 | Batch)
          npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp4.mod6   10 1764.0 1799.1 -872.01    1744.0
exp4.mod4   11 1687.7 1726.3 -832.84    1665.7 78.344  1  < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

#Batch cannot be removed from the model.
#Test removing the interaction term between Input_cat and age:

```
>exp4.mod7<-lmer(Ast353 ~ Input_cat + Age  + (1|Sample) + (1|Batch),
data=exp4, REML=TRUE)

>anova(exp4.mod4, exp4.mod7)

Data: exp4
Models:
exp4.mod7: Ast353 ~ Input_cat + Age + (1 | Sample) + (1 | Batch)
exp4.mod4: Ast353 ~ Input_cat * Age + (1 | Sample) + (1 | Batch)
          npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp4.mod7    8 1688.7 1716.7 -836.33    1672.7
exp4.mod4   11 1687.7 1726.3 -832.84    1665.7 6.9796  3    0.07255 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
#The interaction term between input_cat and age is not significant and can
be removed.
#Test removing age:

>exp4.mod8<-lmer(Ast353 ~ Input_cat  + (1|Sample) + (1|Batch), data=exp4,
REML=TRUE)

>anova(exp4.mod7, exp4.mod8)

Data: exp4
Models:
exp4.mod8: Ast353 ~ Input_cat + (1 | Sample) + (1 | Batch)
exp4.mod7: Ast353 ~ Input_cat + Age + (1 | Sample) + (1 | Batch)
          npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp4.mod8    7 1692.2 1716.8 -839.12    1678.2
exp4.mod7    8 1688.7 1716.7 -836.33    1672.7 5.5806  1    0.01816 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

#Age is significant and cannot be removed from the model.
#Test removing Input_cat:

>exp4.mod9<-lmer(Ast353 ~  Age  + (1|Sample) + (1|Batch), data=exp4,
REML=TRUE)

>anova(exp4.mod7, exp4.mod9)

Data: exp4
Models:
exp4.mod9: Ast353 ~ Age + (1 | Sample) + (1 | Batch)
exp4.mod7: Ast353 ~ Input_cat + Age + (1 | Sample) + (1 | Batch)
          npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp4.mod9    5 1694.7 1712.2 -842.34    1684.7
exp4.mod7    8 1688.7 1716.7 -836.33    1672.7 12.028  3   0.007288 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

#Input_cat is significant and cannot be removed from the model.

>summary(exp4.mod7)

Linear mixed model fit by REML. t-tests use Satterthwaite's method
['lmerModLmerTest']
Formula: Ast353 ~ Input_cat + Age + (1 | Sample) + (1 | Batch)
   Data: exp4

REML criterion at convergence: 1660.5

Scaled residuals:
    Min      1Q  Median      3Q     Max
-5.0240 -0.3915  0.0759  0.4362  3.7849

Random effects:
 Groups   Name        Variance Std.Dev.
 Sample   (Intercept) 134.44   11.595
 Batch    (Intercept)  18.99    4.357
 Residual              19.61    4.428
Number of obs: 247, groups:  Sample, 62; Batch, 16
```

```
Fixed effects:
             Estimate Std. Error      df t value Pr(>|t|)
(Intercept)   54.2857     3.8165 51.3225  14.224   <2e-16 ***
Input_cat2ng  -7.8164     3.1821 10.3235  -2.456   0.0332 *
Input_cat40ng  1.4528     3.1833 10.3389   0.456   0.6576
Input_cat5ng   3.4262     3.1821 10.3235   1.077   0.3061
Age           -0.3231     0.1357 60.3089  -2.381   0.0204 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

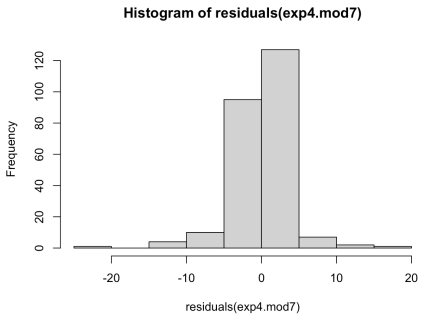
Correlation of Fixed Effects:
            (Intr) Inpt_2 Inp_40 Inpt_5
Input_ct2ng -0.417
Inpt_ct40ng -0.417  0.500
Input_ct5ng -0.417  0.500  0.500
Age         -0.710  0.000  0.000  0.000
```

```
>AIC(exp4.mod7)

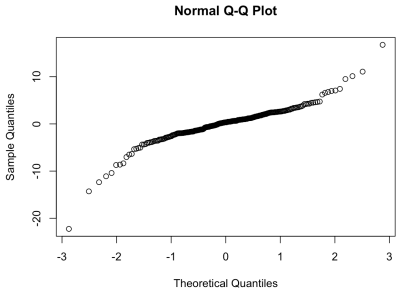
[1] 1676.497
```

#Model residuals are approximately normally distributed according to the histogram and Q-Q plot, but the Shapiro-Wilk test indicates that statistically they are non-normal:

```
>hist(residuals(exp4.mod7))
```




```
>qqnorm(residuals(exp4.mod7))
```




```
>shapiro.test(residuals(exp4.mod7))

Shapiro-Wilk normality test

data:  residuals(exp4.mod7)
W = 0.89471, p-value = 4.142e-12
```

```
#Calculate the variance components:

>vc <- as.data.frame(VarCorr(exp4.mod7))

>vc$percent <- vc$vcov / sum(vc$vcov) * 100

>vc

     grp        var1 var2      vcov     sdcor  percent
1   Sample (Intercept) <NA> 134.44200 11.594913 77.69604
2    Batch (Intercept) <NA>  18.98503  4.357181 10.97173
3 Residual        <NA> <NA>  19.60881  4.428183 11.33222

#The model estimated marginal means were calculated for each library input
treatment:

>emm<-emmeans(exp4.mod7, ~Input_cat, lmer.df="satterthwaite")

>emm

Input_cat emmean   SE   df lower.CL upper.CL
 20ng        47.8 2.69 20.3     42.2     53.5
 2ng         40.0 2.69 20.3     34.4     45.6
 40ng        49.3 2.69 20.3     43.7     54.9
 5ng         51.3 2.69 20.3     45.7     56.9

Degrees-of-freedom method: satterthwaite
Confidence level used: 0.95

#Pairwise contrasts were performed between input treatment groups with a
Tukey correction:

>pairs(emm, adjust="tukey")

contrast    estimate   SE   df t.ratio p.value
 20ng - 2ng      7.82 3.18 10.3   2.456  0.1266
 20ng - 40ng    -1.45 3.18 10.3  -0.456  0.9669
 20ng - 5ng     -3.43 3.18 10.3  -1.077  0.7104
 2ng - 40ng     -9.27 3.18 10.3  -2.912  0.0616
 2ng - 5ng     -11.24 3.18 10.3  -3.533  0.0226
 40ng - 5ng     -1.97 3.18 10.3  -0.620  0.9234

Degrees-of-freedom method: satterthwaite
P value adjustment: tukey method for comparing a family of 4 estimates

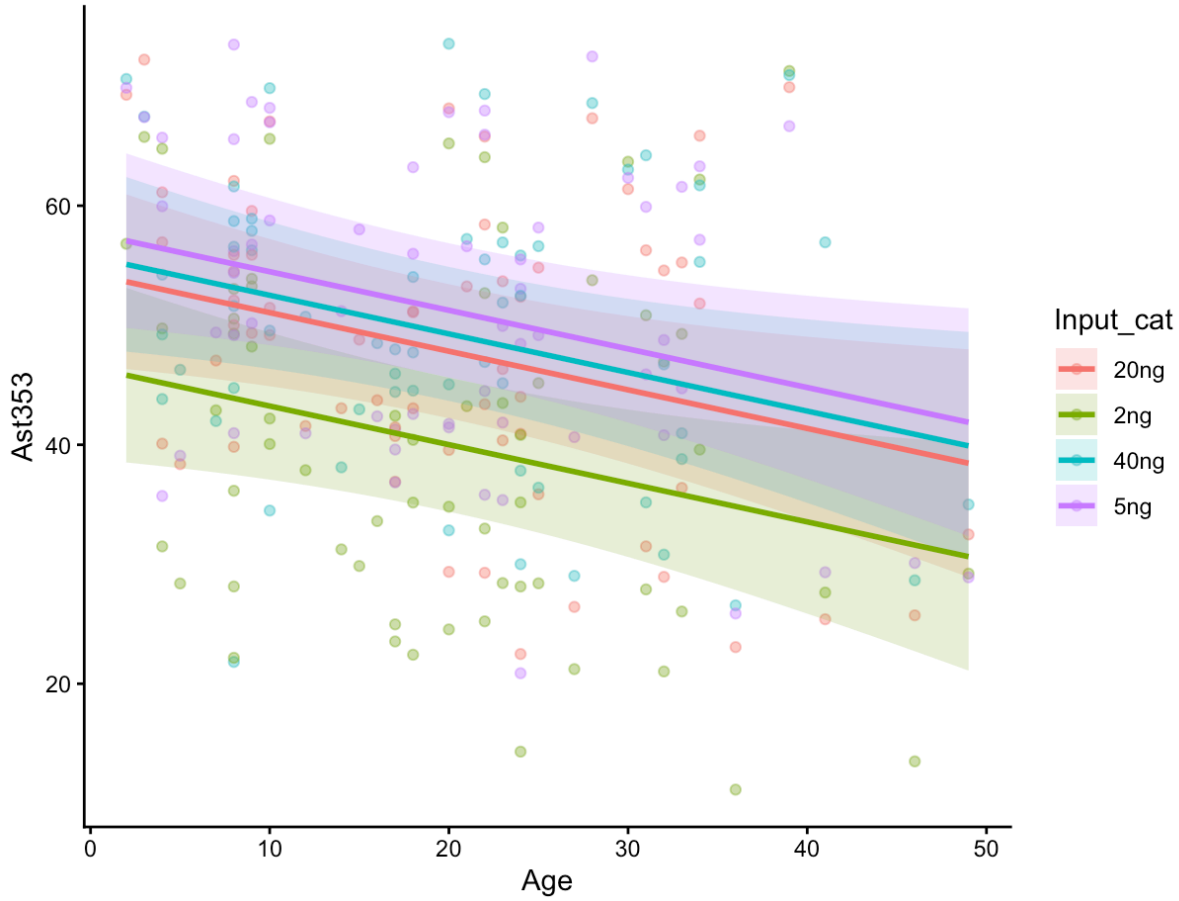
#Plot the model:

>emm<- emmeans(exp4.mod7, ~Age + Input_cat, at = list(Age =
seq(min(exp3$Age), max(exp3$Age), length=100)), lmer.df="satterthwaite")

>emm_df<-as.data.frame(emm)

>ggplot() +
  geom_point(data = exp4,
             aes(x = Age, y = Ast353, colour = Input_cat),
             alpha = 0.4) +
  geom_ribbon(data = emm_df,
```

```
              aes(x = Age, ymin = lower.CL, ymax = upper.CL, fill =
Input_cat),
              alpha = 0.2) +
  geom_line(data = emm_df,
            aes(x = Age, y = emmean, colour = Input_cat),
            linewidth = 1) +
  theme_classic()
```




```
#We also performed a linear mixed model with the percentage target base-
pairs recovered (“Ast353”) as the response variable with “Input” (2, 5,
20, 40) as a fixed, continuous variable, and “Sample” and “Batch” (pool)
as random effects:

>exp4$Input_cont <- as.numeric(exp4$Input_cont)

>exp4.mod10<-lmer(Ast353 ~ Input_cont * Age + (1|Sample)+
(1|Batch),data=exp4, REML=TRUE)

#Sample and batch were presumed to be significant random factors as in the
previous analysis.
#Test removing the interaction between Input_cont and age:

>exp4.mod11<-lmer(Ast353 ~ Input_cont + Age + (1|Sample)+
(1|Batch),data=exp4, REML=TRUE)

>anova(exp4.mod10, exp4.mod11)

ata: exp4
Models:
exp4.mod11: Ast353 ~ Input_cont + Age + (1 | Sample) + (1 | Batch)
exp4.mod10: Ast353 ~ Input_cont * Age + (1 | Sample) + (1 | Batch)
```

```
           npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp4.mod11    6 1695.2 1716.2 -841.57    1683.2
exp4.mod10    7 1696.5 1721.1 -841.27    1682.5 0.5997  1     0.4387

#The interaction between Input_cont and age is non-significant and can be
removed from model.
#Test removing age:

>exp4.mod12<-lmer(Ast353 ~ Input_cont + (1|Sample)+ (1|Batch),data=exp4,
REML=TRUE)

>anova(exp4.mod11, exp4.mod12)

Data: exp4
Models:
exp4.mod12: Ast353 ~ Input_cont + (1 | Sample) + (1 | Batch)
exp4.mod11: Ast353 ~ Input_cont + Age + (1 | Sample) + (1 | Batch)
           npar    AIC    BIC  logLik -2*log(L) Chisq Df Pr(>Chisq)
exp4.mod12    5 1698.7 1716.2 -844.34    1688.7
exp4.mod11    6 1695.2 1716.2 -841.57    1683.2 5.537  1    0.01862 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

#Age is significant and cannot be removed from the model.
#Test removing Input_cont:

>exp4.mod13<-lmer(Ast353 ~ Age + (1|Sample)+ (1|Batch),data=exp4,
REML=TRUE)

>anova(exp4.mod11, exp4.mod13)


Data: exp4
Models:
exp4.mod13: Ast353 ~ Age + (1 | Sample) + (1 | Batch)
exp4.mod11: Ast353 ~ Input_cont + Age + (1 | Sample) + (1 | Batch)
           npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp4.mod13    5 1694.7 1712.2 -842.34    1684.7
exp4.mod11    6 1695.2 1716.2 -841.57    1683.2 1.5402  1     0.2146

#Input_cont is non-significant and can be removed from the model.

>summary(exp4.mod13)

Linear mixed model fit by REML. t-tests use Satterthwaite's method
['lmerModLmerTest']
Formula: Ast353 ~ Age + (1 | Sample) + (1 | Batch)
   Data: exp4

REML criterion at convergence: 1683.5

Scaled residuals:
    Min      1Q  Median      3Q     Max
-4.9685 -0.4000  0.0900  0.4433  3.7792

Random effects:
 Groups   Name        Variance Std.Dev.
 Sample   (Intercept) 134.88   11.614
 Batch    (Intercept)  35.88    5.990
```

```
 Residual              19.61    4.428
Number of obs: 247, groups:  Sample, 62; Batch, 16

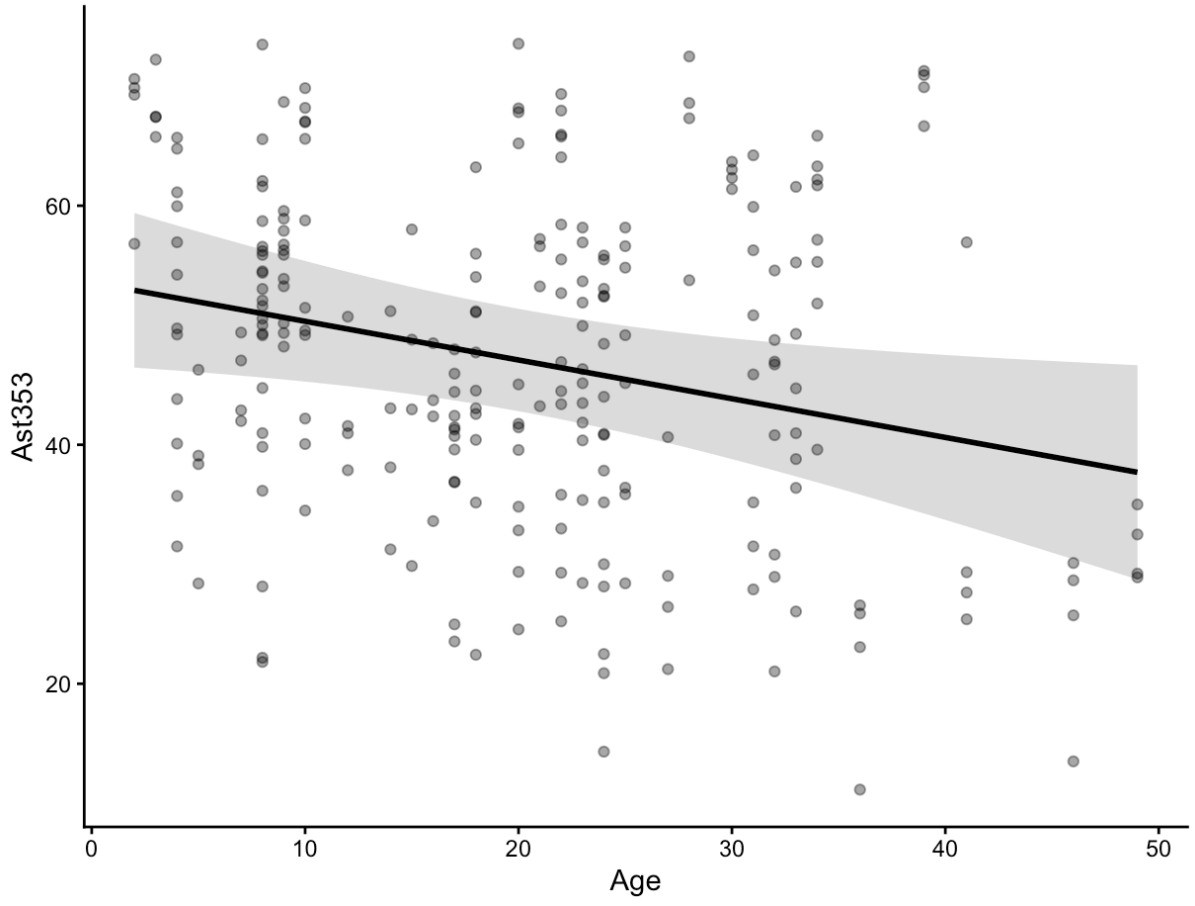
Fixed effects:
            Estimate Std. Error      df t value Pr(>|t|)
(Intercept)  53.5692     3.4517 70.9351  15.520   <2e-16 ***
Age          -0.3239     0.1364 60.1304  -2.375   0.0207 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Correlation of Fixed Effects:
    (Intr)
Age -0.789
```

```
>emm<- emmeans(exp4.mod13, ~Age, at = list(Age = seq(min(exp4$Age),
max(exp4$Age), length=100)), lmer.df="satterthwaite")

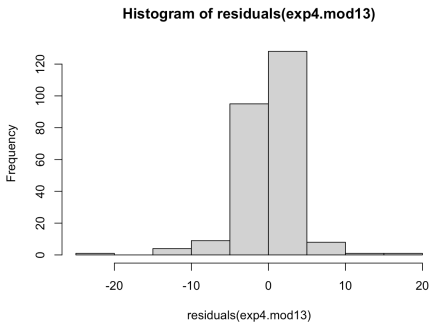
>emm_df<-as.data.frame(emm)

>ggplot() +
  geom_point(data = exp4,
             aes(x = Age, y = Ast353),
             alpha = 0.4) +
  geom_ribbon(data = emm_df,
              aes(x = Age, ymin = lower.CL, ymax = upper.CL),
              alpha = 0.2) +
  geom_line(data = emm_df,
            aes(x = Age, y = emmean),
            linewidth = 1) +
  theme_classic()
```

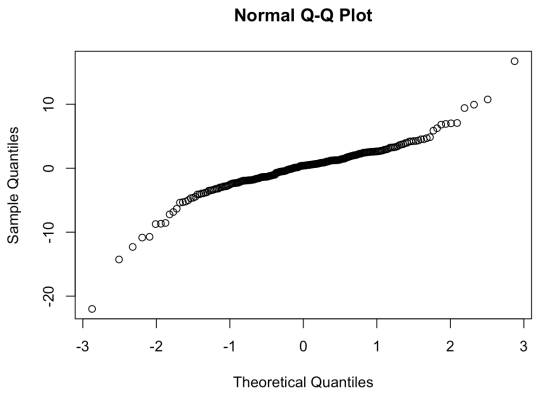


#Model residuals are approximately normally distributed according to the histogram and Q-Q plot, but the Shapiro-Wilk test indicates that statistically they are non-normal:

```
>hist(residuals(exp4.mod13))
```




```
>qqnorm(residuals(exp4.mod13))
```




```
>shapiro.test(residuals(exp4.mod13))

Shapiro-Wilk normality test

data:  residuals(exp4.mod13)
W = 0.89804, p-value = 6.958e-12

#-----------------------------------

#Target recovery
#glmm

#We performed a generalised linear mixed model with the proportion target base-pairs recovered (“Ast353_prop”) as the response variable with “Input” (2 ng, 5 ng, 20 ng, 40 ng) as a fixed, categorical factor, “Tribe”, “Genus within Tribe”, “Sample nested in tribe nested in Genus” and “Batch” (pool) as random effects:

>exp4$Ast353_prop <- exp4$Ast353 / 100

>exp4.mod14<-glmmTMB(Ast353_prop ~ Input_cat * Age + (1|Tribe) + (1|Tribe:Genus) + (1|Tribe:Genus:Sample) + (1|Batch), family=beta_family(link="logit"), data=exp4)

#Test removing Tribe:
```

```
>exp4.mod15<-glmmTMB(Ast353_prop ~ Input_cat * Age + (1|Tribe:Genus) +
(1|Tribe:Genus:Sample) + (1|Batch), family=beta_family(link="logit"),
data=exp4)

>anova(exp4.mod14, exp4.mod15)

Data: exp4
Models:
exp4.mod15: Ast353_prop ~ Input_cat * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + , zi=~0, disp=~1
exp4.mod15:     (1 | Batch), zi=~0, disp=~1
exp4.mod14: Ast353_prop ~ Input_cat * Age + (1 | Tribe) + (1 |
Tribe:Genus) + , zi=~0, disp=~1
exp4.mod14:     (1 | Tribe:Genus:Sample) + (1 | Batch), zi=~0, disp=~1
           Df     AIC     BIC logLik deviance Chisq Chi Df Pr(>Chisq)
exp4.mod15 12 -584.53 -542.42 304.26  -608.53
exp4.mod14 13 -582.53 -536.91 304.26  -608.53     0      1          1

#Tribe is non-significant and can be removed from the model. Test removing
Genus nested within Tribe:

>exp4.mod16<-glmmTMB(Ast353_prop ~ Input_cat * Age +
(1|Tribe:Genus:Sample) + (1|Batch), family=beta_family(link="logit"),
data=exp4)

>anova(exp4.mod15, exp4.mod16)

Data: exp4
Models:
exp4.mod16: Ast353_prop ~ Input_cat * Age + (1 | Tribe:Genus:Sample) + (1
| , zi=~0, disp=~1
exp4.mod16:     Batch), zi=~0, disp=~1
exp4.mod15: Ast353_prop ~ Input_cat * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + , zi=~0, disp=~1
exp4.mod15:     (1 | Batch), zi=~0, disp=~1
           Df     AIC     BIC logLik deviance Chisq Chi Df Pr(>Chisq)
exp4.mod16 11 -586.53 -547.93 304.26  -608.53
exp4.mod15 12 -584.53 -542.42 304.26  -608.53     0      1          1

#Genus nested within Tribe is non-significant and can be removed from the
model. Test removing Sample nested in Genus nested within Tribe:

>exp4.mod17<-glmmTMB(Ast353_prop ~ Input_cat * Age + (1|Batch),
family=beta_family(link="logit"), data=exp4)

>anova(exp4.mod16, exp4.mod17)

Data: exp4
Models:
exp4.mod17: Ast353_prop ~ Input_cat * Age + (1 | Batch), zi=~0, disp=~1
exp4.mod16: Ast353_prop ~ Input_cat * Age + (1 | Tribe:Genus:Sample) + (1
| , zi=~0, disp=~1
exp4.mod16:     Batch), zi=~0, disp=~1
           Df     AIC     BIC logLik deviance  Chisq Chi Df Pr(>Chisq)
exp4.mod17 10 -309.66 -274.56 164.83  -329.66
exp4.mod16 11 -586.53 -547.93 304.26  -608.53 278.88      1  < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
#Sample nested in genus nested within Tribe is significant and cannot be
removed from the model. Test removing Batch:

>exp4.mod18<-glmmTMB(Ast353_prop ~ Input_cat * Age +
(1|Tribe:Genus:Sample), family=beta_family(link="logit"), data=exp4)

>anova(exp4.mod16, exp4.mod18)

Data: exp4
Models:
exp4.mod18: Ast353_prop ~ Input_cat * Age + (1 | Tribe:Genus:Sample),
zi=~0, disp=~1
exp4.mod16: Ast353_prop ~ Input_cat * Age + (1 | Tribe:Genus:Sample) + (1
| , zi=~0, disp=~1
exp4.mod16:     Batch), zi=~0, disp=~1
           Df     AIC     BIC logLik deviance  Chisq Chi Df Pr(>Chisq)
exp4.mod18 10 -511.97 -476.88 265.99  -531.97
exp4.mod16 11 -586.53 -547.93 304.26  -608.53 76.556      1  < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

#Batch is significant and cannot be removed from the model.
#Test removing the interaction between Input_cat and age:

>exp4.mod19<-glmmTMB(Ast353_prop ~ Input_cat + Age +
(1|Tribe:Genus:Sample) + (1|Batch), family=beta_family(link="logit"),
data=exp4)

>anova(exp4.mod16, exp4.mod19)

Data: exp4
Models:
exp4.mod19: Ast353_prop ~ Input_cat + Age + (1 | Tribe:Genus:Sample) + (1
| , zi=~0, disp=~1
exp4.mod19:     Batch), zi=~0, disp=~1
exp4.mod16: Ast353_prop ~ Input_cat * Age + (1 | Tribe:Genus:Sample) + (1
| , zi=~0, disp=~1
exp4.mod16:     Batch), zi=~0, disp=~1
           Df     AIC     BIC logLik deviance  Chisq Chi Df Pr(>Chisq)
exp4.mod19  8 -586.12 -558.05 301.06  -602.12
exp4.mod16 11 -586.53 -547.93 304.26  -608.53 6.4054      3    0.09347 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

#The interaction between Input_cat and age is not significant and can be
removed.
#Test removing Age:

>exp4.mod20<-glmmTMB(Ast353_prop ~ Input_cat + (1|Tribe:Genus:Sample) +
(1|Batch), family=beta_family(link="logit"), data=exp4)

>anova(exp4.mod19, exp4.mod20)

Data: exp4
Models:
exp4.mod20: Ast353_prop ~ Input_cat + (1 | Tribe:Genus:Sample) + (1 |
Batch), zi=~0, disp=~1
```

```
exp4.mod19: Ast353_prop ~ Input_cat + Age + (1 | Tribe:Genus:Sample) + (1
| , zi=~0, disp=~1
exp4.mod19:     Batch), zi=~0, disp=~1
           Df     AIC     BIC logLik deviance  Chisq Chi Df Pr(>Chisq)
exp4.mod20  7 -582.21 -557.64 298.10  -596.21
exp4.mod19  8 -586.12 -558.05 301.06  -602.12 5.9167      1      0.015 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

#Age is significant and cannot be removed from the model.

```
>summary(exp4.mod19)

Family: beta  ( logit )
Formula:          Ast353_prop ~ Input_cat + Age + (1 | Tribe:Genus:Sample)
+ (1 |      Batch)
Data: exp4

      AIC       BIC    logLik -2*log(L)  df.resid
   -586.1    -558.0     301.1    -602.1       239

Random effects:

Conditional model:
 Groups             Name        Variance Std.Dev.
 Tribe:Genus:Sample (Intercept) 0.23476  0.4845
 Batch              (Intercept) 0.02602  0.1613
Number of obs: 247, groups:  Tribe:Genus:Sample, 62; Batch, 16

Dispersion parameter for beta family ():  115

Conditional model:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)    0.189910   0.153961   1.234   0.2174
Input_cat2ng  -0.342231   0.119215  -2.871   0.0041 **
Input_cat40ng  0.058887   0.119278   0.494   0.6215
Input_cat5ng   0.144458   0.119150   1.212   0.2254
Age           -0.014138   0.005676  -2.491   0.0127 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

>AIC(exp4.mod19)

[1] -586.1248

>emm <- emmeans(exp4.mod19, ~Input_cat, type = "response")

>emm

Input_cat response     SE  df asymp.LCL asymp.UCL
 20ng         0.477 0.0260 Inf     0.426     0.528
 2ng          0.393 0.0249 Inf     0.346     0.443
 40ng         0.492 0.0261 Inf     0.441     0.543
 5ng          0.513 0.0261 Inf     0.462     0.564

Confidence level used: 0.95
Intervals are back-transformed from the logit scale

>pairs(emm, adjust = "tukey")
```

```
contrast    odds.ratio     SE  df null z.ratio p.value
 20ng / 2ng       1.408 0.1680 Inf    1   2.871  0.0213
 20ng / 40ng      0.943 0.1120 Inf    1  -0.494  0.9605
 20ng / 5ng       0.865 0.1030 Inf    1  -1.212  0.6189
 2ng / 40ng       0.670 0.0799 Inf    1  -3.361  0.0043
 2ng / 5ng        0.615 0.0733 Inf    1  -4.082  0.0003
 40ng / 5ng       0.918 0.1100 Inf    1  -0.717  0.8902

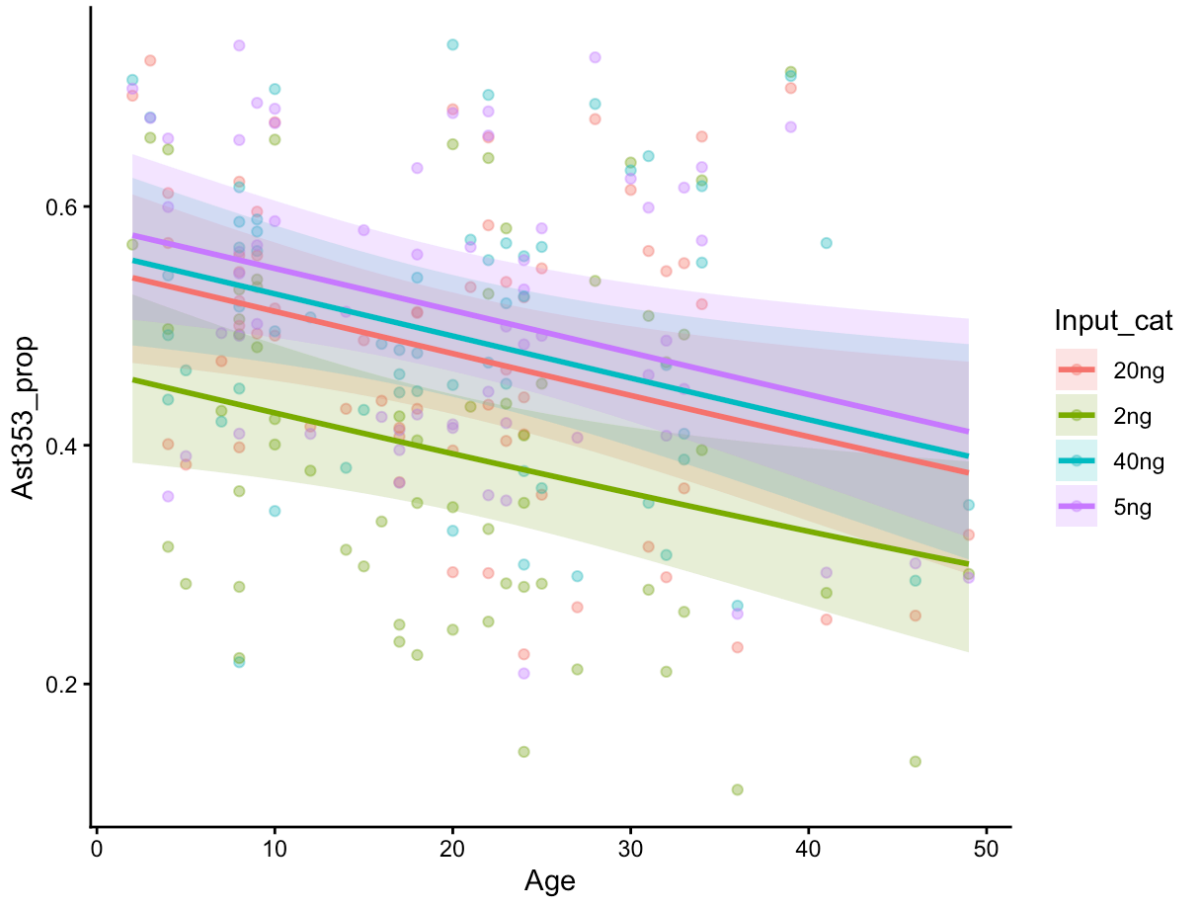
P value adjustment: tukey method for comparing a family of 4 estimates
Tests are performed on the log odds ratio scale
```

```
#Plot model

>emm<- emmeans(exp4.mod19, ~Age | Input_cat, at = list(Age =
seq(min(exp4$Age), max(exp4$Age), length=100)), type="response")

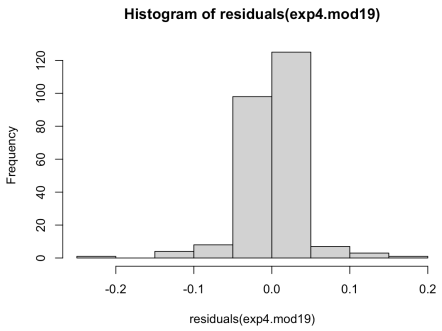
>emm_df<-as.data.frame(emm)

>ggplot() +
  geom_point(data = exp4,
             aes(x = Age, y = Ast353_prop, colour = Input_cat),
             alpha = 0.4) +
  geom_ribbon(data = emm_df,
              aes(x = Age, ymin = asymp.LCL, ymax = asymp.UCL, fill =
Input_cat),
              alpha = 0.2) +
  geom_line(data = emm_df,
            aes(x = Age, y = response, colour = Input_cat),
            linewidth = 1) +
  theme_classic()
```

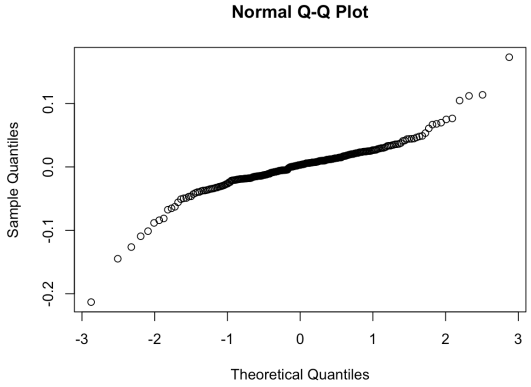


#Model residuals remain approximately normally distributed according to the histogram and Q-Q plot, but the Shapiro-Wilk test indicates that statistically they are still non-normal:

```
>hist(residuals(exp4.mod19))
```




```
>qqnorm(residuals(exp4.mod19))
```




```
>shapiro.test(residuals(exp4.mod19))

Shapiro-Wilk normality test

data:  residuals(exp4.mod19)
W = 0.89916, p-value = 8.305e-12

#-----------------------------------

#Chloroplast recovery
#lmer

#We performed a linear mixed model with the percentage chloroplast base-
pairs recovered (“Chlp”) as the response variable with “Input” (2 ng, 5
ng, 20 ng, 40 ng) as a fixed, categorical factor, “Tribe”, “Genus within
Tribe”, “Sample nested in tribe nested in Genus” and “Batch” (pool) as
random effects:

>exp4.mod21<-lmer(Chlp ~ Input_cat * Age  + (1|Tribe) +(1|Tribe:Genus) +
(1|Tribe:Genus:Sample) + (1|Batch), data=exp4, REML=TRUE)

#Test removing Tribe from the model:

>exp4.mod22<-lmer(Chlp ~ Input_cat * Age + (1|Tribe:Genus) +
(1|Tribe:Genus:Sample) + (1|Batch), data=exp4, REML=TRUE)

>anova(exp4.mod21, exp4.mod22)

Data: exp4
Models:
exp4.mod22: Chlp ~ Input_cat * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Batch)
exp4.mod21: Chlp ~ Input_cat * Age + (1 | Tribe) + (1 | Tribe:Genus) + (1
| Tribe:Genus:Sample) + (1 | Batch)
           npar    AIC    BIC  logLik -2*log(L) Chisq Df Pr(>Chisq)
exp4.mod22   12 1894.3 1936.4 -935.13    1870.3
exp4.mod21   13 1896.3 1941.9 -935.13    1870.3     0  1          1

#Tribe can be removed from the model. Next, test removing Genus within
Tribe:

>exp4.mod23<-lmer(Chlp ~ Input_cat * Age + (1|Tribe:Genus:Sample) +
(1|Batch), data=exp4, REML=TRUE)

>anova(exp4.mod22, exp4.mod23)
```

```
Data: exp4
Models:
exp4.mod23: Chlp ~ Input_cat * Age + (1 | Tribe:Genus:Sample) + (1 |
Batch)
exp4.mod22: Chlp ~ Input_cat * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Batch)
           npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp4.mod23   11 1916.0 1954.7 -947.02    1894.0
exp4.mod22   12 1894.3 1936.4 -935.13    1870.3 23.781  1  1.079e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

#Genus within Tribe cannot be removed from the model. Next test removing Sample from the model:

```
>exp4.mod24<-lmer(Chlp ~ Input_cat * Age + (1|Tribe:Genus) + (1|Batch),
data=exp4, REML=TRUE)

>anova(exp4.mod22, exp4.mod24)
```

```
Data: exp4
Models:
exp4.mod24: Chlp ~ Input_cat * Age + (1 | Tribe:Genus) + (1 | Batch)
exp4.mod22: Chlp ~ Input_cat * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Batch)
           npar    AIC    BIC   logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp4.mod24   11 2028.9 2067.5 -1003.46    2006.9
exp4.mod22   12 1894.3 1936.4  -935.13    1870.3 136.65  1  < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

#Sample cannot be removed from the model. Next test removing Batch from the model:

```
>exp4.mod25<-lmer(Chlp ~ Input_cat * Age + (1|Tribe:Genus) +
(1|Tribe:Genus:Sample), data=exp4, REML=TRUE)

>anova(exp4.mod22, exp4.mod25)
```

```
Data: exp4
Models:
exp4.mod25: Chlp ~ Input_cat * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample)
exp4.mod22: Chlp ~ Input_cat * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Batch)
           npar    AIC    BIC  logLik -2*log(L) Chisq Df Pr(>Chisq)
exp4.mod25   11 1991.8 2030.4 -984.90    1969.8
exp4.mod22   12 1894.3 1936.4 -935.13    1870.3 99.54  1  < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

#Batch cannot be removed from the model.
#Test removing the interaction term between Input_cat and age:

```
>exp4.mod26<-lmer(Chlp ~ Input_cat + Age + (1|Tribe:Genus) +
(1|Tribe:Genus:Sample) + (1|Batch), data=exp4, REML=TRUE)

>anova(exp4.mod22, exp4.mod26)
```

```
Data: exp4
Models:
exp4.mod26: Chlp ~ Input_cat + Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Batch)
exp4.mod22: Chlp ~ Input_cat * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Batch)
           npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp4.mod26    9 1903.4 1935.0 -942.68    1885.4
exp4.mod22   12 1894.3 1936.4 -935.13    1870.3 15.098  3   0.001734 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

#The interaction term between input_cat and age is significant and cannot be removed from the model.

>summary(exp4.mod22)

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method
['lmerModLmerTest']
Formula: Chlp ~ Input_cat * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) +
    (1 | Batch)
   Data: exp4

REML criterion at convergence: 1858.9

Scaled residuals:
    Min      1Q  Median      3Q     Max
-2.5423 -0.4708  0.0195  0.3708  3.5950

Random effects:
 Groups             Name        Variance Std.Dev.
 Tribe:Genus:Sample (Intercept) 122.20   11.054
 Batch              (Intercept)  81.00    9.000
 Tribe:Genus        (Intercept) 171.70   13.103
 Residual                        53.53    7.317
Number of obs: 247, groups:  Tribe:Genus:Sample, 62; Batch, 16;
Tribe:Genus, 5

Fixed effects:
                    Estimate Std. Error         df t value Pr(>|t|)
(Intercept)        27.939271   9.079925   8.380062   3.077 0.014363 *
Input_cat2ng       -1.549973   6.931708  12.971221  -0.224 0.826547
Input_cat40ng      -9.293846   6.932650  12.978066  -1.341 0.203049
Input_cat5ng        9.823312   6.931708  12.971221   1.417 0.180007
Age                 0.006088   0.160941  90.274234   0.038 0.969907
Input_cat2ng:Age   -0.178142   0.120631 170.722286  -1.477 0.141586
Input_cat40ng:Age  -0.011382   0.120640 170.721427  -0.094 0.924947
Input_cat5ng:Age   -0.411637   0.120631 170.722286  -3.412 0.000804 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Correlation of Fixed Effects:
            (Intr) Inpt_2 Inp_40 Inpt_5 Age    In_2:A I_40:A
Input_ct2ng -0.382
Inpt_ct40ng -0.382  0.500
Input_ct5ng -0.382  0.500  0.500
Age         -0.406  0.130  0.131  0.130
```

```
Inpt_ct2n:A  0.133 -0.348 -0.174 -0.174 -0.375
Inpt_ct40:A  0.133 -0.174 -0.348 -0.174 -0.375  0.500
Inpt_ct5n:A  0.133 -0.174 -0.174 -0.348 -0.375  0.500  0.500
```

#The model estimated marginal means were calculated for each library input treatment:

```
>emm<- emmeans(exp4.mod22, ~Input_cat, lmer.df="satterthwaite")
```

```
>emm
```

```
Input_cat emmean   SE   df lower.CL upper.CL
 20ng        28.1 8.31 6.29     7.95     48.2
 2ng         23.0 8.31 6.29     2.85     43.1
 40ng        18.5 8.31 6.29    -1.57     38.7
 5ng         29.7 8.31 6.29     9.57     49.8

Degrees-of-freedom method: satterthwaite
Confidence level used: 0.95
```

#Pairwise contrasts were performed between input treatment groups with a Tukey correction:

```
>pairs(emm, adjust="tukey")
```

```
contrast    estimate  SE df t.ratio p.value
 20ng - 2ng      5.10 6.5 10   0.785  0.8597
 20ng - 40ng     9.52 6.5 10   1.465  0.4912
 20ng - 5ng     -1.62 6.5 10  -0.249  0.9942
 2ng - 40ng      4.42 6.5 10   0.680  0.9023
 2ng - 5ng      -6.72 6.5 10  -1.034  0.7342
 40ng - 5ng    -11.14 6.5 10  -1.714  0.3659

Degrees-of-freedom method: satterthwaite
P value adjustment: tukey method for comparing a family of 4 estimates
```

#Calculate the variance components:

```
>vc <- as.data.frame(VarCorr(exp4.mod22))
```

```
>vc$percent <- vc$vcov / sum(vc$vcov) * 100
```

```
>vc
```

```
              grp        var1 var2      vcov     sdcor  percent
1 Tribe:Genus:Sample (Intercept) <NA> 122.19895 11.054364 28.52267
2              Batch (Intercept) <NA>  80.99678  8.999821 18.90560
3        Tribe:Genus (Intercept) <NA> 171.69766 13.103346 40.07626
4           Residual        <NA> <NA>  53.53401  7.316694 12.49547
```

```
>AIC(exp4.mod22)
```

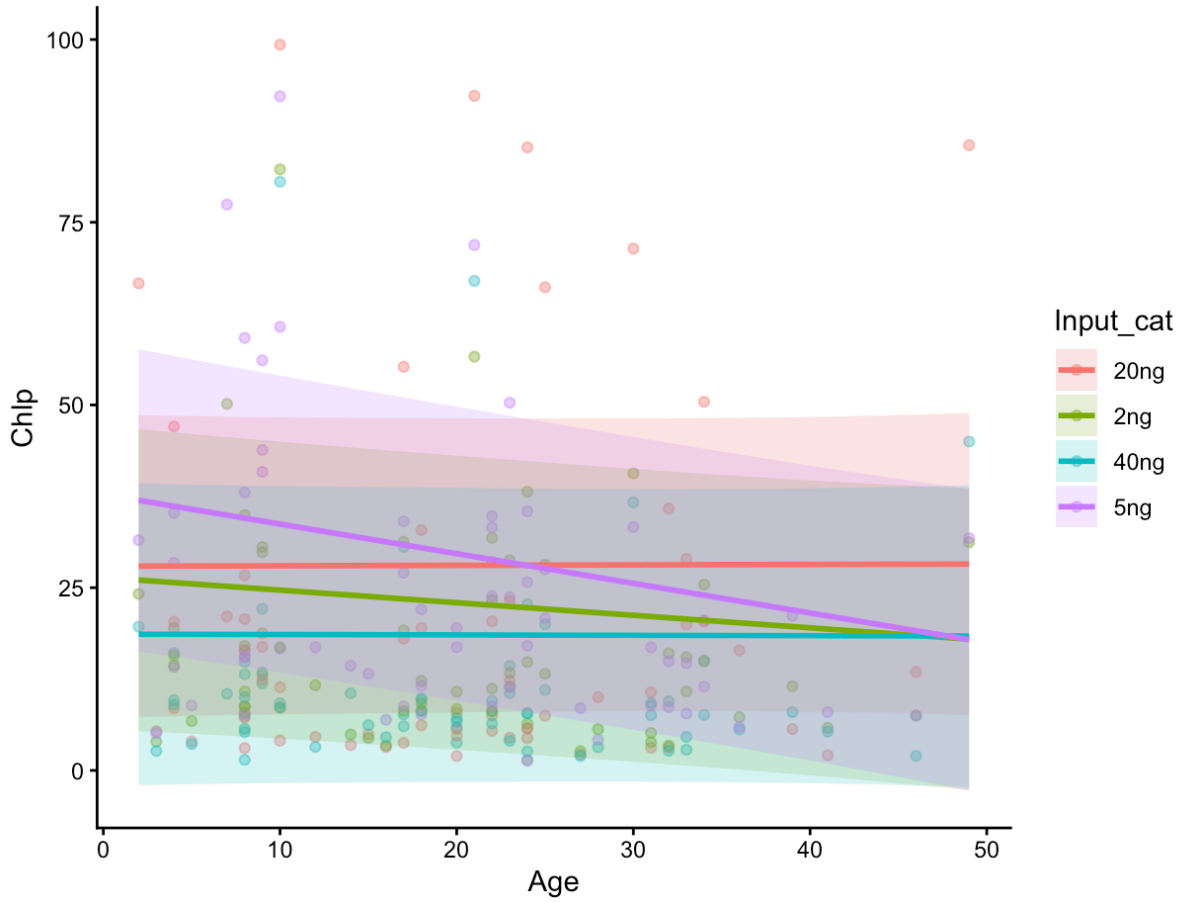
```
[1] 1882.936
```

#Plot model

```
>emm<- emmeans(exp4.mod22, ~Age | Input_cat, at = list(Age =
seq(min(exp4$Age), max(exp4$Age), length=100)), lmer.df="satterthwaite")
```

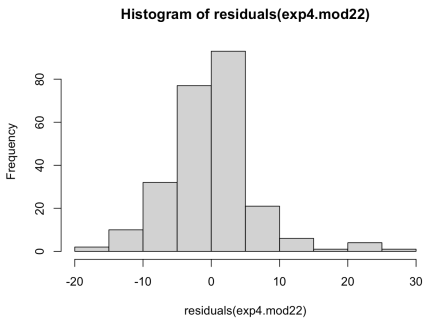
```
>emm_df<-as.data.frame(emm)

>ggplot() +
  geom_point(data = exp4,
             aes(x = Age, y = Chlp, colour = Input_cat),
             alpha = 0.4) +
  geom_ribbon(data = emm_df,
              aes(x = Age, ymin = lower.CL, ymax = upper.CL, fill =
Input_cat),
              alpha = 0.2) +
  geom_line(data = emm_df,
            aes(x = Age, y = emmean, colour = Input_cat),
            linewidth = 1) +
  theme_classic()
```

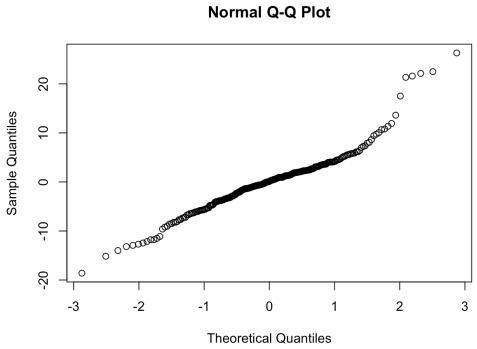



```
#Model residuals are approximately normally distributed according to the
histogram and Q-Q plot, but the Shapiro-Wilk test indicates that
statistically they are non-normal:

>hist(residuals(exp4.mod22))
```




```
>qqnorm(residuals(exp4.mod22))
```




```
>shapiro.test(residuals(exp4.mod22))

Shapiro-Wilk normality test

data:  residuals(exp4.mod22)
W = 0.94031, p-value = 1.755e-08

#-----------------------------------

#Chloroplast recovery
#glmm

#We performed a generalised linear mixed model with the proportion
chloroplast base-pairs recovered (“Chlp_prop”) as the response variable
with “Input” (2 ng, 5 ng, 20 ng, 40 ng) as a fixed, categorical factor,
“Tribe”, “Genus within Tribe”, “Sample nested in tribe nested in Genus”
and “Batch” (pool) as random effects:

>exp4$Chlp_prop <- exp4$Chlp / 100

>exp4.mod27<-glmmTMB(Chlp_prop ~ Input_cat * Age  + (1|Tribe)
+(1|Tribe:Genus) + (1|Tribe:Genus:Sample) + (1|Batch),
family=beta_family(link="logit"), data=exp4)

#Test removing Tribe as a factor:

>exp4.mod28<-glmmTMB(Chlp_prop ~ Input_cat * Age +(1|Tribe:Genus) +
(1|Tribe:Genus:Sample) + (1|Batch), family=beta_family(link="logit"),
data=exp4)

>anova(exp4.mod27, exp4.mod28)
```

```
Data: exp4
Models:
exp4.mod28: Chlp_prop ~ Input_cat * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + , zi=~0, disp=~1
exp4.mod28:     (1 | Batch), zi=~0, disp=~1
exp4.mod27: Chlp_prop ~ Input_cat * Age + (1 | Tribe) + (1 | Tribe:Genus)
+ , zi=~0, disp=~1
exp4.mod27:     (1 | Tribe:Genus:Sample) + (1 | Batch), zi=~0, disp=~1
           Df     AIC     BIC logLik deviance Chisq Chi Df Pr(>Chisq)
exp4.mod28 12 -673.01 -630.90  348.5  -697.01
exp4.mod27 13 -671.01 -625.39  348.5  -697.01     0      1          1
```

#Tribe is non-significant and can be removed as a factor. Test removing Genus nested Tribe:

```
>exp4.mod29<-glmmTMB(Chlp_prop ~ Input_cat * Age + (1|Tribe:Genus:Sample)
+ (1|Batch), family=beta_family(link="logit"), data=exp4)

>anova(exp4.mod28, exp4.mod29)
```

```
Data: exp4
Models:
exp4.mod29: Chlp_prop ~ Input_cat * Age + (1 | Tribe:Genus:Sample) + (1 |
, zi=~0, disp=~1
exp4.mod29:     Batch), zi=~0, disp=~1
exp4.mod28: Chlp_prop ~ Input_cat * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + , zi=~0, disp=~1
exp4.mod28:     (1 | Batch), zi=~0, disp=~1
           Df     AIC     BIC logLik deviance  Chisq Chi Df Pr(>Chisq)
exp4.mod29 11 -649.99 -611.39  336.0  -671.99
exp4.mod28 12 -673.01 -630.90  348.5  -697.01 25.017      1  5.683e-07 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

#Genus nested in Tribe is significant and cannot be removed as a factor. #Test removing Sample nested in Genus nested Tribe:

```
>exp4.mod30<-glmmTMB(Chlp_prop ~ Input_cat * Age +(1|Tribe:Genus)+
(1|Batch), family=beta_family(link="logit"), data=exp4)

>anova(exp4.mod28, exp4.mod30)
```

```
Data: exp4
Models:
exp4.mod30: Chlp_prop ~ Input_cat * Age + (1 | Tribe:Genus) + (1 | Batch),
zi=~0, disp=~1
exp4.mod28: Chlp_prop ~ Input_cat * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + , zi=~0, disp=~1
exp4.mod28:     (1 | Batch), zi=~0, disp=~1
           Df     AIC     BIC logLik deviance  Chisq Chi Df Pr(>Chisq)
exp4.mod30 11 -410.24 -371.63 216.12  -432.24
exp4.mod28 12 -673.01 -630.90 348.50  -697.01 264.77      1  < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

#Sample nested in Genus nested in Tribe is significant and cannot be removed as a factor. Test removing Batch:

```
>exp4.mod31<-glmmTMB(Chlp_prop ~ Input_cat  * Age+(1|Tribe:Genus) +
(1|Tribe:Genus:Sample), family=beta_family(link="logit"), data=exp4)

>anova(exp4.mod28, exp4.mod31)

Data: exp4
Models:
exp4.mod31: Chlp_prop ~ Input_cat * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample), zi=~0, disp=~1
exp4.mod28: Chlp_prop ~ Input_cat * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + , zi=~0, disp=~1
exp4.mod28:     (1 | Batch), zi=~0, disp=~1
           Df     AIC     BIC logLik deviance Chisq Chi Df Pr(>Chisq)
exp4.mod31 11 -460.31 -421.71 241.16  -482.31
exp4.mod28 12 -673.01 -630.90 348.50  -697.01 214.7      1  < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

#Batch significant and cannot be removed as a factor.
#Test removing the interacton between Input_cat and age:

>exp4.mod32<-glmmTMB(Chlp_prop ~ Input_cat + Age +(1|Tribe:Genus) +
(1|Tribe:Genus:Sample) + (1|Batch), family=beta_family(link="logit"),
data=exp4)

>anova(exp4.mod28, exp4.mod32)

Data: exp4
Models:
exp4.mod32: Chlp_prop ~ Input_cat + Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + , zi=~0, disp=~1
exp4.mod32:     (1 | Batch), zi=~0, disp=~1
exp4.mod28: Chlp_prop ~ Input_cat * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + , zi=~0, disp=~1
exp4.mod28:     (1 | Batch), zi=~0, disp=~1
           Df     AIC     BIC logLik deviance  Chisq Chi Df Pr(>Chisq)
exp4.mod32  9 -662.09 -630.51 340.05  -680.09
exp4.mod28 12 -673.01 -630.90 348.50  -697.01 16.917      3  0.0007352 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

#The interaction between input_cat and age is significant and cannot be
removed from the model.

>summary(exp4.mod28)

Family: beta  ( logit )
Formula:
Chlp_prop ~ Input_cat * Age + (1 | Tribe:Genus) + (1 | Tribe:Genus:Sample)
+      (1 | Batch)
Data: exp4

      AIC       BIC    logLik -2*log(L)  df.resid
   -673.0    -630.9     348.5    -697.0       235

Random effects:

Conditional model:
 Groups             Name        Variance Std.Dev.
```

```
 Tribe:Genus        (Intercept) 0.5057   0.7111
 Tribe:Genus:Sample (Intercept) 0.5530   0.7436
 Batch              (Intercept) 0.3429   0.5856
Number of obs: 247, groups:  Tribe:Genus, 5; Tribe:Genus:Sample, 62;
Batch, 16

Dispersion parameter for beta family (): 73.6

Conditional model:
                  Estimate Std. Error z value Pr(>|z|)
(Intercept)       -1.110174   0.549546  -2.020 0.043366 *
Input_cat2ng      -0.071192   0.431129  -0.165 0.868842
Input_cat40ng     -0.725453   0.434227  -1.671 0.094786 .
Input_cat5ng       0.609374   0.429610   1.418 0.156064
Age               -0.005520   0.009976  -0.553 0.580046
Input_cat2ng:Age  -0.007180   0.005328  -1.348 0.177739
Input_cat40ng:Age  0.002978   0.005700   0.522 0.601340
Input_cat5ng:Age  -0.018237   0.005184  -3.518 0.000435 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
>emm <- emmeans(exp4.mod28, ~Input_cat, type="response")
```

```
>emm
```

```
Input_cat response     SE  df asymp.LCL asymp.UCL
 20ng         0.228 0.0874 Inf    0.1004     0.439
 2ng          0.192 0.0771 Inf    0.0826     0.387
 40ng         0.132 0.0568 Inf    0.0541     0.287
 5ng          0.274 0.0987 Inf    0.1250     0.499

Confidence level used: 0.95
Intervals are back-transformed from the logit scale
```

```
>AIC(exp4.mod28)
```

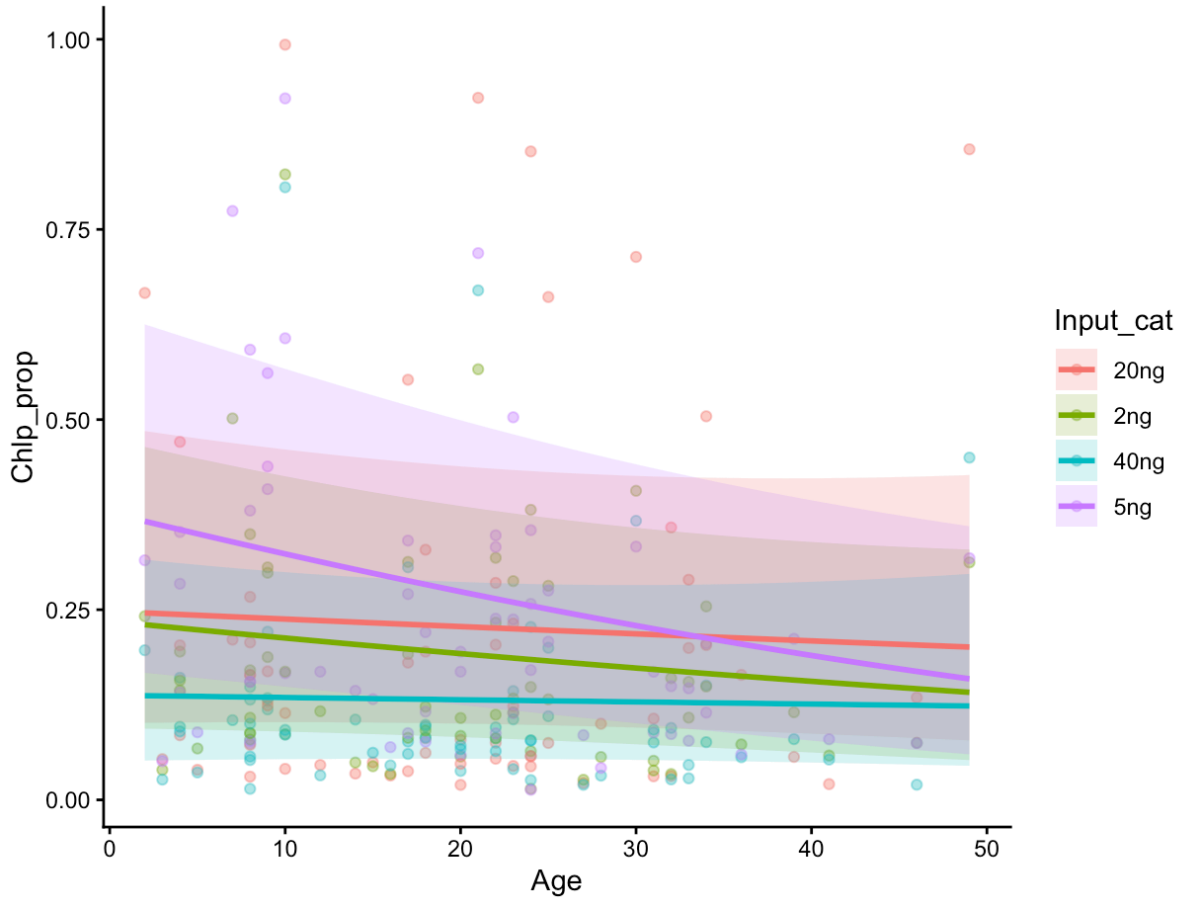
```
[1] -673.0094
```

```
#Plot model

>emm<- emmeans(exp4.mod28, ~Age | Input_cat, at = list(Age =
seq(min(exp4$Age), max(exp4$Age), length=100)), type="response")

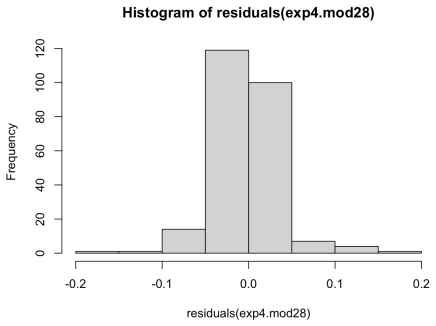
>emm_df<-as.data.frame(emm)

>ggplot() +
  geom_point(data = exp4,
             aes(x = Age, y = Chlp_prop, colour = Input_cat),
             alpha = 0.4) +
  geom_ribbon(data = emm_df,
              aes(x = Age, ymin = asymp.LCL, ymax = asymp.UCL, fill =
Input_cat),
              alpha = 0.2) +
  geom_line(data = emm_df,
            aes(x = Age, y = response, colour = Input_cat),
            linewidth = 1) +
  theme_classic()
```

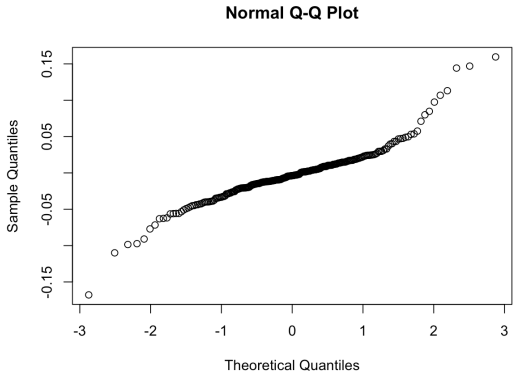


#Model residuals remain approximately normally distributed according to the histogram and Q-Q plot, but the Shapiro-Wilk test indicates that statistically they are still non-normal:

```
>hist(residuals(exp4.mod28))
```




```
>qqnorm(residuals(exp4.mod28))
```




```
>shapiro.test(residuals(exp4.mod28))

Shapiro-Wilk normality test

data:  residuals(exp4.mod28)
W = 0.91719, p-value = 1.746e-10

#-------------------------------------
#Between treatments (Aim 5)
#-------------------------------------

>exp5<-read.csv("exp5_w_year.csv", header=TRUE)

>head(exp5)

                     Sample Year Age         Genus       Tribe
Pool   Plex Bait
1 Anaphalioides_bellidioides_AG440 2000  20 Anaphalioides Gnaphalieae
AG_P5P13C 16plex   1X
2 Anaphalioides_bellidioides_AG440 2000  20 Anaphalioides Gnaphalieae
AG_P5P14/15C 32plex   1X
3      Anaphalioides_papuana_AG442 1998  22 Anaphalioides Gnaphalieae
AG_P5P6C 16plex   1X
4      Anaphalioides_papuana_AG442 1998  22 Anaphalioides Gnaphalieae
AG_P5P7C 16plex 0.5X
5      Anaphalioides_papuana_AG442 1998  22 Anaphalioides Gnaphalieae
AG_P5P4/5C 32plex   1X
6       Anemocarpa_calcicola_AG577 1997  23    Anemocarpa Gnaphalieae
AG_P5P8C 16plex   1X
  Treatment CompositionA CompositionB     Chlp      Ast353     logChlp
logAst353
1     16rxn            L            L 93.64823 73.48825144 1.971499555
1.866217914
2     32rxn          L/M            L 99.85981 66.25784296 1.999390727
1.821237293
3     16rxn            H            H  0.00000  0.04217276       #NUM! -
1.374967986
4 half_bait            H            H  0.00000  0.15759294       #NUM! -
0.802463242
5     32rxn          G/H            H  0.00000  0.06658857       #NUM! -
1.176600338
6     16rxn            I            I 99.15346 49.10018908 1.996307857
1.691083165

#Target recovery
```

```
#lmer

#We performed a linear mixed model with the percentage target base-pairs
recovered (“Ast353”) as the response variable with “Treatment” (16rxn,
32rxn, half_bait) as a fixed, categorical factor, and “Sample” and “Pool”
(batch) as random effects:

>exp5.mod<-lmer(Ast353 ~ Treatment * Age + (1|Tribe) + (1|Tribe:Genus) +
(1|Tribe:Genus:Sample) + (1|Pool), data=exp5, REML=TRUE)

#Remove Tribe as a random factor:

>exp5.mod2<-lmer(Ast353 ~ Treatment* Age + (1|Tribe:Genus) +
(1|Tribe:Genus:Sample) + (1|Pool), data=exp5, REML=TRUE)

>anova(exp5.mod, exp5.mod2)

Data: exp5
Models:
exp5.mod2: Ast353 ~ Treatment * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Pool)
exp5.mod: Ast353 ~ Treatment * Age + (1 | Tribe) + (1 | Tribe:Genus) + (1
| Tribe:Genus:Sample) + (1 | Pool)
          npar    AIC    BIC  logLik -2*log(L) Chisq Df Pr(>Chisq)
exp5.mod2   10 1265.7 1298.7 -622.86    1245.7
exp5.mod    11 1267.7 1304.0 -622.86    1245.7     0  1          1

#Tribe is nonsignificant as a factor. Remove Genus nested within Tribe.

>exp5.mod3<-lmer(Ast353 ~ Treatment * Age + (1|Tribe:Genus:Sample) +
(1|Pool), data=exp5, REML=TRUE)

>anova(exp5.mod2, exp5.mod3)

Data: exp5
Models:
exp5.mod3: Ast353 ~ Treatment * Age + (1 | Tribe:Genus:Sample) + (1 |
Pool)
exp5.mod2: Ast353 ~ Treatment * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Pool)
          npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp5.mod3    9 1267.5 1297.2 -624.75    1249.5
exp5.mod2   10 1265.7 1298.7 -622.86    1245.7 3.7725  1     0.0521 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

#Genus is a nonsignificant factor. Simplify the terms:

>exp5.mod4<-lmer(Ast353 ~ Treatment * Age + (1|Sample) + (1|Pool),
data=exp5, REML=TRUE)

>anova(exp5.mod3, exp5.mod4)

Data: exp5
Models:
exp5.mod3: Ast353 ~ Treatment * Age + (1 | Tribe:Genus:Sample) + (1 |
Pool)
exp5.mod4: Ast353 ~ Treatment * Age + (1 | Sample) + (1 | Pool)
          npar    AIC    BIC  logLik -2*log(L) Chisq Df Pr(>Chisq)
```

```
exp5.mod3    9 1267.5 1297.2 -624.75    1249.5
exp5.mod4    9 1267.5 1297.2 -624.75    1249.5     0  0
```

#Model 3 = Model 4. Remove Sample:

```
>exp5.mod5<-lmer(Ast353 ~ Treatment * Age+ (1|Pool), data=exp5, REML=TRUE)

>anova(exp5.mod4, exp5.mod5)

Data: exp5
Models:
exp5.mod5: Ast353 ~ Treatment * Age + (1 | Pool)
exp5.mod4: Ast353 ~ Treatment * Age + (1 | Sample) + (1 | Pool)
          npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp5.mod5    8 1789.2 1815.5 -886.58    1773.2
exp5.mod4    9 1267.5 1297.2 -624.75    1249.5 523.66  1  < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

#Sample is a significant factor and can’t be removed. Remove Pool as a factor:

```
>exp5.mod6<-lmer(Ast353 ~ Treatment * Age + (1|Pool), data=exp5,
REML=TRUE)

>anova(exp5.mod4, exp5.mod6)

Data: exp5
Models:
exp5.mod6: Ast353 ~ Treatment * Age + (1 | Pool)
exp5.mod4: Ast353 ~ Treatment * Age + (1 | Sample) + (1 | Pool)
          npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp5.mod6    8 1789.2 1815.5 -886.58    1773.2
exp5.mod4    9 1267.5 1297.2 -624.75    1249.5 523.66  1  < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

#Pool is a significant factor and can’t be removed.
#Remove the interaction between treatment and age:

```
>exp5.mod7<-lmer(Ast353 ~ Treatment + Age + (1|Sample) + (1|Pool),
data=exp5, REML=TRUE)

>anova(exp5.mod4, exp5.mod7)

Data: exp5
Models:
exp5.mod7: Ast353 ~ Treatment + Age + (1 | Sample) + (1 | Pool)
exp5.mod4: Ast353 ~ Treatment * Age + (1 | Sample) + (1 | Pool)
          npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp5.mod7    7 1270.8 1293.9 -628.40    1256.8
exp5.mod4    9 1267.5 1297.2 -624.75    1249.5 7.2997  2      0.026 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

#The interaction between treatment and age is significant and cannot be removed.

```
>summary(exp5.mod4)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method
['lmerModLmerTest']
Formula: Ast353 ~ Treatment * Age + (1 | Sample) + (1 | Pool)
   Data: exp5

REML criterion at convergence: 1254.7

Scaled residuals:
     Min       1Q   Median       3Q      Max
-2.58789 -0.37736  0.01807  0.38320  2.53666

Random effects:
 Groups   Name        Variance Std.Dev.
 Sample   (Intercept) 411.014  20.273
 Pool     (Intercept)   5.735   2.395
 Residual               2.224   1.491
Number of obs: 200, groups:  Sample, 81; Pool, 15

Fixed effects:
                    Estimate Std. Error        df t value Pr(>|t|)
(Intercept)         68.60739    4.27842  85.45896  16.036  < 2e-16 ***
Treatment32rxn      -2.77732    1.55983  10.98820  -1.781   0.1026
Treatmenthalf_bait  -4.76734    1.76003  10.46310  -2.709   0.0212 *
Age                 -0.65332    0.15598  79.94876  -4.189 7.17e-05 ***
Treatment32rxn:Age   0.04342    0.01722 109.01761   2.521   0.0131 *
Treatmenthalf_bait:Age   0.04355    0.02370 107.64747   1.838   0.0689 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Correlation of Fixed Effects:
            (Intr) Trtm32 Trtmn_ Age    Tr32:A
Trtmnt32rxn -0.142
Trtmnthlf_b -0.125  0.345
Age         -0.820  0.005  0.009
Trtmnt32r:A  0.040 -0.206 -0.112 -0.055
Trtmnthl_:A  0.028 -0.067 -0.258 -0.039  0.363

>AIC(exp5.mod4)

[1] 1272.709

>vc <- as.data.frame(VarCorr(exp5.mod4))

>vc$percent <- vc$vcov / sum(vc$vcov) * 100

>vc

    grp        var1 var2       vcov     sdcor    percent
1   Sample (Intercept) <NA> 411.013792 20.273475 98.1003033
2     Pool (Intercept) <NA>   5.735499  2.394890  1.3689423
3 Residual        <NA> <NA>   2.223718  1.491214  0.5307544

>emm<-emmeans(exp5.mod4, ~Treatment, lmer.df="satterthwaite")
pairs(emm, adjust="tukey")

contrast          estimate   SE   df t.ratio p.value
 16rxn - 32rxn         1.82 1.53 9.99   1.190  0.4851
 16rxn - half_bait     3.81 1.70 9.13   2.236  0.1172
```

```
 32rxn - half_bait     1.99 1.86 9.82   1.070  0.5532

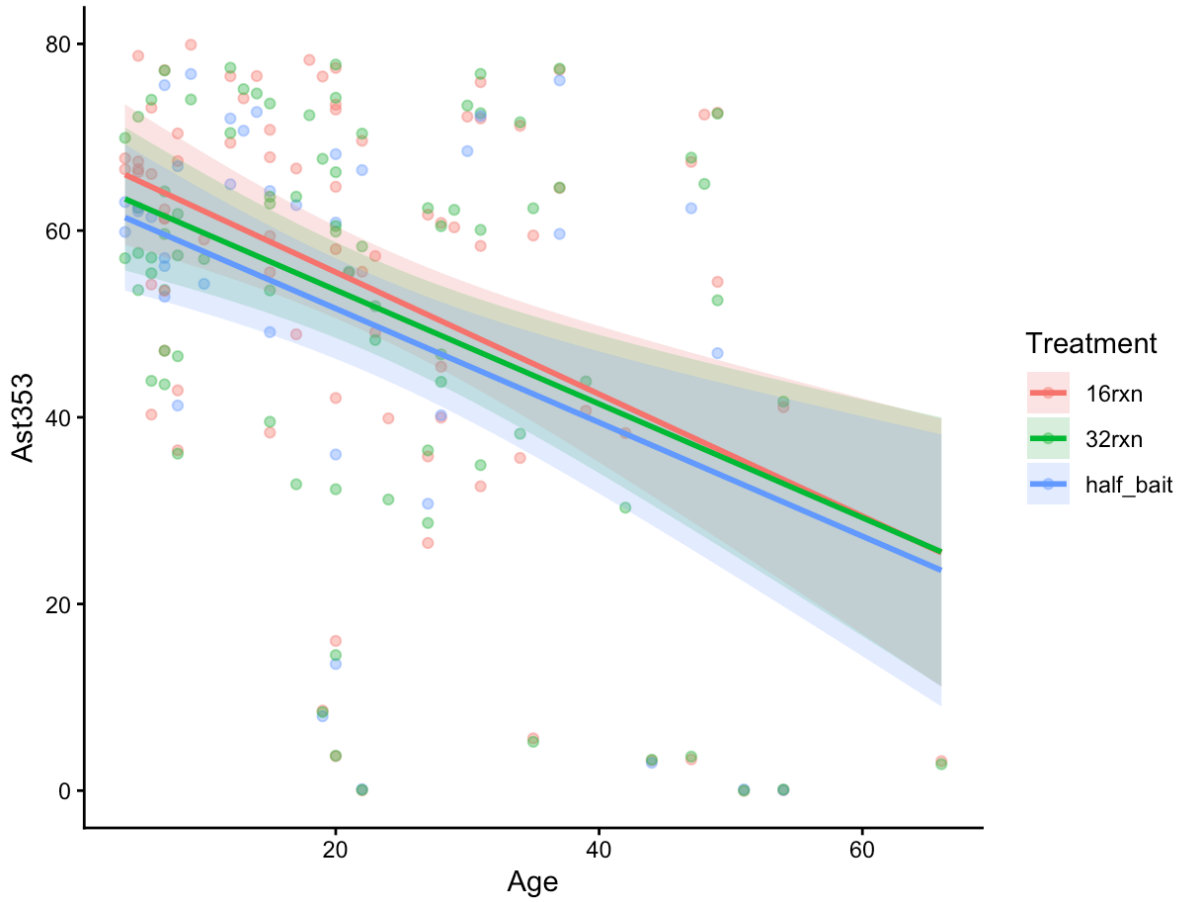
Degrees-of-freedom method: satterthwaite
P value adjustment: tukey method for comparing a family of 3 estimates
```

#Plot model

```
>emm<- emmeans(exp5.mod4, ~Age | Treatment, at = list(Age =
seq(min(exp5$Age), max(exp5$Age), length=100)), lmer.df="satterthwaite")

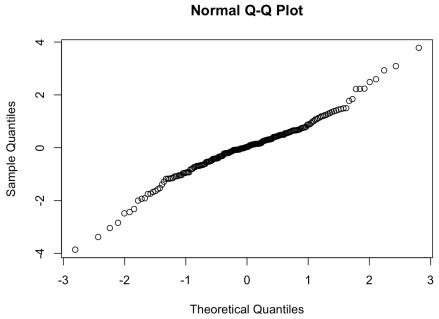
>emm_df<-as.data.frame(emm)

>ggplot() +
  geom_point(data = exp5,
             aes(x = Age, y = Ast353, colour = Treatment),
             alpha = 0.4) +
  geom_ribbon(data = emm_df,
              aes(x = Age, ymin = lower.CL, ymax = upper.CL, fill =
Treatment),
              alpha = 0.2) +
  geom_line(data = emm_df,
            aes(x = Age, y = emmean, colour = Treatment),
            linewidth = 1) +
  theme_classic()
```

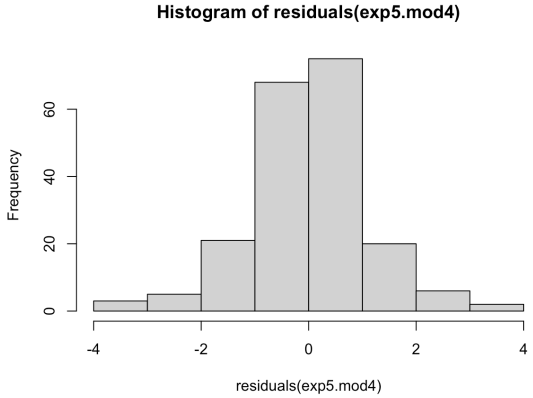


#Model residuals are approximately normally distributed according to the histogram and Q-Q plot, but the Shapiro-Wilk test indicates that statistically they are non-normal:

```
>qqnorm(residuals(exp5.mod4))
```




```
>hist(residuals(exp5.mod4))
```




```
>shapiro.test(residuals(exp5.mod4))

      Shapiro-Wilk normality test

data:  residuals(exp5.mod4)
W = 0.97135, p-value = 0.0004161

#-----------------------------------

#Target recovery
#glmm

#We performed a generalised linear mixed model with the proportion of
target base-pairs recovered (“Ast353_prop”) as the response variable with
“Treatment” (16rxn, 32rxn, half_bait) as a fixed, categorical factor, and
“Sample” and “Pool” (batch) as random effects:

>exp5$Ast353_prop <- exp5$Ast353/100

>exp5.mod8<-glmmTMB(Ast353_prop ~ Treatment * Age + (1|Tribe) +
(1|Tribe:Genus) + (1|Tribe:Genus:Sample) + (1|Pool),
family=beta_family(link="logit"), data=exp5, subset = Ast353_prop != 0)

#Remove Tribe as a factor:

>exp5.mod9<-glmmTMB(Ast353_prop ~ Treatment * Age+ (1|Tribe:Genus) +
(1|Tribe:Genus:Sample) + (1|Pool), family=beta_family(link="logit"),
data=exp5, subset = Ast353_prop != 0)
```

```
>anova(exp5.mod8, exp5.mod9)

Data: exp5
Subset: Ast353_prop != 0
Models:
exp5.mod9: Ast353_prop ~ Treatment * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + , zi=~0, disp=~1
exp5.mod9:     (1 | Pool), zi=~0, disp=~1
exp5.mod8: Ast353_prop ~ Treatment * Age + (1 | Tribe) + (1 | Tribe:Genus)
+ , zi=~0, disp=~1
exp5.mod8:     (1 | Tribe:Genus:Sample) + (1 | Pool), zi=~0, disp=~1
          Df     AIC     BIC logLik deviance Chisq Chi Df Pr(>Chisq)
exp5.mod9 10 -638.21 -605.32  329.1  -658.21
exp5.mod8 11 -636.21 -600.04  329.1  -658.21     0      1          1

#Tribe is non-significant and can be removed as a factor. Remove Genus
nested within Tribe:

>exp5.mod10<-glmmTMB(Ast353_prop ~ Treatment * Age+ (1|Tribe:Genus:Sample)
+ (1|Pool), family=beta_family(link="logit"), data=exp5, subset =
Ast353_prop != 0)

>anova(exp5.mod9, exp5.mod10)

Data: exp5
Subset: Ast353_prop != 0
Models:
exp5.mod10: Ast353_prop ~ Treatment * Age + (1 | Tribe:Genus:Sample) + (1
| , zi=~0, disp=~1
exp5.mod10:     Pool), zi=~0, disp=~1
exp5.mod9: Ast353_prop ~ Treatment * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + , zi=~0, disp=~1
exp5.mod9:     (1 | Pool), zi=~0, disp=~1
          Df     AIC     BIC logLik deviance  Chisq Chi Df Pr(>Chisq)
exp5.mod10  9 -636.51 -606.91 327.25  -654.51
exp5.mod9  10 -638.21 -605.32 329.10  -658.21 3.7021      1    0.05435 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

#Genus is non-significant and can be removed as a factor. Remove Sample
nested in Genus and Tribe:

>exp5.mod11<-glmmTMB(Ast353_prop ~ Treatment * Age+ (1|Pool),
family=beta_family(link="logit"), data=exp5, subset = Ast353_prop != 0)

>anova(exp5.mod10, exp5.mod11)

Data: exp5
Subset: Ast353_prop != 0
Models:
exp5.mod11: Ast353_prop ~ Treatment * Age + (1 | Pool), zi=~0, disp=~1
exp5.mod10: Ast353_prop ~ Treatment * Age + (1 | Tribe:Genus:Sample) + (1
| , zi=~0, disp=~1
exp5.mod10:     Pool), zi=~0, disp=~1
           Df     AIC     BIC logLik deviance  Chisq Chi Df Pr(>Chisq)
exp5.mod11  8  -45.13  -18.83  30.57   -61.13
exp5.mod10  9 -636.51 -606.91 327.25  -654.51 593.37      1  < 2.2e-16 ***
---
```

```
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

#Sample is significant and cannot be removed as a factor. Remove Pool:

>exp5.mod12<-glmmTMB(Ast353_prop ~ Treatment * Age +
(1|Tribe:Genus:Sample), family=beta_family(link="logit"), data=exp5,
subset = Ast353_prop != 0)

>anova(exp5.mod10, exp5.mod12)

Data: exp5
Subset: Ast353_prop != 0
Models:
exp5.mod12: Ast353_prop ~ Treatment * Age + (1 | Tribe:Genus:Sample),
zi=~0, disp=~1
exp5.mod10: Ast353_prop ~ Treatment * Age + (1 | Tribe:Genus:Sample) + (1
| , zi=~0, disp=~1
exp5.mod10:     Pool), zi=~0, disp=~1
           Df     AIC     BIC logLik deviance  Chisq Chi Df Pr(>Chisq)
exp5.mod12  8 -475.73 -449.43 245.87  -491.73
exp5.mod10  9 -636.51 -606.91 327.25  -654.51 162.77      1  < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

#Pool is significant and cannot be removed as a factor.
#Remove the interaction between treatment and age:

>exp5.mod13<-glmmTMB(Ast353_prop ~ Treatment + Age+ (1|Tribe:Genus:Sample)
+ (1|Pool), family=beta_family(link="logit"), data=exp5, subset =
Ast353_prop != 0)

>anova(exp5.mod10, exp5.mod13)

Data: exp5
Subset: Ast353_prop != 0
Models:
exp5.mod13: Ast353_prop ~ Treatment + Age + (1 | Tribe:Genus:Sample) + (1
| , zi=~0, disp=~1
exp5.mod13:     Pool), zi=~0, disp=~1
exp5.mod10: Ast353_prop ~ Treatment * Age + (1 | Tribe:Genus:Sample) + (1
| , zi=~0, disp=~1
exp5.mod10:     Pool), zi=~0, disp=~1
           Df     AIC     BIC logLik deviance  Chisq Chi Df Pr(>Chisq)
exp5.mod13  7 -635.78 -612.76 324.89  -649.78
exp5.mod10  9 -636.51 -606.91 327.25  -654.51 4.7275      2    0.09407 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

#The interaction between treatment and age is non-significant and can be
removed.
#Remove age:

>exp5.mod14<-glmmTMB(Ast353_prop ~ Treatment + (1|Tribe:Genus:Sample) +
(1|Pool), family=beta_family(link="logit"), data=exp5, subset =
Ast353_prop != 0)

>anova(exp5.mod13, exp5.mod14)

Data: exp5
```

```
Subset: Ast353_prop != 0
Models:
exp5.mod14: Ast353_prop ~ Treatment + (1 | Tribe:Genus:Sample) + (1 |
Pool), zi=~0, disp=~1
exp5.mod13: Ast353_prop ~ Treatment + Age + (1 | Tribe:Genus:Sample) + (1
| , zi=~0, disp=~1
exp5.mod13:     Pool), zi=~0, disp=~1
           Df     AIC     BIC logLik deviance Chisq Chi Df Pr(>Chisq)
exp5.mod14  6 -620.86 -601.13 316.43  -632.86
exp5.mod13  7 -635.78 -612.76 324.89  -649.78 16.92      1    3.9e-05 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

#Age is significant and cannot be removed from the model.
#Remove treatment:

```
>exp5.mod15<-glmmTMB(Ast353_prop ~ Age+ (1|Tribe:Genus:Sample) + (1|Pool),
family=beta_family(link="logit"), data=exp5, subset = Ast353_prop != 0)
```

```
>anova(exp5.mod13, exp5.mod15)
```

```
Data: exp5
Subset: Ast353_prop != 0
Models:
exp5.mod15: Ast353_prop ~ Age + (1 | Tribe:Genus:Sample) + (1 | Pool),
zi=~0, disp=~1
exp5.mod13: Ast353_prop ~ Treatment + Age + (1 | Tribe:Genus:Sample) + (1
| , zi=~0, disp=~1
exp5.mod13:     Pool), zi=~0, disp=~1
           Df     AIC     BIC logLik deviance  Chisq Chi Df Pr(>Chisq)
exp5.mod15  5 -633.84 -617.40 321.92  -643.84
exp5.mod13  7 -635.78 -612.76 324.89  -649.78 5.9416      2    0.05126 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

#Treatment is non-significant and can be removed from the model.

```
>summary(exp5.mod13)
```

```
Family: beta  ( logit )
Formula:          Ast353_prop ~ Treatment + Age + (1 | Tribe:Genus:Sample)
+ (1 |      Pool)
Data: exp5

      AIC       BIC    logLik -2*log(L)  df.resid
   -635.8    -612.8     324.9    -649.8       191

Random effects:

Conditional model:
 Groups             Name        Variance Std.Dev.
 Tribe:Genus:Sample (Intercept) 2.21160  1.4871
 Pool               (Intercept) 0.01075  0.1037
Number of obs: 198, groups:  Tribe:Genus:Sample, 81; Pool, 15

Dispersion parameter for beta family (): 1.75e+03

Conditional model:
                  Estimate Std. Error z value Pr(>|z|)
```

```
(Intercept)         1.04697    0.30871   3.391 0.000695 ***
Treatment32rxn     -0.09149    0.06495  -1.409 0.158933
Treatmenthalf_bait -0.19897    0.07303  -2.725 0.006436 **
Age                -0.04951    0.01144  -4.327 1.51e-05 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

>emm <- emmeans(exp5.mod13, ~Treatment, type="response")

>emm
Treatment response     SE  df asymp.LCL asymp.UCL
 16rxn        0.492 0.0426 Inf     0.409     0.575
 32rxn        0.469 0.0431 Inf     0.386     0.554
 half_bait    0.442 0.0435 Inf     0.360     0.529

Confidence level used: 0.95
Intervals are back-transformed from the logit scale

>pairs(emm, adjust="tukey")

contrast          odds.ratio     SE  df null z.ratio p.value
 16rxn / 32rxn           1.10 0.0712 Inf    1   1.409  0.3364
 16rxn / half_bait       1.22 0.0891 Inf    1   2.725  0.0177
 32rxn / half_bait       1.11 0.0882 Inf    1   1.358  0.3634

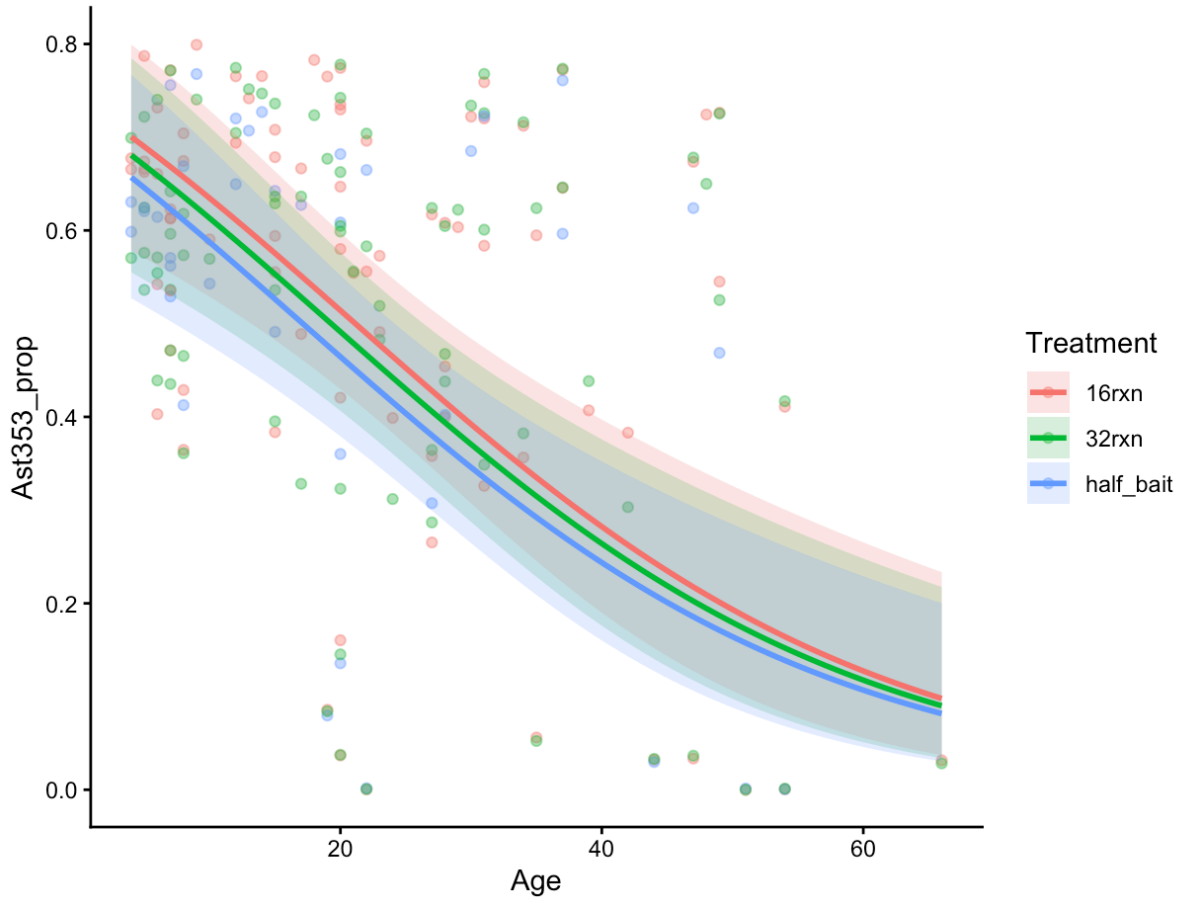
P value adjustment: tukey method for comparing a family of 3 estimates
Tests are performed on the log odds ratio scale

#Plot model

>emm<- emmeans(exp5.mod13, ~Age | Treatment, at = list(Age =
seq(min(exp5$Age), max(exp5$Age), length=100)), type="response")

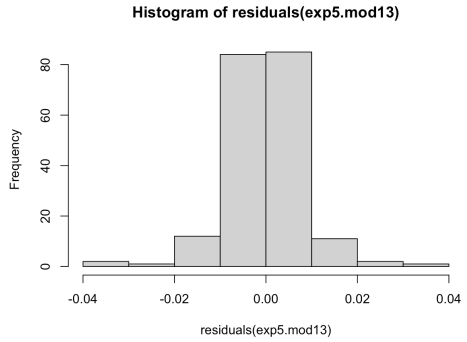
>emm_df<-as.data.frame(emm)

>ggplot() +
  geom_point(data = exp5,
             aes(x = Age, y = Ast353_prop, colour = Treatment),
             alpha = 0.4) +
  geom_ribbon(data = emm_df,
              aes(x = Age, ymin = asymp.LCL, ymax = asymp.UCL, fill =
Treatment),
              alpha = 0.2) +
  geom_line(data = emm_df,
            aes(x = Age, y = response, colour = Treatment),
            linewidth = 1) +
  theme_classic()
```

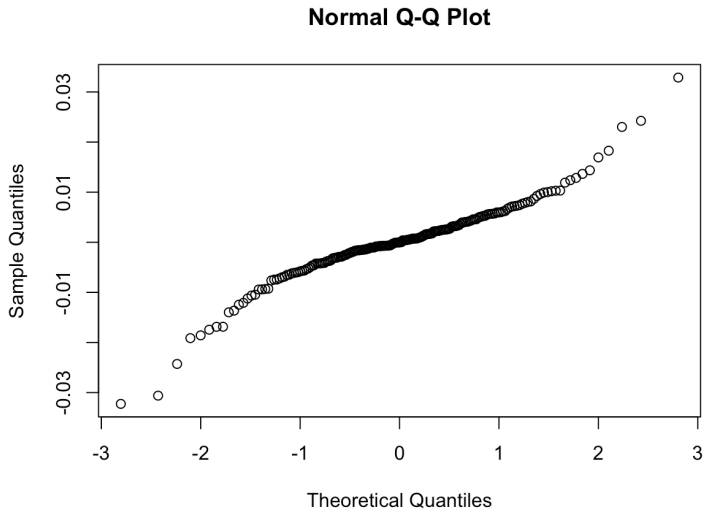


#Model residuals remain approximately normally distributed according to the histogram and Q-Q plot, but the Shapiro-Wilk test indicates that statistically they are still non-normal:

```
>hist(residuals(exp5.mod13))
```




```
>qqnorm(residuals(exp5.mod13))
```




```
>shapiro.test(residuals(exp5.mod13))

Shapiro-Wilk normality test

data:  residuals(exp5.mod13)
W = 0.93323, p-value = 6.972e-08

>AIC(exp5.mod13)

[1] -635.778

#-----------------------------------

#Chloroplast recovery
#lmer

#We performed a linear mixed model with the percentage of chloroplast
base-pairs recovered ("Chlp") as the response variable with "Treatment"
(16rxn, 32rxn, half_bait) as a fixed, categorical factor, and "Sample" and
"Pool" (batch) as random effects:

>exp5.mod16<-lmer(Chlp ~ Treatment * Age + (1|Tribe) + (1|Tribe:Genus) +
(1|Tribe:Genus:Sample) + (1|Pool), data=exp5, REML=TRUE)

#Test removing Tribe from the model:

>exp5.mod17<-lmer(Chlp ~ Treatment  * Age + (1|Tribe:Genus) +
(1|Tribe:Genus:Sample) + (1|Pool), data=exp5, REML=TRUE)

>anova(exp5.mod16, exp5.mod17)

Data: exp5
Models:
exp5.mod17: Chlp ~ Treatment * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Pool)
exp5.mod16: Chlp ~ Treatment * Age + (1 | Tribe) + (1 | Tribe:Genus) + (1
| Tribe:Genus:Sample) + (1 | Pool)
           npar    AIC    BIC  logLik -2*log(L) Chisq Df Pr(>Chisq)
exp5.mod17   10 1743.9 1776.9 -861.95    1723.9
exp5.mod16   11 1745.9 1782.2 -861.95    1723.9     0  1          1
```

#Tribe is non-significant and can be removed from the model. Test removing Genus nested in Tribe:

```
>exp5.mod18<-lmer(Chlp ~ Treatment * Age +  (1|Tribe:Genus:Sample) + (1|Pool), data=exp5, REML=TRUE)

>anova(exp5.mod17, exp5.mod18)

Data: exp5
Models:
exp5.mod18: Chlp ~ Treatment * Age + (1 | Tribe:Genus:Sample) + (1 | Pool)
exp5.mod17: Chlp ~ Treatment * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Pool)
           npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp5.mod18    9 1750.8 1780.5 -866.41    1732.8
exp5.mod17   10 1743.9 1776.9 -861.95    1723.9 8.9185  1   0.002823 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

#Genus nested in Tribe is significant and cannot be removed from the model. Test removing Sample nested in Genus nested in Tribe:

```
>exp5.mod19<-lmer(Chlp ~ Treatment  * Age + (1|Tribe:Genus) + (1|Pool), data=exp5, REML=TRUE)

>anova(exp5.mod17, exp5.mod19)

Data: exp5
Models:
exp5.mod19: Chlp ~ Treatment * Age + (1 | Tribe:Genus) + (1 | Pool)
exp5.mod17: Chlp ~ Treatment * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Pool)
           npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp5.mod19    9 1876.2 1905.9 -929.13    1858.2
exp5.mod17   10 1743.9 1776.9 -861.95    1723.9 134.35  1  < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

#Sample nested in Genus nested in Tribe is significant and cannot be removed from the model. Test removing Pool:

```
>exp5.mod20<-lmer(Chlp ~ Treatment  * Age + (1|Tribe:Genus) + (1|Tribe:Genus:Sample), data=exp5, REML=TRUE)

>anova(exp5.mod17, exp5.mod20)

Data: exp5
Models:
exp5.mod20: Chlp ~ Treatment * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample)
exp5.mod17: Chlp ~ Treatment * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Pool)
           npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp5.mod20    9 1853.1 1882.8 -917.53    1835.1
exp5.mod17   10 1743.9 1776.9 -861.95    1723.9 111.16  1  < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

#Pool is significant and cannot be removed from the model.

```
#Test removing the interaction between treatment and age:

>exp5.mod21<-lmer(Chlp ~ Treatment  + Age + (1|Tribe:Genus) +
(1|Tribe:Genus:Sample) + (1|Pool), data=exp5, REML=TRUE)

>anova(exp5.mod17, exp5.mod21)

Data: exp5
Models:
exp5.mod21: Chlp ~ Treatment + Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Pool)
exp5.mod17: Chlp ~ Treatment * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Pool)
           npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp5.mod21    8 1741.1 1767.5 -862.54    1725.1
exp5.mod17   10 1743.9 1776.9 -861.95    1723.9 1.1798  2     0.5544

#The interaction between treatment and age is not significant and can be
removed from the model.
#Remove Age:

>exp5.mod22<-lmer(Chlp ~ Treatment + (1|Tribe:Genus) +
(1|Tribe:Genus:Sample) + (1|Pool), data=exp5, REML=TRUE)

>anova(exp5.mod21, exp5.mod22)

Data: exp5
Models:
exp5.mod22: Chlp ~ Treatment + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Pool)
exp5.mod21: Chlp ~ Treatment + Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Pool)
           npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp5.mod22    7 1752.4 1775.5 -869.19    1738.4
exp5.mod21    8 1741.1 1767.5 -862.54    1725.1 13.302  1  0.0002651 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

#Age is significant and cannot be removed from the model.
#Remove Treatment:

>exp5.mod23<-lmer(Chlp ~ Age + (1|Tribe:Genus) + (1|Tribe:Genus:Sample) +
(1|Pool), data=exp5, REML=TRUE)

>anova(exp5.mod21, exp5.mod23)

Data: exp5
Models:
exp5.mod23: Chlp ~ Age + (1 | Tribe:Genus) + (1 | Tribe:Genus:Sample) + (1
| Pool)
exp5.mod21: Chlp ~ Treatment + Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Pool)
           npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp5.mod23    6 1737.2 1757.0 -862.62    1725.2
exp5.mod21    8 1741.1 1767.5 -862.54    1725.1 0.1537  2      0.926

#Treatment is not significant and can be removed from the model.

>summary(exp5.mod21)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method
['lmerModLmerTest']
Formula: Chlp ~ Treatment + Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) +
    (1 | Pool)
   Data: exp5

REML criterion at convergence: 1707.8

Scaled residuals:
     Min       1Q   Median       3Q      Max
-2.00944 -0.34533  0.06961  0.38374  2.07902

Random effects:
 Groups              Name        Variance Std.Dev.
 Tribe:Genus:Sample (Intercept) 444.08   21.073
 Tribe:Genus        (Intercept) 266.67   16.330
 Pool               (Intercept) 328.37   18.121
 Residual                        84.58    9.197
Number of obs: 200, groups:  Tribe:Genus:Sample, 81; Tribe:Genus, 19;
Pool, 15

Fixed effects:
                  Estimate Std. Error      df t value Pr(>|t|)
(Intercept)        81.4830     9.3187 23.4890   8.744 7.56e-09 ***
Treatment32rxn     -1.0507    11.2322  9.5843  -0.094 0.927395
Treatmenthalf_bait -4.2250    12.7354  8.6053  -0.332 0.748014
Age                -0.7072     0.1874 70.3852  -3.773 0.000333 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Correlation of Fixed Effects:
            (Intr) Trtm32 Trtmn_
Trtmnt32rxn -0.478
Trtmnthlf_b -0.418  0.344
Age         -0.404 -0.005 -0.001
```

```
>AIC(exp5.mod21)

[1] 1723.785

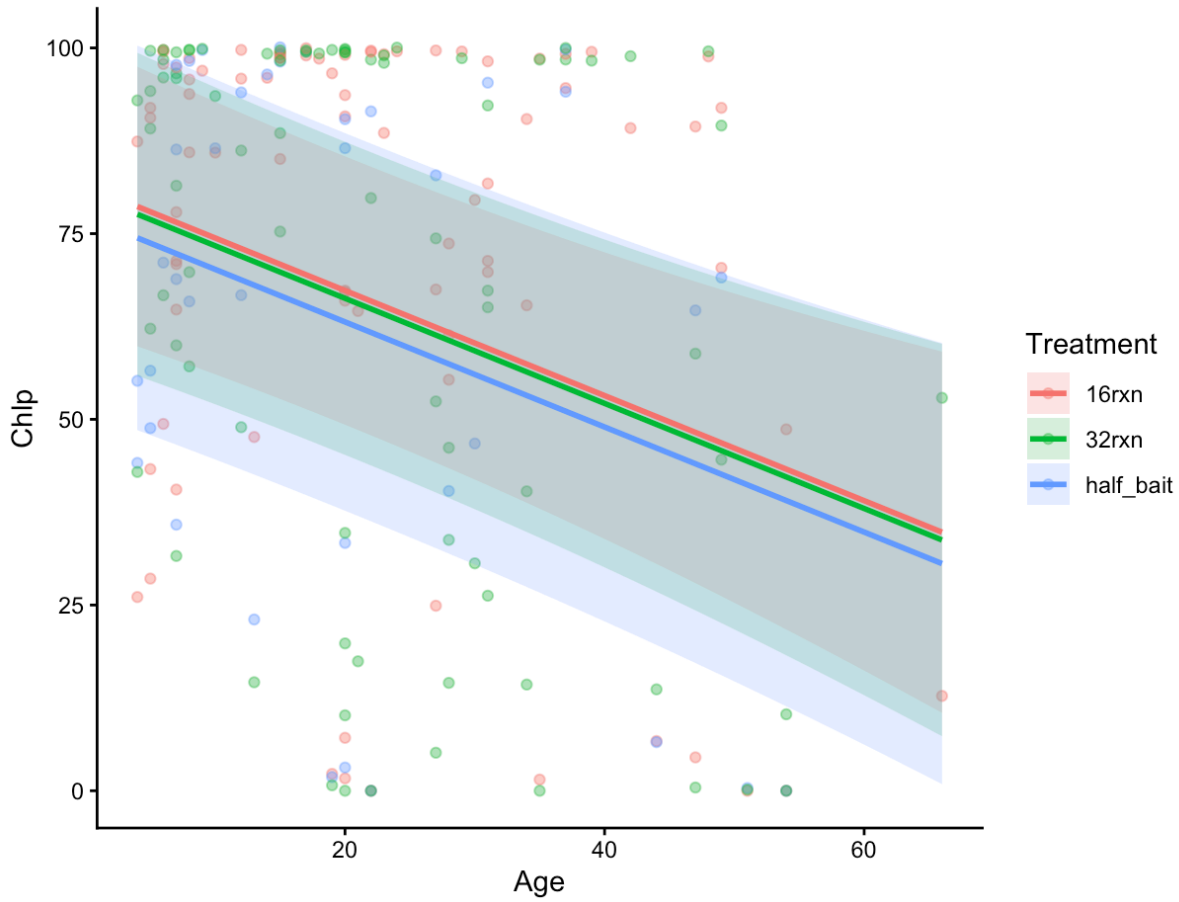
#Plot model

>emm<- emmeans(exp5.mod21, ~Age | Treatment, at = list(Age =
seq(min(exp5$Age), max(exp5$Age), length=100)), lmer.df="satterthwaite")

>emm_df<-as.data.frame(emm)

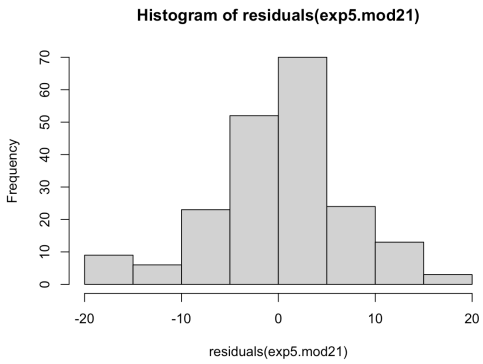
>ggplot() +
  geom_point(data = exp5,
             aes(x = Age, y = Chlp, colour = Treatment),
             alpha = 0.4) +
  geom_ribbon(data = emm_df,
              aes(x = Age, ymin = lower.CL, ymax = upper.CL, fill =
Treatment),
              alpha = 0.2) +
  geom_line(data = emm_df,
            aes(x = Age, y = emmean, colour = Treatment),
```

```
          linewidth = 1) +
  theme_classic()
```

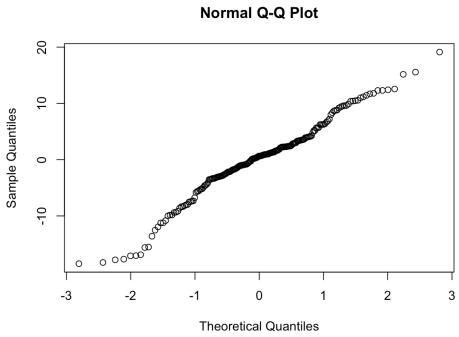


#Model residuals are approximately normally distributed according to the histogram and Q-Q plot, but the Shapiro-Wilk test indicates that statistically they are non-normal:

```
>hist(residuals(exp5.mod21))
```




```
>qqnorm(residuals(exp5.mod21))
```




```
>shapiro.test(residuals(exp5.mod21))

Shapiro-Wilk normality test

data:  residuals(exp5.mod21)
W = 0.97639, p-value = 0.001883

#-----------------------------------

#Chloroplast recovery
#glmm

#We performed a generalised linear mixed model with the proportion of
chloroplast base-pairs recovered (“Chlp_prop”) as the response variable
with “Treatment” (16rxn, 32rxn, half_bait) as a fixed, categorical factor,
and “Sample” and “Pool” (batch) as random effects:

>exp5_nz_n1 <- exp5[exp5$Chlp > 0 & exp5$Chlp <= 1, ]

>exp5_nz_n1$Chlp_prop <- exp5_nz_n1$Chlp / 100

>exp5.mod24<-glmmTMB(Chlp_prop ~ Treatment * Age + (1|Tribe) +
(1|Tribe:Genus) + (1|Tribe:Genus:Sample) + (1|Pool),
family=beta_family(link="logit"), data=exp5_nz_n1)

#Note this model is rank-deficient.

#Test removing Tribe as a factor:

>exp5.mod25<-glmmTMB(Chlp_prop ~ Treatment* Age + (1|Tribe:Genus) +
(1|Tribe:Genus:Sample) + (1|Pool), family=beta_family(link="logit"),
data=exp5_nz_n1)

>anova(exp5.mod24, exp5.mod25)

Data: exp5_nz_n1
Models:
exp5.mod25: Chlp_prop ~ Treatment * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + , zi=~0, disp=~1
exp5.mod25:     (1 | Pool), zi=~0, disp=~1
exp5.mod24: Chlp_prop ~ Treatment * Age + (1 | Tribe) + (1 | Tribe:Genus)
+ , zi=~0, disp=~1
exp5.mod24:     (1 | Tribe:Genus:Sample) + (1 | Pool), zi=~0, disp=~1
           Df     AIC     BIC logLik deviance Chisq Chi Df Pr(>Chisq)
exp5.mod25  7
exp5.mod24  8 -27.889 -32.799 21.945  -43.889           1
```

```
#Tribe is non-significant and can be removed as a factor. Test removing
Genus within Tribe:

>exp5.mod26<-glmmTMB(Chlp_prop ~ Treatment* Age  + (1|Tribe:Genus:Sample)
+ (1|Pool), family=beta_family(link="logit"), data=exp5_nz_n1)

>anova(exp5.mod25, exp5.mod26)

Data: exp5_nz_n1
Models:
exp5.mod26: Chlp_prop ~ Treatment * Age + (1 | Tribe:Genus:Sample) + (1 |
, zi=~0, disp=~1
exp5.mod26:     Pool), zi=~0, disp=~1
exp5.mod25: Chlp_prop ~ Treatment * Age + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + , zi=~0, disp=~1
exp5.mod25:     (1 | Pool), zi=~0, disp=~1
           Df AIC BIC logLik deviance Chisq Chi Df Pr(>Chisq)
exp5.mod26  6
exp5.mod25  7                                       1

#Genus within Tribe is non-significant and can be removed as a factor.
Test removing Sample within Genus within Tribe:

>exp5.mod27<-glmmTMB(Chlp_prop ~ Treatment* Age  + (1|Pool),
family=beta_family(link="logit"), data=exp5_nz_n1)

>anova(exp5.mod26, exp5.mod27)

Data: exp5_nz_n1
Models:
exp5.mod27: Chlp_prop ~ Treatment * Age + (1 | Pool), zi=~0, disp=~1
exp5.mod26: Chlp_prop ~ Treatment * Age + (1 | Tribe:Genus:Sample) + (1 |
, zi=~0, disp=~1
exp5.mod26:     Pool), zi=~0, disp=~1
           Df     AIC     BIC logLik deviance Chisq Chi Df Pr(>Chisq)
exp5.mod27  5 -33.889 -36.958 21.945  -43.889
exp5.mod26  6                                         1

#Sample within Genus within Tribe is non-significant and can be removed as
a factor. Test removing Pool:

>exp5.mod28<-glmmTMB(Chlp_prop ~ Treatment* Age,
family=beta_family(link="logit"), data=exp5_nz_n1)

>anova(exp5.mod27, exp5.mod28)

Data: exp5_nz_n1
Models:
exp5.mod28: Chlp_prop ~ Treatment * Age, zi=~0, disp=~1
exp5.mod27: Chlp_prop ~ Treatment * Age + (1 | Pool), zi=~0, disp=~1
           Df     AIC     BIC logLik deviance Chisq Chi Df Pr(>Chisq)
exp5.mod28  4 -35.889 -38.344 21.945  -43.889
exp5.mod27  5 -33.889 -36.958 21.945  -43.889     0      1          1

#Pool is non-significant and cannot be removed as a factor.
#Remove the interaction between treatment and age:

>exp5.mod29<-glmmTMB(Chlp_prop ~ Treatment+ Age,
family=beta_family(link="logit"), data=exp5_nz_n1)
```

```
>anova(exp5.mod28, exp5.mod29)

Data: exp5_nz_n1
Models:
exp5.mod28: Chlp_prop ~ Treatment * Age, zi=~0, disp=~1
exp5.mod29: Chlp_prop ~ Treatment + Age, zi=~0, disp=~1
           Df     AIC     BIC logLik deviance Chisq Chi Df Pr(>Chisq)
exp5.mod28  4 -35.889 -38.344 21.945  -43.889
exp5.mod29  4 -35.889 -38.344 21.945  -43.889     0      0          1

#The interaction between treatment and age is non-signficant and can be
removed.
#Remove Age:

>exp5.mod30<-glmmTMB(Chlp_prop ~ Treatment,
family=beta_family(link="logit"), data=exp5_nz_n1)

>anova(exp5.mod29, exp5.mod30)

Data: exp5_nz_n1
Models:
exp5.mod30: Chlp_prop ~ Treatment, zi=~0, disp=~1
exp5.mod29: Chlp_prop ~ Treatment + Age, zi=~0, disp=~1
           Df     AIC     BIC logLik deviance  Chisq Chi Df Pr(>Chisq)
exp5.mod30  3 -32.329 -34.170 19.165  -38.329
exp5.mod29  4 -35.889 -38.344 21.945  -43.889 5.5597      1    0.01838 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

#Age is significant and cannot be removed from the model.
#Remove Treatment:

>exp5.mod31<-glmmTMB(Chlp_prop ~  Age, family=beta_family(link="logit"),
data=exp5_nz_n1)

>anova(exp5.mod29, exp5.mod31)

Data: exp5_nz_n1
Models:
exp5.mod31: Chlp_prop ~ Age, zi=~0, disp=~1
exp5.mod29: Chlp_prop ~ Treatment + Age, zi=~0, disp=~1
           Df     AIC     BIC logLik deviance  Chisq Chi Df Pr(>Chisq)
exp5.mod31  3 -36.882 -38.723 21.441  -42.882
exp5.mod29  4 -35.889 -38.344 21.945  -43.889 1.0074      1     0.3155

#Treatment is nonsignificant and can be removed from the model.

>summary(exp5.mod29)

Family: beta  ( logit )
Formula:          Chlp_prop ~ Treatment + Age
Data: exp5_nz_n1

      AIC       BIC    logLik -2*log(L)  df.resid
    -35.9     -38.3      21.9     -43.9         0


Dispersion parameter for beta family (): 3.53e+03
```

```
Conditional model:
                   Estimate Std. Error z value Pr(>|z|)
(Intercept)       -4.208858   0.336039 -12.525  < 2e-16 ***
Treatmenthalf_bait  0.395308   0.358092   1.104 0.269624
Age               -0.034668   0.009585  -3.617 0.000298 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
>AIC(exp5.mod29)
```

```
[1] -35.889
```

```
>emm <- emmeans(exp5.mod29, ~Treatment, type="response")
```

```
>emm
```

```
Treatment response       SE  df asymp.LCL asymp.UCL
 32rxn      0.00345 0.000613 Inf   0.00244   0.00489
 half_bait  0.00512 0.001450 Inf   0.00294   0.00890

Confidence level used: 0.95
Intervals are back-transformed from the logit scale
```

```
>pairs(emm, adjust="tukey")
```

```
contrast          odds.ratio    SE  df null z.ratio p.value
 32rxn / half_bait      0.673 0.241 Inf    1  -1.104  0.2696

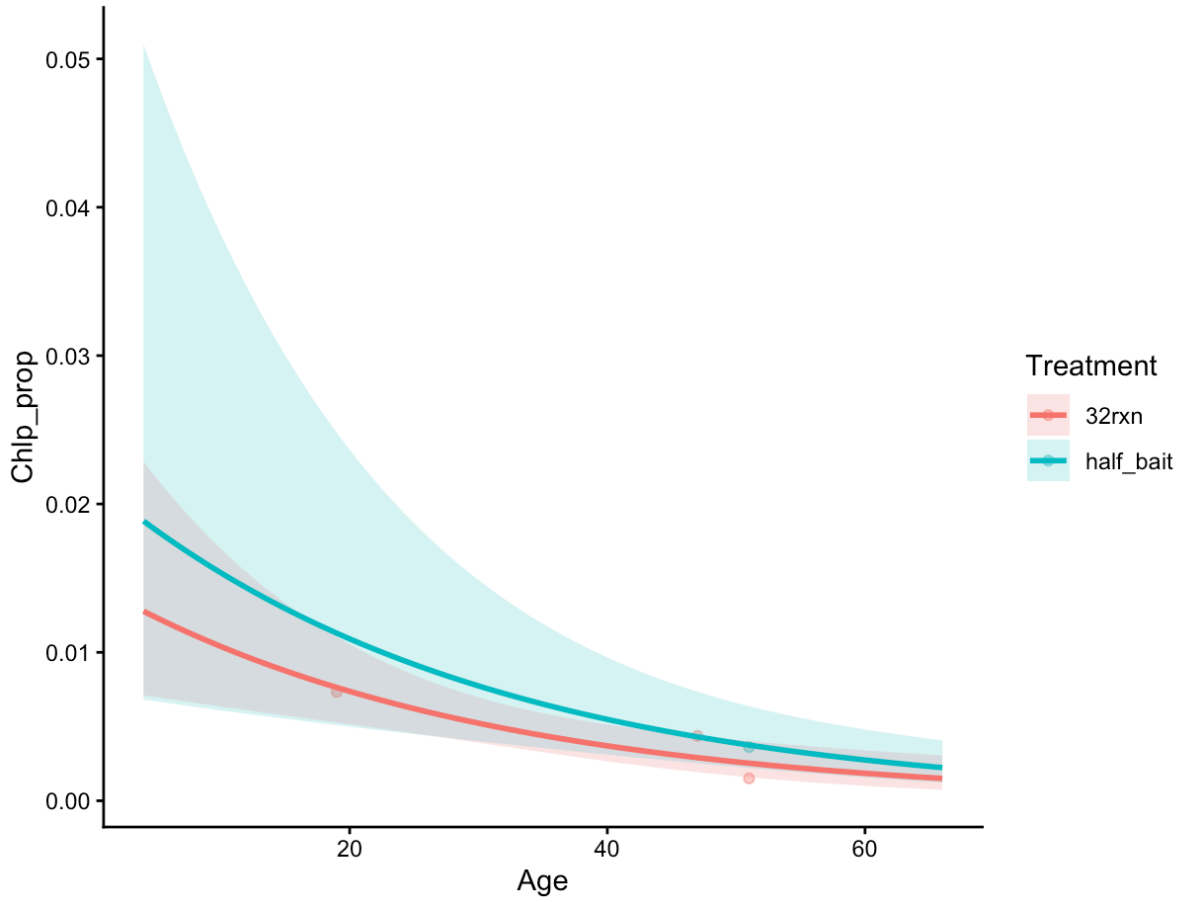
Tests are performed on the log odds ratio scale
```

```
#Plot model

>emm<- emmeans(exp5.mod29, ~Age | Treatment, at = list(Age =
seq(min(exp5$Age), max(exp5$Age), length=100)), type="response")

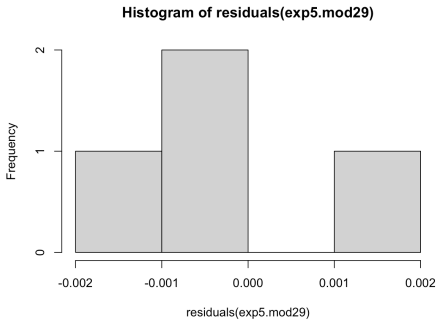
>emm_df<-as.data.frame(emm)

>ggplot() +
  geom_point(data = exp5_nz_n1,
             aes(x = Age, y = Chlp_prop, colour = Treatment),
             alpha = 0.4) +
  geom_ribbon(data = emm_df,
              aes(x = Age, ymin = asymp.LCL, ymax = asymp.UCL, fill =
Treatment),
              alpha = 0.2) +
  geom_line(data = emm_df,
            aes(x = Age, y = response, colour = Treatment),
            linewidth = 1) +
  theme_classic()
```

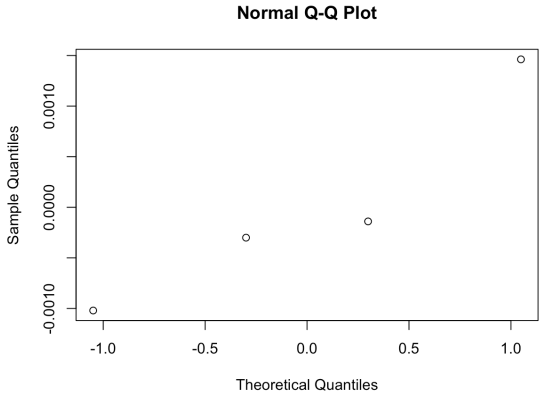


#Model residuals remain approximately normally distributed according to the histogram and Q-Q plot, but the Shapiro-Wilk test indicates that statistically they are still non-normal:

```
>hist(residuals(exp5.mod29))
```




```
>qqnorm(residuals(exp5.mod29))
```




```
>shapiro.test(residuals(exp5.mod29))

Shapiro-Wilk normality test

data:  residuals(exp5.mod29)
W = 0.91308, p-value = 0.4989

#-----------------------------------
#Target recovery vs chloroplast recovery
#-----------------------------------

#lmer

>exp2345<-read.csv("exp2_3_4_5_w_year.csv", header=TRUE)

>head(exp2345)
                          Sample Year Age        Batch         Genus
Tribe Size
1 Anaphalioides_bellidioides_AG440 2000  20    AG_P5P13C Anaphalioides
Gnaphalieae mini
2 Anaphalioides_bellidioides_AG440 2000  20 AG_P5P14/15C Anaphalioides
Gnaphalieae mini
3      Anaphalioides_papuana_AG442 1998  22     AG_P5P6C Anaphalioides
Gnaphalieae mini
4      Anaphalioides_papuana_AG442 1998  22     AG_P5P7C Anaphalioides
Gnaphalieae mini
5      Anaphalioides_papuana_AG442 1998  22   AG_P5P4/5C Anaphalioides
Gnaphalieae mini
6       Anemocarpa_calcicola_AG577 1997  23     AG_P5P8C    Anemocarpa
Gnaphalieae mini
  Input_cat Input_cont Treatment Experiment     Chlp      Ast353
1      5 ng          5        5M       exp5 93.64823 73.48825144
2      5 ng          5     32rxn       exp5 99.85981 66.25784296
3      5 ng          5        5M       exp5  0.00000  0.04217276
4      5 ng          5 half_bait       exp5  0.00000  0.15759294
5      5 ng          5     32rxn       exp5  0.00000  0.06658857
6      5 ng          5        5M       exp5 99.15346 49.10018908

#We performed a linear mixed model with the percentage chloroplast base-
pairs recovered (“Chlp”) as the response variable with the percentage of
target base-pairs recovered (“Ast353”) as a continuous predictor,
“Experiment” (2, 3_4, and 5) as a fixed, categorical factor, and “Sample”
and “Pool” (batch) as random effects:
```

```
>exp2345.mod<-lmer(Chlp ~ Ast353 * Age * Experiment + Treatment +
(1|Tribe) + (1|Tribe:Genus) + (1|Tribe:Genus:Sample) + (1|Batch),
data=exp2345, REML=TRUE)

#Test removing Tribe as a factor:

>exp2345.mod2<-lmer(Chlp ~ Ast353 * Age* Experiment + Treatment +
(1|Tribe:Genus) + (1|Tribe:Genus:Sample) + (1|Batch), data=exp2345,
REML=TRUE)

>anova(exp2345.mod, exp2345.mod2)

Data: exp2345
Models:
exp2345.mod2: Chlp ~ Ast353 * Age * Experiment + Treatment + (1 |
Tribe:Genus) + (1 | Tribe:Genus:Sample) + (1 | Batch)
exp2345.mod: Chlp ~ Ast353 * Age * Experiment + Treatment + (1 | Tribe) +
(1 | Tribe:Genus) + (1 | Tribe:Genus:Sample) + (1 | Batch)
             npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp2345.mod2   22 5507.7 5607.5 -2731.8    5463.7
exp2345.mod    23 5509.5 5613.9 -2731.8    5463.5 0.1314  1      0.717

#Tribe is non-significant and can be removed from the model. Test removing
Genus within Tribe:

>exp2345.mod3<-lmer(Chlp ~ Ast353 * Age* Experiment + Treatment +
(1|Tribe:Genus:Sample) + (1|Batch), data=exp2345, REML=TRUE)

>anova(exp2345.mod2, exp2345.mod3)

Data: exp2345
Models:
exp2345.mod3: Chlp ~ Ast353 * Age * Experiment + Treatment + (1 |
Tribe:Genus:Sample) + (1 | Batch)
exp2345.mod2: Chlp ~ Ast353 * Age * Experiment + Treatment + (1 |
Tribe:Genus) + (1 | Tribe:Genus:Sample) + (1 | Batch)
             npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp2345.mod3   21 5530.8 5626.0 -2744.4    5488.8
exp2345.mod2   22 5507.7 5607.5 -2731.8    5463.7 25.081  1  5.497e-07 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

#Genus within Tribe is significant and cannot be removed from the model.
Test removing Sample within Genus within Tribe:

>exp2345.mod4<-lmer(Chlp ~ Ast353 * Age* Experiment + Treatment +
(1|Tribe:Genus)  + (1|Batch), data=exp2345, REML=TRUE)

>anova(exp2345.mod2, exp2345.mod4)

Data: exp2345
Models:
exp2345.mod4: Chlp ~ Ast353 * Age * Experiment + Treatment + (1 |
Tribe:Genus) + (1 | Batch)
exp2345.mod2: Chlp ~ Ast353 * Age * Experiment + Treatment + (1 |
Tribe:Genus) + (1 | Tribe:Genus:Sample) + (1 | Batch)
             npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp2345.mod4   21 5955.9 6051.1 -2956.9    5913.9
exp2345.mod2   22 5507.7 5607.5 -2731.8    5463.7 450.19  1  < 2.2e-16 ***
```

```
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

#Sample within Genus within Tribe is significant and cannot be removed from the model. Test removing Batch:

```
>exp2345.mod5<-lmer(Chlp ~ Ast353 * Age* Experiment + Treatment + (1|Tribe:Genus) + (1|Tribe:Genus:Sample), data=exp2345, REML=TRUE)

>anova(exp2345.mod2, exp2345.mod5)

Data: exp2345
Models:
exp2345.mod5: Chlp ~ Ast353 * Age * Experiment + Treatment + (1 | Tribe:Genus) + (1 | Tribe:Genus:Sample)
exp2345.mod2: Chlp ~ Ast353 * Age * Experiment + Treatment + (1 | Tribe:Genus) + (1 | Tribe:Genus:Sample) + (1 | Batch)
             npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp2345.mod5   21 5839.2 5934.4 -2898.6    5797.2
exp2345.mod2   22 5507.7 5607.5 -2731.8    5463.7 333.47  1  < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

#Batch is significant and cannot be removed from the model. Test the interaction between target recovery/Age and experiment:

```
>exp2345.mod6<-lmer(Chlp ~ Ast353 * Age + Experiment + Treatment + (1|Tribe:Genus) + (1|Tribe:Genus:Sample) + (1|Batch), data=exp2345, REML=TRUE)

>anova(exp2345.mod2, exp2345.mod6)

Data: exp2345
Models:
exp2345.mod6: Chlp ~ Ast353 * Age + Experiment + Treatment + (1 | Tribe:Genus) + (1 | Tribe:Genus:Sample) + (1 | Batch)
exp2345.mod2: Chlp ~ Ast353 * Age * Experiment + Treatment + (1 | Tribe:Genus) + (1 | Tribe:Genus:Sample) + (1 | Batch)
             npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp2345.mod6   16 5519.1 5591.7 -2743.6    5487.1
exp2345.mod2   22 5507.7 5607.5 -2731.8    5463.7 23.418  6  0.0006679 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

#The interaction between target recovery/Age and experiment is significant and cannot be removed from the model. Test removing Treatment:

```
>exp2345.mod7<-lmer(Chlp ~ Ast353 * Age* Experiment + (1|Tribe:Genus) + (1|Tribe:Genus:Sample) + (1|Batch), data=exp2345, REML=TRUE)

>anova(exp2345.mod2, exp2345.mod7)

Data: exp2345
Models:
exp2345.mod7: Chlp ~ Ast353 * Age * Experiment + (1 | Tribe:Genus) + (1 | Tribe:Genus:Sample) + (1 | Batch)
exp2345.mod2: Chlp ~ Ast353 * Age * Experiment + Treatment + (1 | Tribe:Genus) + (1 | Tribe:Genus:Sample) + (1 | Batch)
             npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
```

```
exp2345.mod7   16 5499.0 5571.6 -2733.5    5467.0
exp2345.mod2   22 5507.7 5607.5 -2731.8    5463.7 3.3123  6     0.7688

#Treatment is non-significant and can be removed from the model.
#Test removing the interaction between target recovery/experiment and age:

>exp2345.mod8<-lmer(Chlp ~ Ast353 + Age* Experiment + (1|Tribe:Genus) +
(1|Tribe:Genus:Sample) + (1|Batch), data=exp2345, REML=TRUE)

>anova(exp2345.mod7, exp2345.mod8)

Data: exp2345
Models:
exp2345.mod8: Chlp ~ Ast353 + Age * Experiment + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Batch)
exp2345.mod7: Chlp ~ Ast353 * Age * Experiment + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) + (1 | Batch)
             npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
exp2345.mod8   11 5510.4 5560.3 -2744.2    5488.4
exp2345.mod7   16 5499.0 5571.6 -2733.5    5467.0 21.445  5  0.0006674 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

#The interaction between target recovery/experiment and age is significant
and cannot be removed from the model.

>summary(exp2345.mod7)

Linear mixed model fit by REML. t-tests use Satterthwaite's method
['lmerModLmerTest']
Formula: Chlp ~ Ast353 * Age * Experiment + (1 | Tribe:Genus) + (1 |
Tribe:Genus:Sample) +
    (1 | Batch)
   Data: exp2345

REML criterion at convergence: 5491.2

Scaled residuals:
    Min      1Q  Median      3Q     Max
-4.2128 -0.3120  0.0267  0.3312  3.6802

Random effects:
 Groups             Name        Variance Std.Dev.
 Tribe:Genus:Sample (Intercept) 280.17   16.738
 Batch              (Intercept) 117.25   10.828
 Tribe:Genus        (Intercept)  88.41    9.403
 Residual                        52.82    7.268
Number of obs: 689, groups:  Tribe:Genus:Sample, 233; Batch, 47;
Tribe:Genus, 40

Fixed effects:
                            Estimate Std. Error         df t value
Pr(>|t|)
(Intercept)                53.218170   7.442049 363.866510   7.151
4.76e-12 ***
Ast353                      0.917778   0.148783 647.820865   6.169
1.21e-09 ***
Age                         0.747799   0.330251 337.790042   2.264
0.02419 *
```

```
Experimentexp3_4           -43.000973  11.399693 524.706443  -3.772
0.00018 ***
Experimentexp5              -7.693628  14.693806 347.354386  -0.524
0.60089
Ast353:Age                  -0.018042   0.007964 409.695882  -2.266
0.02400 *
Ast353:Experimentexp3_4     -0.552004   0.201477 651.189072  -2.740
0.00632 **
Ast353:Experimentexp5       -0.504295   0.257342 393.727880  -1.960
0.05074 .
Age:Experimentexp3_4        -0.718941   0.456625 438.625343  -1.574
0.11610
Age:Experimentexp5          -1.464429   0.458708 320.857200  -3.193
0.00155 **
Ast353:Age:Experimentexp3_4  0.017039   0.009430 545.074806   1.807
0.07135 .
Ast353:Age:Experimentexp5    0.031059   0.010033 365.801251   3.096
0.00212 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Correlation of Fixed Effects:
            (Intr) Ast353 Age    Exp3_4 Exprm5 As353:A A353:E3 A353:E5
A:E3_4 Ag:Ex5 A353:A:E3
Ast353      -0.748
Age         -0.584  0.579
Exprmntx3_4 -0.561  0.436  0.437
Expermntxp5 -0.488  0.375  0.297  0.285
Ast353:Age   0.458 -0.668 -0.847 -0.342 -0.239
Ast353:E3_4  0.550 -0.736 -0.430 -0.747 -0.275  0.494
Ast353:Exp5  0.441 -0.584 -0.334 -0.253 -0.866  0.389   0.431
Ag:Exprm3_4  0.419 -0.418 -0.724 -0.727 -0.216  0.615   0.597   0.243
Ag:Exprmnt5  0.425 -0.422 -0.713 -0.310 -0.746  0.608   0.314   0.734
0.517
A353:A:E3_4 -0.386  0.563  0.717  0.518  0.202 -0.845  -0.677  -0.329  -
0.802 -0.515
Ast353:A:E5 -0.363  0.532  0.669  0.270  0.584 -0.790  -0.395  -0.743  -
0.488 -0.852  0.669
```

```
>vc <- as.data.frame(VarCorr(exp2345.mod7))

>vc$percent <- vc$vcov / sum(vc$vcov) * 100

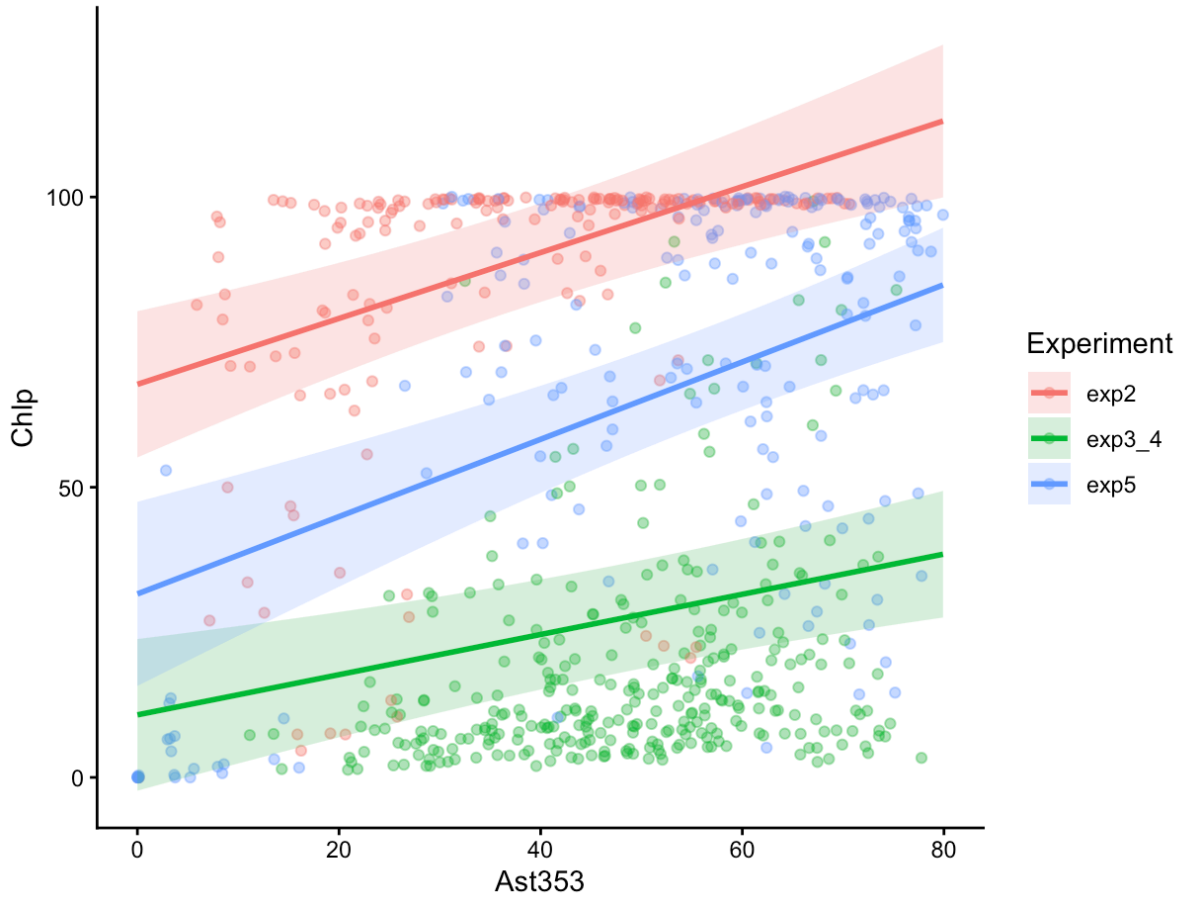
>vc
                 grp        var1 var2      vcov     sdcor   percent
1 Tribe:Genus:Sample (Intercept) <NA> 280.17025 16.738287 52.012811
2              Batch (Intercept) <NA> 117.25068 10.828235 21.767256
3        Tribe:Genus (Intercept) <NA>  88.41217  9.402775 16.413468
4           Residual        <NA> <NA>  52.82314  7.267953  9.806465

>emm<-emmeans(exp2345.mod7, ~Ast353 | Experiment | Age, at = list (Ast353
= seq (min(exp2345$Ast353), max(exp2345$Ast353), length=100)),
lmer.df="satterthwaite")

>emm_df<-as.data.frame(emm)

>ggplot() +
  geom_point(data = exp2345,
             aes(x = Ast353, y = Chlp, colour = Experiment),
```

```
            alpha = 0.4) +
  geom_ribbon(data = emm_df,
              aes(x = Ast353, ymin = lower.CL, ymax = upper.CL, fill =
Experiment),
              alpha = 0.2) +
  geom_line(data = emm_df,
            aes(x = Ast353, y = emmean, colour = Experiment),
            linewidth = 1) +
  theme_classic()
```

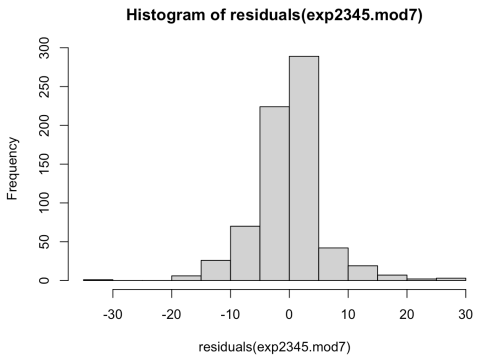



```
>AIC(exp2345.mod7)
```

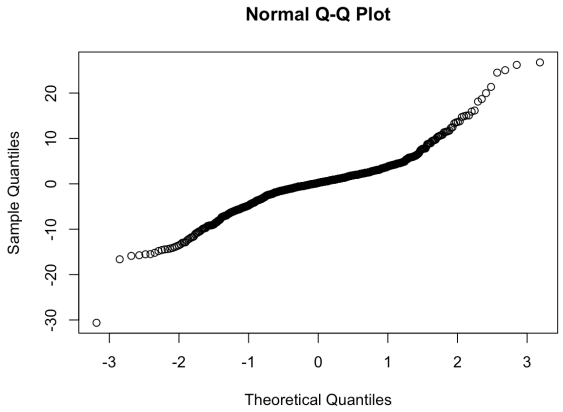
```
[1] 5523.217
```

```
#Model residuals are approximately normally distributed according to the
histogram and Q-Q plot, but the Shapiro-Wilk test indicates that
statistically they are non-normal:

>hist(residuals(exp2345.mod7))
```




```
>qqnorm(residuals(exp2345.mod7))
```




```
>shapiro.test(residuals(exp2345.mod7))
Shapiro-Wilk normality test

data:  residuals(exp2345.mod7)
W = 0.92747, p-value < 2.2e-16

#-----------------------------------

#glmm

#We performed a linear mixed model with the proportion of chloroplast
base-pairs (“Chlp_prop”) as the response variable with the proportion of
target base-pairs recovered (“Ast353_prop”) as a continuous predictor,
“Experiment” (2, 3_4, and 5) as a fixed, categorical factor, and “Sample”
and “Pool” (batch) as random effects:

>exp2345$Chlp_prop <- exp2345$Chlp / 100

>exp2345$Ast353_prop <- exp2345$Ast353 / 100

>exp2345_nz_n1 <- exp2345[exp2345$Chlp_prop > 0 & exp2345$Chlp_prop <= 1,]

>exp2345_n0_n1 <- exp2345_nz_n1[exp2345_nz_n1$Ast353_prop > 0 &

>exp2345_nz_n1$Ast353_prop <= 1,]

>head(exp2345_n0_n1)
```

```
                          Sample Year Age        Batch         Genus
Tribe Size
1 Anaphalioides_bellidioides_AG440 2000  20    AG_P5P13C Anaphalioides
Gnaphalieae mini
2 Anaphalioides_bellidioides_AG440 2000  20 AG_P5P14/15C Anaphalioides
Gnaphalieae mini
6       Anemocarpa_calcicola_AG577 1997  23     AG_P5P8C    Anemocarpa
Gnaphalieae mini
7       Anemocarpa_calcicola_AG577 1997  23  AG_P5P9/10C    Anemocarpa
Gnaphalieae mini
8     Anemocarpa_podolepidum_AG578 1992  28     AG_P5P8C    Anemocarpa
Gnaphalieae mini
9     Anemocarpa_podolepidum_AG578 1992  28  AG_P5P9/10C    Anemocarpa
Gnaphalieae mini
  Input_cat Input_cont Treatment Experiment     Chlp   Ast353 Chlp_prop
Ast353_prop
1      5 ng          5        5M       exp5 93.64823 73.48825 0.9364823
0.7348825
2      5 ng          5     32rxn       exp5 99.85981 66.25784 0.9985981
0.6625784
6      5 ng          5        5M       exp5 99.15346 49.10019 0.9915346
0.4910019
7      5 ng          5     32rxn       exp5 97.99418 51.90468 0.9799418
0.5190468
8      5 ng          5        5M       exp5 55.33269 39.96424 0.5533269
0.3996424
9      5 ng          5     32rxn       exp5 46.18247 43.79641 0.4618247
0.4379641
```

```
>exp2345.mod9<-glmmTMB(Chlp_prop ~ Ast353_prop *Age * Experiment +
Treatment + (1|Tribe) + (1|Tribe:Genus) + (1|Tribe:Genus:Sample) +
(1|Batch), family=beta_family(link="logit"), data=exp2345_n0_n1)

#Test removing batch from the model:

>exp2345.mod10<-glmmTMB(Chlp_prop ~ Ast353_prop *Age * Experiment +
Treatment + (1|Tribe) + (1|Tribe:Genus) + (1|Tribe:Genus:Sample),
family=beta_family(link="logit"), data=exp2345_n0_n1)

>anova(exp2345.mod9,exp2345.mod10)
```

```
Data: exp2345_n0_n1
Models:
exp2345.mod10: Chlp_prop ~ Ast353_prop * Age * Experiment + Treatment + (1
| , zi=~0, disp=~1
exp2345.mod10:     Tribe) + (1 | Tribe:Genus) + (1 | Tribe:Genus:Sample),
zi=~0, disp=~1
exp2345.mod9: Chlp_prop ~ Ast353_prop * Age * Experiment + Treatment + (1
| , zi=~0, disp=~1
exp2345.mod9:     Tribe) + (1 | Tribe:Genus) + (1 | Tribe:Genus:Sample) +
(1 | , zi=~0, disp=~1
exp2345.mod9:     Batch), zi=~0, disp=~1
             Df     AIC     BIC  logLik deviance  Chisq Chi Df Pr(>Chisq)
exp2345.mod10 22 -1455.1 -1355.8  749.57  -1499.1
exp2345.mod9  23 -2231.4 -2127.6 1138.72  -2277.4 778.31      1  < 2.2e-16
***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
#Batch is significant and cannot be removed from the model.
#Test removing Sample:

>exp2345.mod11<-glmmTMB(Chlp_prop ~ Ast353_prop *Age * Experiment +
Treatment + (1|Tribe) + (1|Tribe:Genus) + (1|Batch),
family=beta_family(link="logit"), data=exp2345_n0_n1)

>anova(exp2345.mod9,exp2345.mod11)

Data: exp2345_n0_n1
Models:
exp2345.mod11: Chlp_prop ~ Ast353_prop * Age * Experiment + Treatment + (1
| , zi=~0, disp=~1
exp2345.mod11:     Tribe) + (1 | Tribe:Genus) + (1 | Batch), zi=~0,
disp=~1
exp2345.mod9: Chlp_prop ~ Ast353_prop * Age * Experiment + Treatment + (1
| , zi=~0, disp=~1
exp2345.mod9:     Tribe) + (1 | Tribe:Genus) + (1 | Tribe:Genus:Sample) +
(1 | , zi=~0, disp=~1
exp2345.mod9:     Batch), zi=~0, disp=~1
             Df     AIC     BIC  logLik deviance  Chisq Chi Df Pr(>Chisq)
exp2345.mod11 22 -1351.6 -1252.2  697.79  -1395.6
exp2345.mod9  23 -2231.4 -2127.6 1138.72  -2277.4 881.87      1  < 2.2e-16
***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

#Sample is significant and cannot be removed from the model.
#Test removing Genus:

>exp2345.mod12<-glmmTMB(Chlp_prop ~ Ast353_prop *Age * Experiment +
Treatment + (1|Tribe)  + (1|Tribe:Genus:Sample) + (1|Batch),
family=beta_family(link="logit"), data=exp2345_n0_n1)

>anova(exp2345.mod9,exp2345.mod12)

Data: exp2345_n0_n1
Models:
exp2345.mod12: Chlp_prop ~ Ast353_prop * Age * Experiment + Treatment + (1
| , zi=~0, disp=~1
exp2345.mod12:     Tribe) + (1 | Tribe:Genus:Sample) + (1 | Batch), zi=~0,
disp=~1
exp2345.mod9: Chlp_prop ~ Ast353_prop * Age * Experiment + Treatment + (1
| , zi=~0, disp=~1
exp2345.mod9:     Tribe) + (1 | Tribe:Genus) + (1 | Tribe:Genus:Sample) +
(1 | , zi=~0, disp=~1
exp2345.mod9:     Batch), zi=~0, disp=~1
             Df     AIC     BIC logLik deviance  Chisq Chi Df Pr(>Chisq)
exp2345.mod12 22 -2195.4 -2096.0 1119.7  -2239.4
exp2345.mod9  23 -2231.4 -2127.6 1138.7  -2277.4 38.057      1  6.869e-10
***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

#Genus is significant and cannot be removed from the model.
#Test removing Tribe:
```

```
>exp2345.mod13<-glmmTMB(Chlp_prop ~ Ast353_prop *Age * Experiment +
Treatment  + (1|Tribe:Genus) + (1|Tribe:Genus:Sample) + (1|Batch),
family=beta_family(link="logit"), data=exp2345_n0_n1)

>anova(exp2345.mod9,exp2345.mod13)

Data: exp2345_n0_n1
Models:
exp2345.mod13: Chlp_prop ~ Ast353_prop * Age * Experiment + Treatment + (1
| , zi=~0, disp=~1
exp2345.mod13:     Tribe:Genus) + (1 | Tribe:Genus:Sample) + (1 | Batch),
zi=~0, disp=~1
exp2345.mod9: Chlp_prop ~ Ast353_prop * Age * Experiment + Treatment + (1
| , zi=~0, disp=~1
exp2345.mod9:     Tribe) + (1 | Tribe:Genus) + (1 | Tribe:Genus:Sample) +
(1 | , zi=~0, disp=~1
exp2345.mod9:     Batch), zi=~0, disp=~1
             Df     AIC     BIC logLik deviance Chisq Chi Df Pr(>Chisq)
exp2345.mod13 22 -2233.4 -2134.1 1138.7  -2277.4
exp2345.mod9  23 -2231.4 -2127.6 1138.7  -2277.4     0      1          1

#Tribe is insignificant and can be removed from the model.
#Test removing Treatment:

>exp2345.mod14<-glmmTMB(Chlp_prop ~ Ast353_prop *Age * Experiment +
(1|Tribe:Genus) + (1|Tribe:Genus:Sample) + (1|Batch),
family=beta_family(link="logit"), data=exp2345_n0_n1)

>anova(exp2345.mod13,exp2345.mod14)

Data: exp2345_n0_n1
Models:
exp2345.mod14: Chlp_prop ~ Ast353_prop * Age * Experiment + (1 |
Tribe:Genus) + , zi=~0, disp=~1
exp2345.mod14:     (1 | Tribe:Genus:Sample) + (1 | Batch), zi=~0, disp=~1
exp2345.mod13: Chlp_prop ~ Ast353_prop * Age * Experiment + Treatment + (1
| , zi=~0, disp=~1
exp2345.mod13:     Tribe:Genus) + (1 | Tribe:Genus:Sample) + (1 | Batch),
zi=~0, disp=~1
             Df     AIC     BIC logLik deviance Chisq Chi Df Pr(>Chisq)
exp2345.mod14 16 -2243.0 -2170.7 1137.5  -2275.0
exp2345.mod13 22 -2233.4 -2134.1 1138.7  -2277.4 2.455      6     0.8735

#Treatment is not significant and can be removed from the model.
#Test the interaction between target recovery/age and experiment:

>exp2345.mod15<-glmmTMB(Chlp_prop ~ Ast353_prop *Age + Experiment +
(1|Tribe:Genus) + (1|Tribe:Genus:Sample) + (1|Batch),
family=beta_family(link="logit"), data=exp2345_n0_n1)

>anova(exp2345.mod14,exp2345.mod15)

Data: exp2345_n0_n1
Models:
exp2345.mod15: Chlp_prop ~ Ast353_prop * Age + Experiment + (1 |
Tribe:Genus) + , zi=~0, disp=~1
exp2345.mod15:     (1 | Tribe:Genus:Sample) + (1 | Batch), zi=~0, disp=~1
exp2345.mod14: Chlp_prop ~ Ast353_prop * Age * Experiment + (1 |
Tribe:Genus) + , zi=~0, disp=~1
```

```
exp2345.mod14:     (1 | Tribe:Genus:Sample) + (1 | Batch), zi=~0, disp=~1
             Df   AIC     BIC logLik deviance  Chisq Chi Df Pr(>Chisq)
exp2345.mod15 10 -2236 -2190.8 1128.0    -2256
exp2345.mod14 16 -2243 -2170.7 1137.5    -2275 19.034      6   0.004106 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

#The interaction between target recovery/age and experiment is significant and cannot be removed from the model.
#Test the interaction between target recovery/experiment and age:

```
>exp2345.mod16<-glmmTMB(Chlp_prop ~ Ast353_prop + Age * Experiment +
(1|Tribe:Genus) + (1|Tribe:Genus:Sample) + (1|Batch),
family=beta_family(link="logit"), data=exp2345_n0_n1)

>anova(exp2345.mod14,exp2345.mod16)
```

```
Data: exp2345_n0_n1
Models:
exp2345.mod16: Chlp_prop ~ Ast353_prop + Age * Experiment + (1 |
Tribe:Genus) + , zi=~0, disp=~1
exp2345.mod16:     (1 | Tribe:Genus:Sample) + (1 | Batch), zi=~0, disp=~1
exp2345.mod14: Chlp_prop ~ Ast353_prop * Age * Experiment + (1 |
Tribe:Genus) + , zi=~0, disp=~1
exp2345.mod14:     (1 | Tribe:Genus:Sample) + (1 | Batch), zi=~0, disp=~1
             Df     AIC     BIC logLik deviance  Chisq Chi Df Pr(>Chisq)
exp2345.mod16 11 -2231.7 -2182.1 1126.9  -2253.7
exp2345.mod14 16 -2243.0 -2170.7 1137.5  -2275.0 21.258      5  0.0007242
***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

#The interaction between target recovery/experiment and age is significant and cannot be removed from the model.

```
>summary(exp2345.mod14)
```

```
Family: beta  ( logit )
Formula:          Chlp_prop ~ Ast353_prop * Age * Experiment + (1 |
Tribe:Genus) +
    (1 | Tribe:Genus:Sample) + (1 | Batch)
Data: exp2345_n0_n1

      AIC       BIC    logLik -2*log(L)  df.resid
  -2243.0   -2170.7    1137.5   -2275.0       660

Random effects:

Conditional model:
 Groups             Name        Variance Std.Dev.
 Tribe:Genus        (Intercept) 0.7043   0.8392
 Tribe:Genus:Sample (Intercept) 1.6977   1.3030
 Batch              (Intercept) 0.9188   0.9585
Number of obs: 676, groups:  Tribe:Genus, 40; Tribe:Genus:Sample, 231;
Batch, 47

Dispersion parameter for beta family ():  111

Conditional model:
```

```
                                Estimate Std. Error z value Pr(>|z|)
(Intercept)                      2.04964    0.64196   3.193  0.00141 **
Ast353_prop                      4.10910    1.30172   3.157  0.00160 **
Age                              0.01141    0.02503   0.456  0.64855
Experimentexp3_4                -3.91337    0.79392  -4.929 8.26e-07 ***
Experimentexp5                  -4.57222    1.17069  -3.906 9.40e-05 ***
Ast353_prop:Age                 -0.04287    0.06113  -0.701  0.48316
Ast353_prop:Experimentexp3_4    -1.55446    1.42870  -1.088  0.27659
Ast353_prop:Experimentexp5       2.73724    2.02411   1.352  0.17627
Age:Experimentexp3_4            -0.02695    0.03149  -0.856  0.39204
Age:Experimentexp5              -0.01762    0.03446  -0.511  0.60901
Ast353_prop:Age:Experimentexp3_4  0.07260    0.06543   1.110  0.26717
Ast353_prop:Age:Experimentexp5    0.05693    0.07458   0.763  0.44524
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
>AIC(exp2345.mod14)

[1] -2242.989

>emm <- emmeans(exp2345.mod14, ~Ast353_prop | Experiment | Age,
type="response")

>emm

Experiment = exp2, Age = 19.1:
 Ast353_prop response      SE  df asymp.LCL asymp.UCL
       0.477    0.979 0.00783 Inf     0.957     0.990

Experiment = exp3_4, Age = 19.1:
 Ast353_prop response      SE  df asymp.LCL asymp.UCL
       0.477    0.338 0.08890 Inf     0.190     0.527

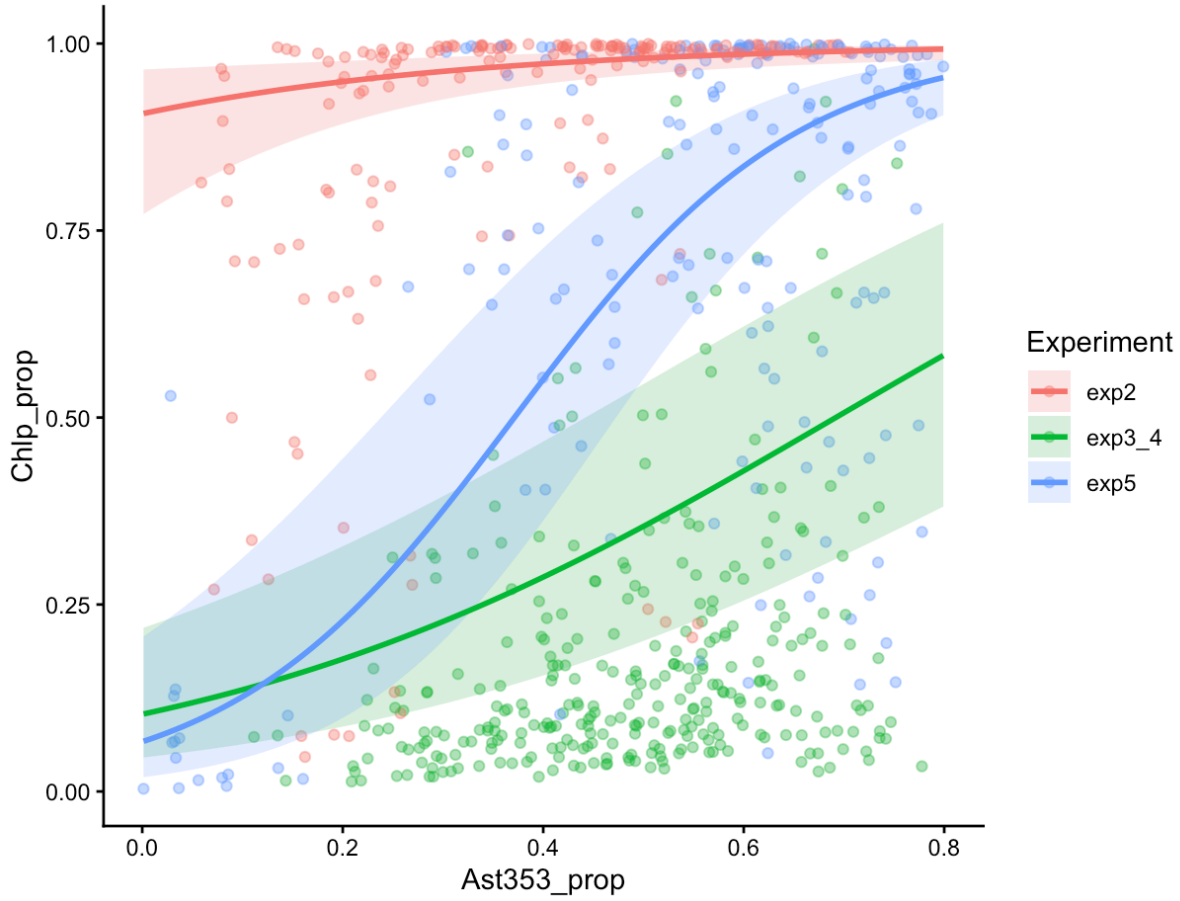
Experiment = exp5, Age = 19.1:
 Ast353_prop response      SE  df asymp.LCL asymp.UCL
       0.477    0.679 0.07990 Inf     0.508     0.813

Confidence level used: 0.95
Intervals are back-transformed from the logit scale

>emm<-emmeans(exp2345.mod14, ~Ast353_prop | Experiment | Age,
type="response", at = list (Ast353_prop = seq
(min(exp2345_n0_n1$Ast353_prop), max(exp2345_n0_n1$Ast353_prop),
length=100)))

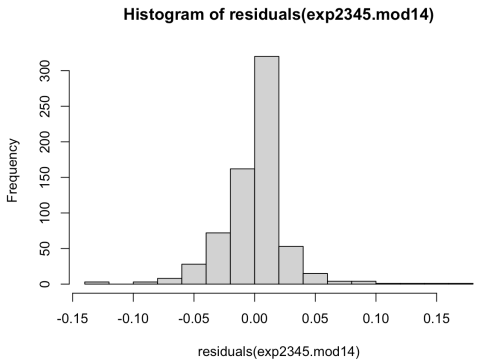
>emm_df<-as.data.frame(emm)

>ggplot() +
  geom_point(data = exp2345_n0_n1,
             aes(x = Ast353_prop, y = Chlp_prop, colour = Experiment),
             alpha = 0.4) +
  geom_ribbon(data = emm_df,
              aes(x = Ast353_prop, ymin = asymp.LCL, ymax = asymp.UCL,
fill = Experiment),
              alpha = 0.2) +
  geom_line(data = emm_df,
            aes(x = Ast353_prop, y = response, colour = Experiment),
            linewidth = 1) +
  theme_classic()
```

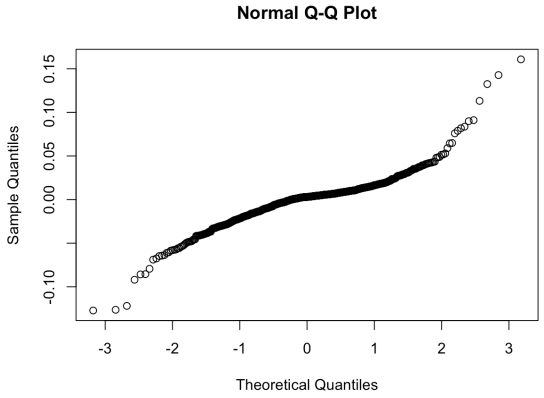


#Model residuals remain approximately normally distributed according to the histogram and Q-Q plot, but the Shapiro-Wilk test indicates that statistically they are still non-normal:

```
>hist(residuals(exp2345.mod14))
```




```
>qqnorm(residuals(exp2345.mod14))
```




```
>shapiro.test(residuals(exp2345.mod14))
Shapiro-Wilk normality test

data:  residuals(exp2345.mod14)
W = 0.89533, p-value < 2.2e-16
```